Configuration Summary

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FORECAST ANALYSIS — CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-09-02 15:30:51
======================================================================

── Data Preparation ──

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

── Model Training ──

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 10
rollout_epoch_end...................... 60
rollout_loss_decay..................... 0.92
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

── Model Inference ──

  model_path............................. checkpoints_64x128/best_model.pt
  input_data_path........................ data/processed_64x128/sst_final.nc
  run_validation......................... True
  run_test............................... True
  forecast_lead_steps.................... 90
  chain_stride_steps..................... 90
  output_dir............................. results_64x128/
  val_predictions_file................... predictions_val.nc
  test_predictions_file.................. predictions_test.nc

── Results Analysis ──
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val_predictions_path..................... results_64x128/predictions_val.nc
test_predictions_path.................... results_64x128/predictions_test.nc
processed_data_path...................... data/processed_64x128/sst_final.nc
analyse_validation....................... True
analyse_test............................. True
std_params_path.......................... data/processed_64x128/std_params.nc
trend_params_path........................ data/processed_64x128/trend_params.nc
compute_persistence...................... True
compute_climatology...................... True
acc_clim_years........................... (1991, 2020)
acc_clim_smoothing_window................ 5
n_sample_chains.......................... 9
spatial_lead_steps....................... [1, 3, 7, 14, 21, 30, 45, 60, 90]
compute_enso34........................... True
enso34_lat_bounds........................ (-5.0, 5.0)
enso34_lon_bounds........................ (190.0, 240.0)
enso34_clim_years........................ (1991, 2020)
enso34_n_sample_chains................... 9
compute_dhw.............................. True
dhw_baseline_years....................... (1985, 2012)
dhw_output_val........................... dhw_val.nc
dhw_output_test.......................... dhw_test.nc
output_dir............................... results_64x128/
report_file.............................. results_64x128/analysis_report.txt
metrics_csv.............................. results_64x128/metrics_by_lead_step.csv
figures_dir.............................. results_64x128/figures/
analysis_pdf............................. results_64x128/forecast_analysis.pdf
generate_plots........................... True
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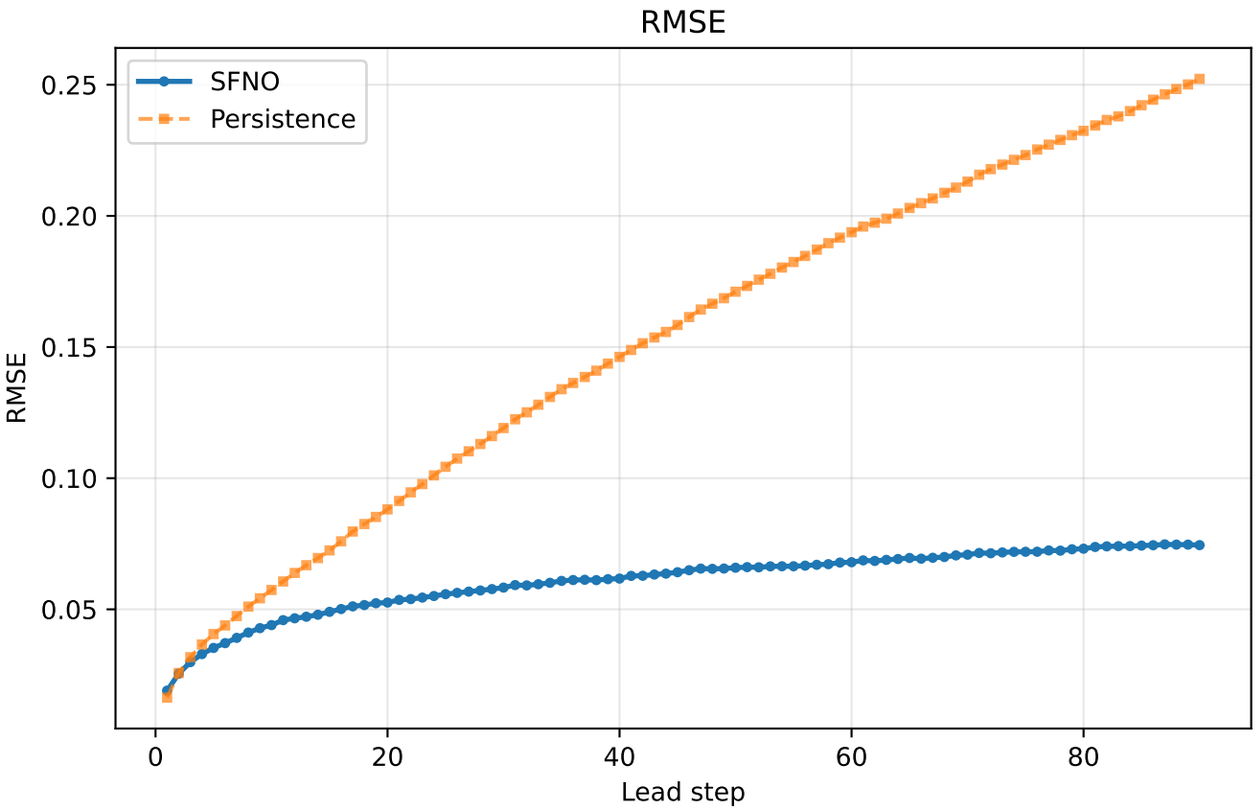
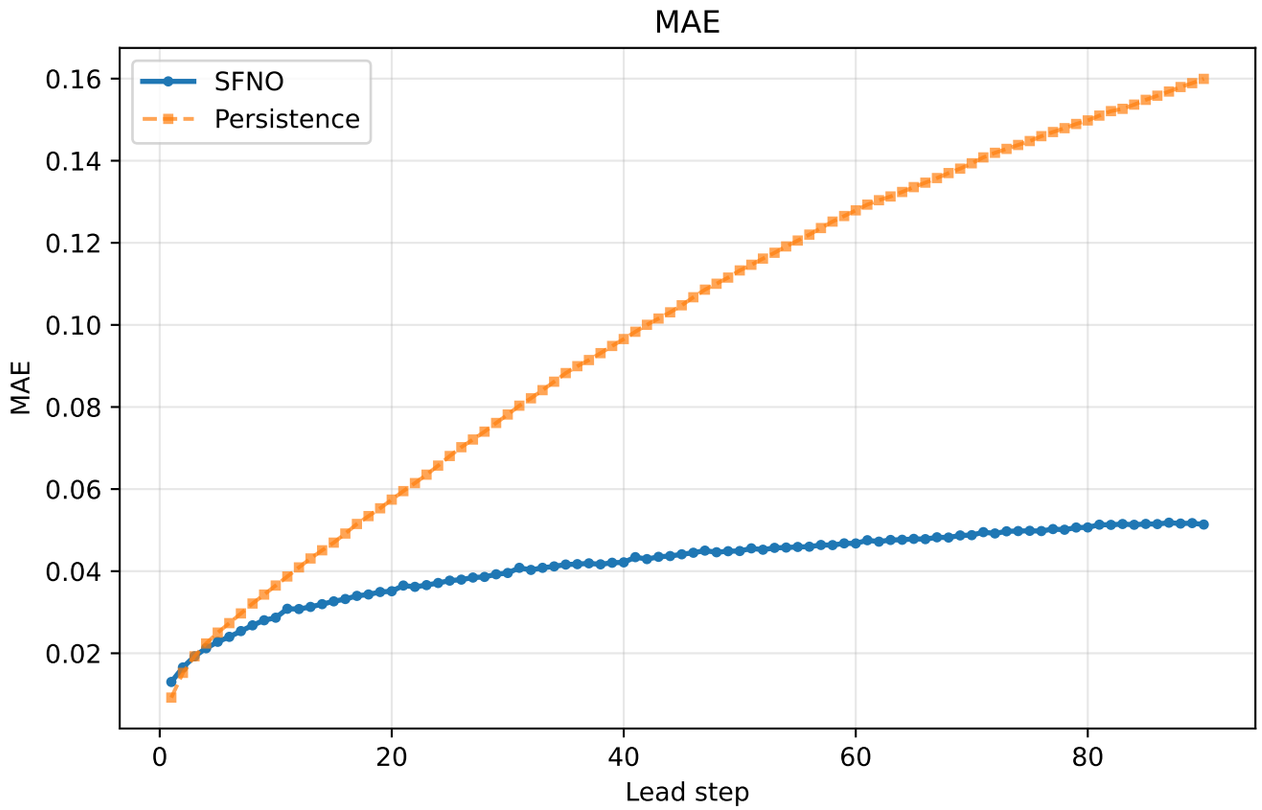
Metrics by Lead Step — Validation

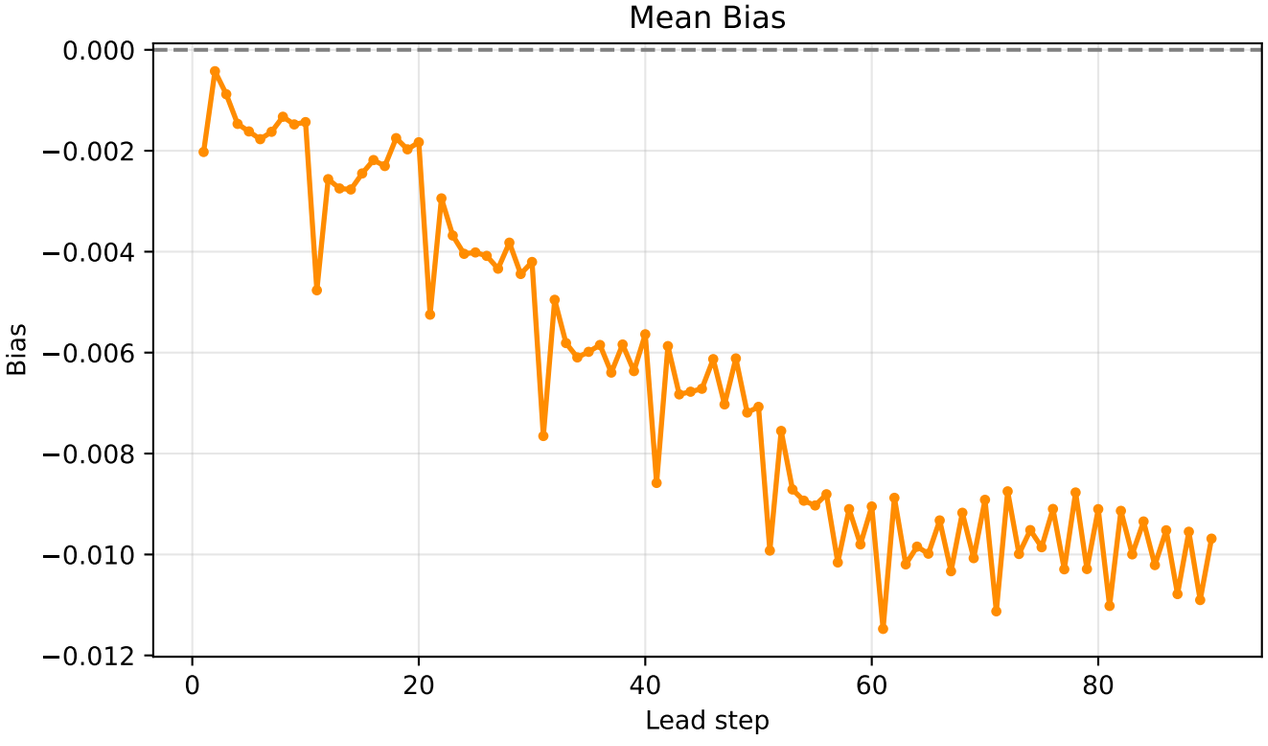
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0190	0.0164	-16.2%	0.0130	-0.0020	0.9482	0.9592	0.9996
2	0.0255	0.0257	0.9%	0.0166	-0.0004	0.9013	0.8991	0.9994
3	0.0299	0.0318	5.9%	0.0193	-0.0009	0.8593	0.8481	0.9991
4	0.0329	0.0366	10.0%	0.0212	-0.0015	0.8255	0.8008	0.9989
5	0.0353	0.0406	13.0%	0.0228	-0.0016	0.7948	0.7574	0.9988
6	0.0371	0.0439	15.4%	0.0240	-0.0018	0.7691	0.7202	0.9987
7	0.0391	0.0474	17.5%	0.0254	-0.0016	0.7430	0.6815	0.9985
8	0.0411	0.0510	19.4%	0.0268	-0.0013	0.7160	0.6425	0.9984
9	0.0428	0.0542	21.0%	0.0280	-0.0015	0.6896	0.6072	0.9982
10	0.0440	0.0574	23.4%	0.0287	-0.0014	0.6685	0.5705	0.9981
11	0.0459	0.0606	24.3%	0.0308	-0.0048	0.6355	0.5345	0.9980
12	0.0466	0.0639	27.1%	0.0308	-0.0026	0.6189	0.4984	0.9979
13	0.0472	0.0668	29.4%	0.0313	-0.0027	0.5996	0.4642	0.9978
14	0.0479	0.0695	31.1%	0.0320	-0.0028	0.5795	0.4363	0.9978
15	0.0490	0.0725	32.3%	0.0326	-0.0024	0.5595	0.4112	0.9977
16	0.0501	0.0759	34.0%	0.0332	-0.0022	0.5419	0.3812	0.9976
17	0.0511	0.0797	35.8%	0.0340	-0.0023	0.5237	0.3482	0.9975
18	0.0516	0.0825	37.4%	0.0343	-0.0018	0.5037	0.3242	0.9974
19	0.0523	0.0852	38.6%	0.0349	-0.0020	0.4833	0.3058	0.9973
20	0.0526	0.0881	40.3%	0.0351	-0.0018	0.4685	0.2821	0.9973
21	0.0536	0.0913	41.3%	0.0365	-0.0052	0.4477	0.2589	0.9972
22	0.0539	0.0946	43.0%	0.0362	-0.0029	0.4415	0.2377	0.9972
23	0.0545	0.0978	44.3%	0.0366	-0.0037	0.4262	0.2157	0.9971
24	0.0551	0.1011	45.5%	0.0371	-0.0040	0.4101	0.1932	0.9970
25	0.0558	0.1043	46.5%	0.0377	-0.0040	0.3903	0.1725	0.9970
26	0.0563	0.1075	47.6%	0.0379	-0.0041	0.3735	0.1551	0.9969
27	0.0568	0.1103	48.5%	0.0384	-0.0043	0.3560	0.1418	0.9969
28	0.0572	0.1130	49.4%	0.0386	-0.0038	0.3423	0.1313	0.9968
29	0.0577	0.1161	50.3%	0.0392	-0.0044	0.3284	0.1164	0.9968
30	0.0583	0.1191	51.1%	0.0396	-0.0042	0.3156	0.1038	0.9967
31	0.0592	0.1225	51.6%	0.0408	-0.0077	0.2975	0.0889	0.9966
32	0.0591	0.1251	52.7%	0.0403	-0.0050	0.2960	0.0821	0.9966
33	0.0596	0.1280	53.5%	0.0408	-0.0058	0.2842	0.0694	0.9965
34	0.0601	0.1310	54.1%	0.0412	-0.0061	0.2778	0.0582	0.9965
35	0.0608	0.1339	54.6%	0.0416	-0.0060	0.2689	0.0464	0.9964
36	0.0612	0.1363	55.1%	0.0417	-0.0058	0.2591	0.0400	0.9964
37	0.0613	0.1386	55.8%	0.0419	-0.0064	0.2520	0.0378	0.9963
38	0.0611	0.1410	56.6%	0.0417	-0.0058	0.2422	0.0334	0.9964
39	0.0615	0.1437	57.2%	0.0421	-0.0064	0.2314	0.0265	0.9963
40	0.0617	0.1462	57.8%	0.0421	-0.0056	0.2196	0.0227	0.9963
41	0.0627	0.1489	57.9%	0.0434	-0.0086	0.1963	0.0178	0.9962
42	0.0628	0.1514	58.5%	0.0429	-0.0059	0.1930	0.0118	0.9962
43	0.0633	0.1536	58.8%	0.0435	-0.0068	0.1761	0.0084	0.9961
44	0.0636	0.1558	59.2%	0.0437	-0.0068	0.1647	0.0082	0.9961
45	0.0642	0.1584	59.5%	0.0441	-0.0067	0.1515	-0.0016	0.9960
46	0.0649	0.1614	59.8%	0.0445	-0.0061	0.1378	-0.0150	0.9959
47	0.0655	0.1643	60.1%	0.0450	-0.0070	0.1299	-0.0262	0.9958
48	0.0654	0.1665	60.7%	0.0446	-0.0061	0.1297	-0.0296	0.9958
49	0.0656	0.1686	61.1%	0.0449	-0.0072	0.1285	-0.0316	0.9958
50	0.0659	0.1711	61.5%	0.0449	-0.0071	0.1254	-0.0359	0.9958
51	0.0661	0.1733	61.9%	0.0456	-0.0099	0.1154	-0.0392	0.9957
52	0.0660	0.1757	62.4%	0.0452	-0.0076	0.1094	-0.0454	0.9958
53	0.0664	0.1779	62.7%	0.0457	-0.0087	0.1012	-0.0488	0.9957
54	0.0664	0.1803	63.2%	0.0458	-0.0089	0.1064	-0.0525	0.9957
55	0.0664	0.1824	63.6%	0.0459	-0.0090	0.1043	-0.0540	0.9957
56	0.0667	0.1847	63.9%	0.0460	-0.0088	0.0989	-0.0585	0.9957
57	0.0670	0.1871	64.2%	0.0464	-0.0102	0.0978	-0.0631	0.9956
58	0.0672	0.1896	64.5%	0.0464	-0.0091	0.0933	-0.0696	0.9956
59	0.0678	0.1917	64.6%	0.0468	-0.0098	0.0860	-0.0708	0.9955
60	0.0680	0.1938	64.9%	0.0468	-0.0090	0.0866	-0.0716	0.9955
61	0.0687	0.1959	65.0%	0.0475	-0.0115	0.0772	-0.0755	0.9954
62	0.0684	0.1974	65.3%	0.0472	-0.0089	0.0699	-0.0678	0.9954
63	0.0689	0.1989	65.4%	0.0476	-0.0102	0.0648	-0.0586	0.9954
64	0.0692	0.2009	65.6%	0.0477	-0.0098	0.0608	-0.0567	0.9953
65	0.0696	0.2030	65.7%	0.0479	-0.0100	0.0543	-0.0588	0.9953
66	0.0693	0.2049	66.2%	0.0478	-0.0093	0.0519	-0.0583	0.9953
67	0.0697	0.2067	66.3%	0.0483	-0.0103	0.0506	-0.0556	0.9953
68	0.0700	0.2088	66.5%	0.0482	-0.0092	0.0466	-0.0542	0.9952
69	0.0706	0.2108	66.5%	0.0487	-0.0101	0.0413	-0.0508	0.9952
70	0.0708	0.2131	66.8%	0.0488	-0.0089	0.0370	-0.0529	0.9951
71	0.0715	0.2157	66.9%	0.0495	-0.0111	0.0301	-0.0598	0.9950
72	0.0714	0.2178	67.2%	0.0492	-0.0087	0.0270	-0.0609	0.9950
73	0.0717	0.2196	67.3%	0.0497	-0.0100	0.0250	-0.0564	0.9950
74	0.0719	0.2214	67.5%	0.0498	-0.0095	0.0254	-0.0508	0.9950
75	0.0719	0.2232	67.8%	0.0498	-0.0099	0.0242	-0.0483	0.9950
76	0.0720	0.2253	68.1%	0.0498	-0.0091	0.0185	-0.0507	0.9950
77	0.0724	0.2272	68.1%	0.0503	-0.0103	0.0131	-0.0482	0.9949
78	0.0724	0.2290	68.4%	0.0501	-0.0088	0.0057	-0.0477	0.9949
79	0.0729	0.2308	68.4%	0.0506	-0.0103	-0.0025	-0.0476	0.9948
80	0.0731	0.2324	68.5%	0.0507	-0.0091	-0.0112	-0.0445	0.9948
81	0.0738	0.2345	68.5%	0.0513	-0.0110	-0.0176	-0.0448	0.9947
82	0.0740	0.2366	68.7%	0.0513	-0.0091	-0.0230	-0.0443	0.9947
83	0.0741	0.2380	68.9%	0.0515	-0.0100	-0.0236	-0.0426	0.9947
84	0.0741	0.2399	69.1%	0.0513	-0.0093	-0.0216	-0.0439	0.9947
85	0.0743	0.2422	69.3%	0.0515	-0.0102	-0.0159	-0.0456	0.9946
86	0.0744	0.2443	69.5%	0.0515	-0.0095	-0.0146	-0.0445	0.9946
87	0.0748	0.2464	69.7%	0.0518	-0.0108	-0.0146	-0.0446	0.9946
88	0.0747	0.2484	69.9%	0.0516	-0.0095	-0.0157	-0.0447	0.9946
89	0.0747	0.2501	70.1%	0.0517	-0.0109	-0.0140	-0.0431	0.9946
90	0.0745	0.2522	70.5%	0.0514	-0.0097	-0.0121	-0.0441	0.9946

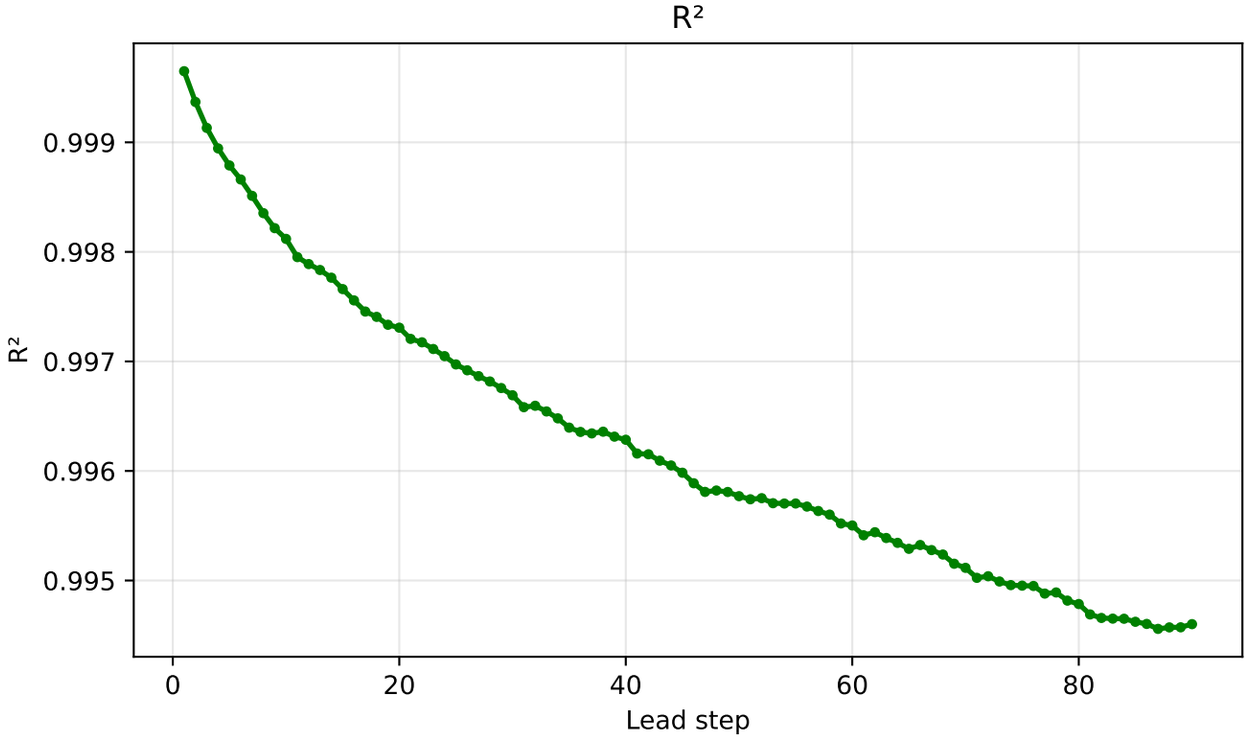
Forecast Skill vs Lead Step — Validation





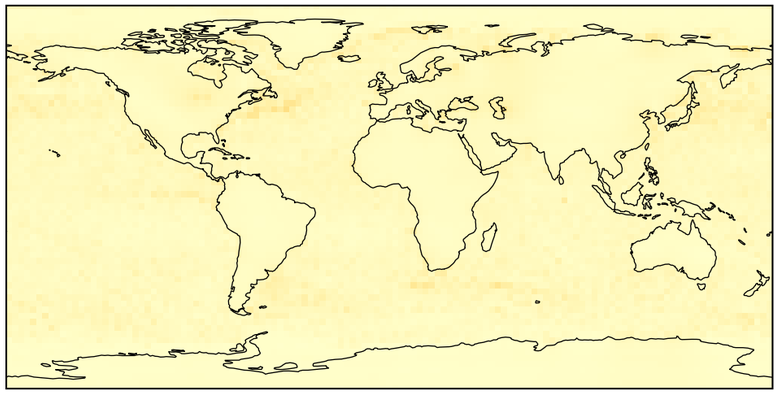
Bias and R² — Validation



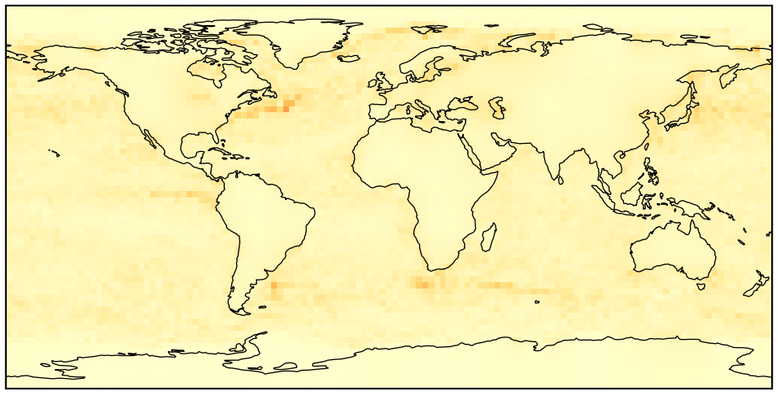


Spatial RMSE (all grid) — Validation

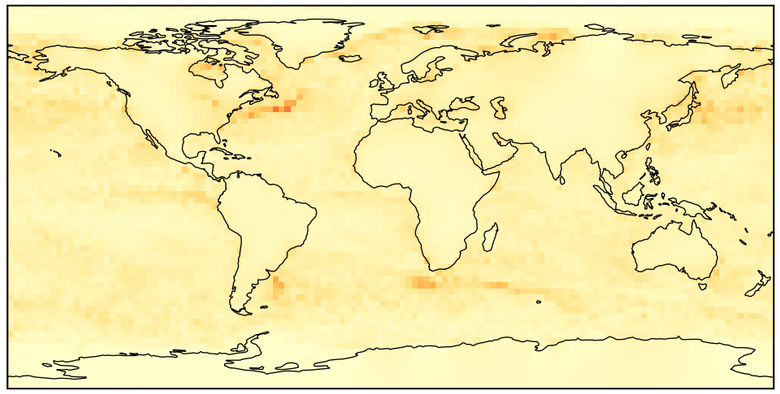
Step 1



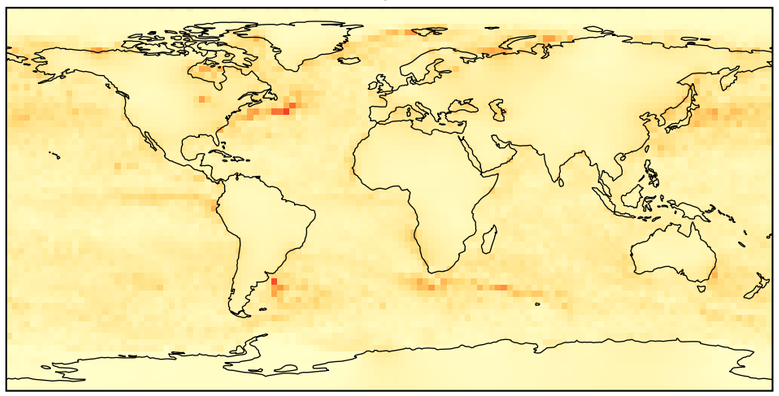
Step 3



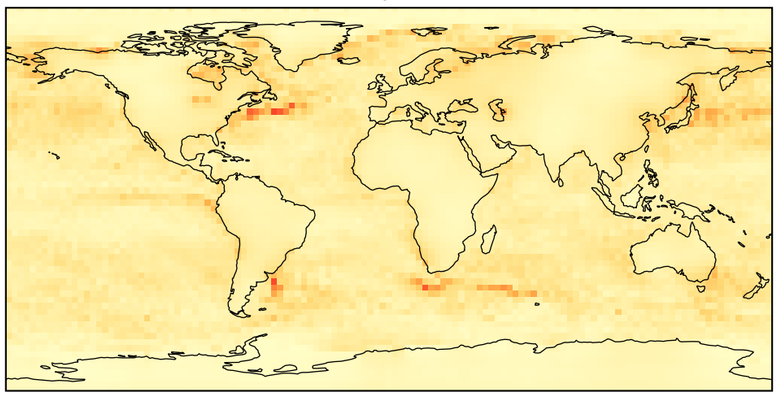
Step 7



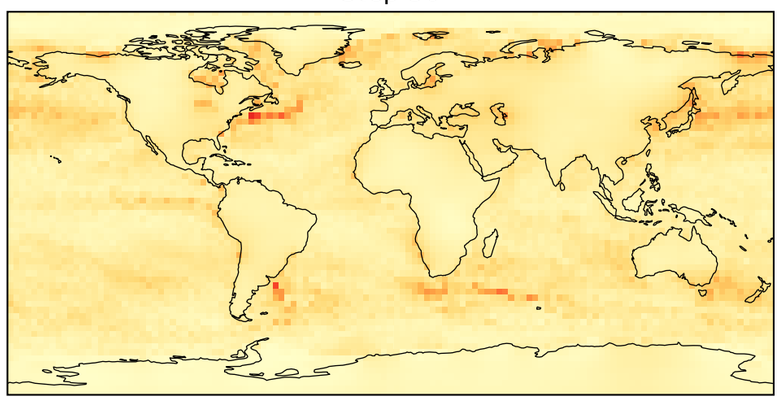
Step 14



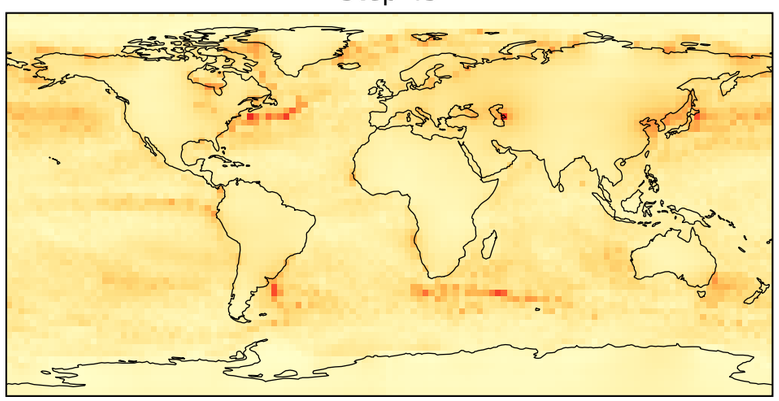
Step 21



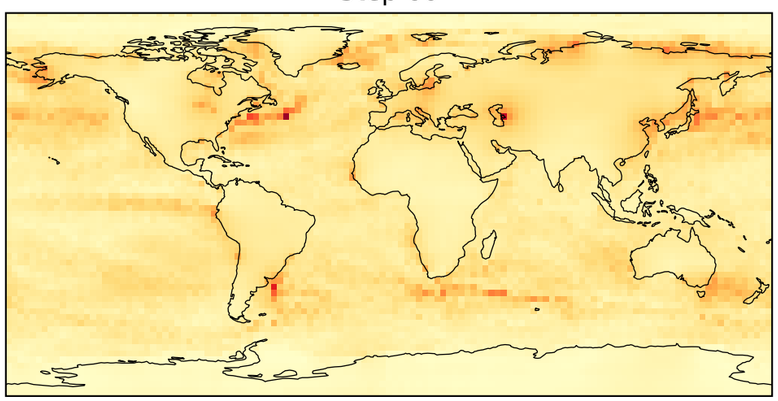
Step 30



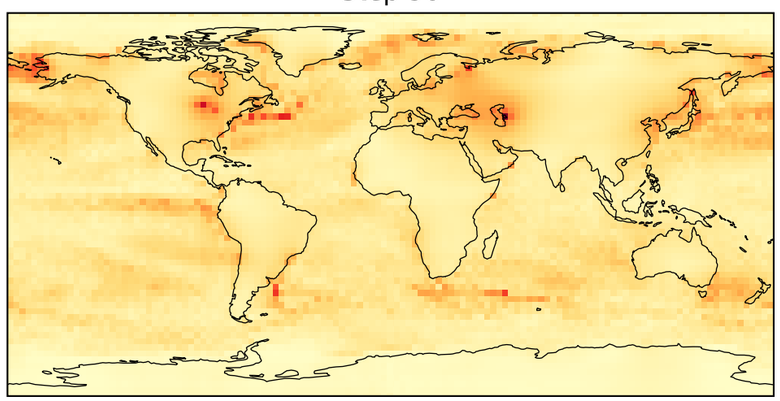
Step 45

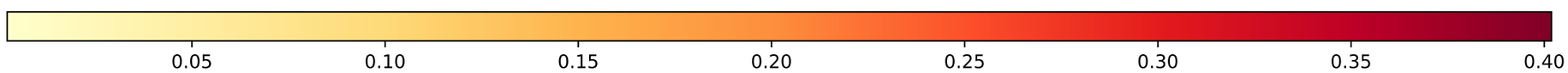


Step 60



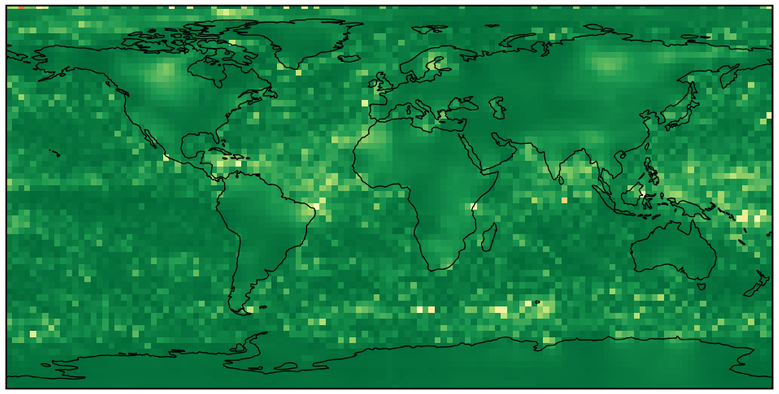
Step 90



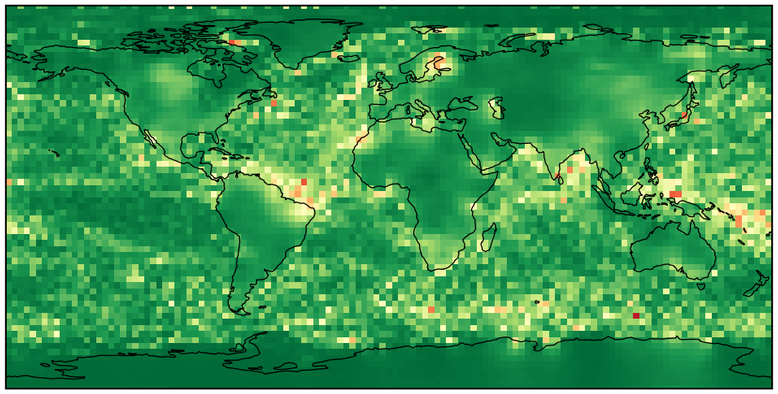


Spatial ACC (all grid) — Validation

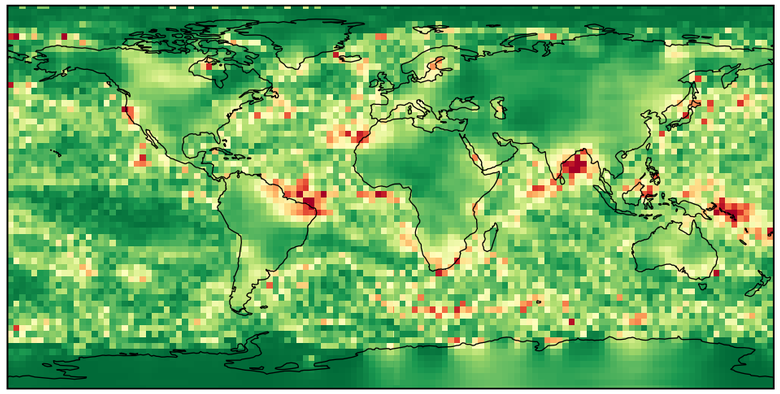
Step 1



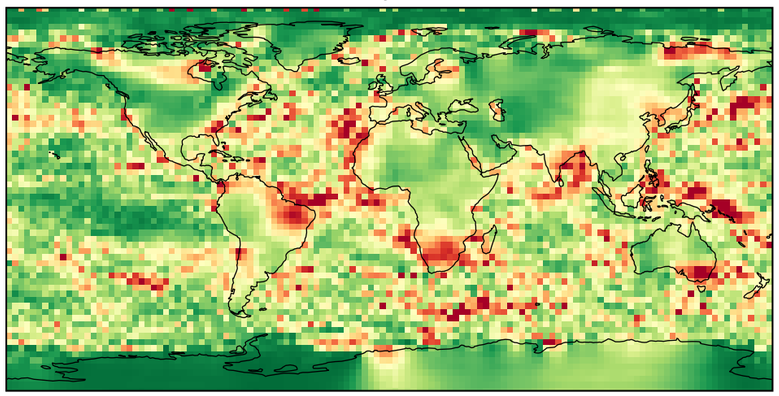
Step 3



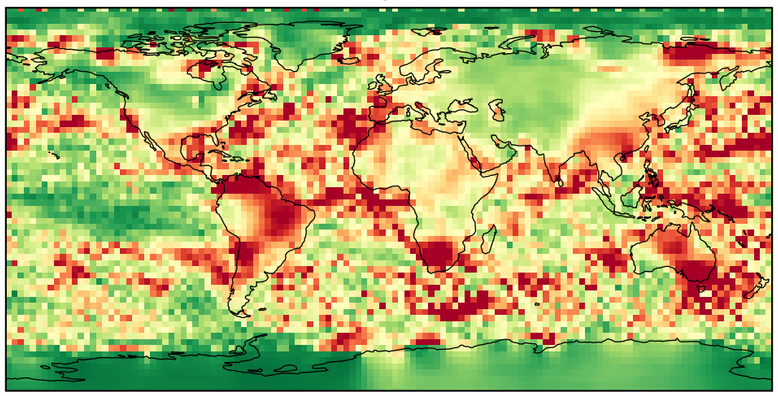
Step 7



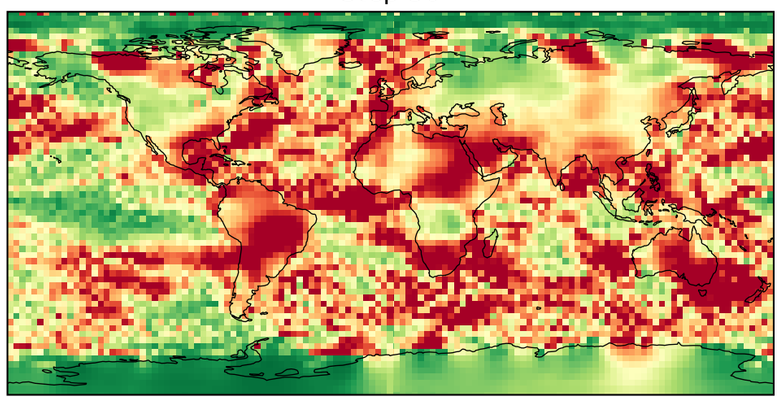
Step 14



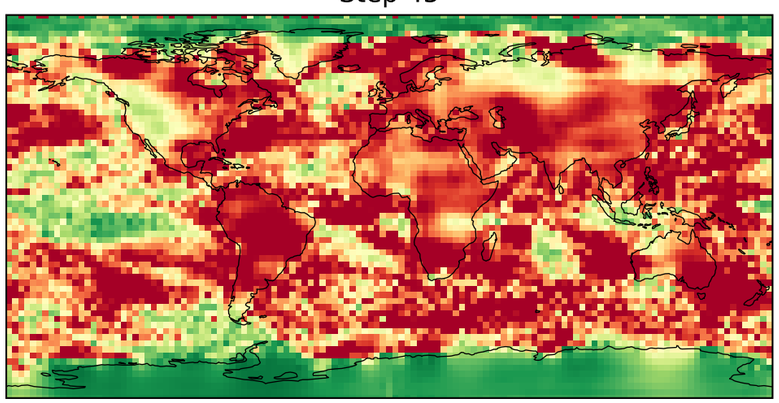
Step 21



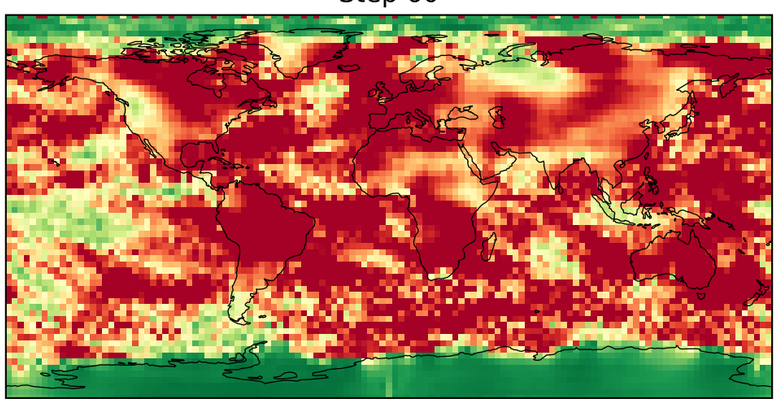
Step 30



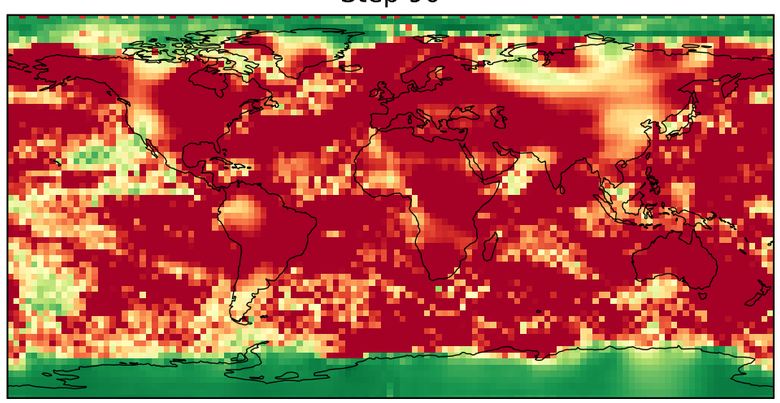
Step 45

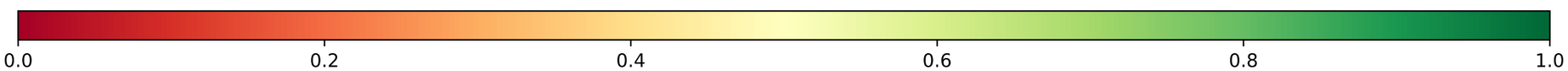


Step 60



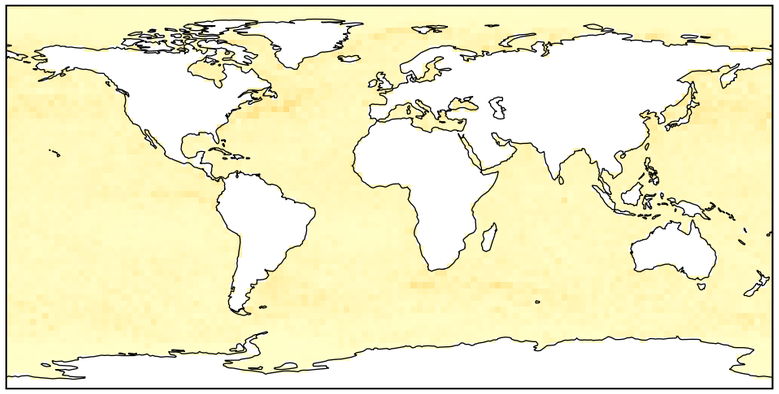
Step 90



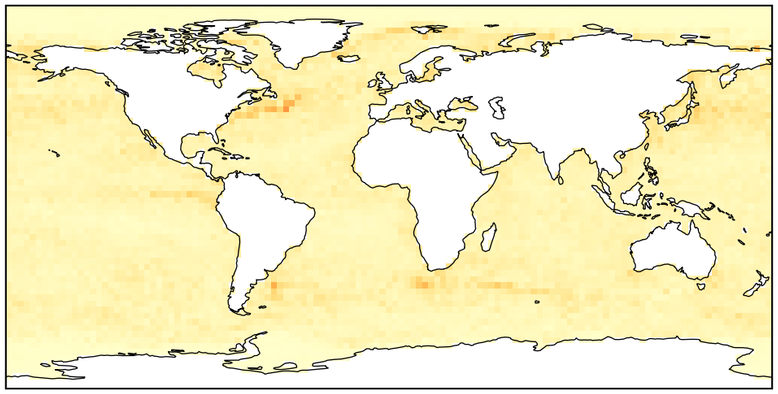


Spatial RMSE (ocean only) — Validation

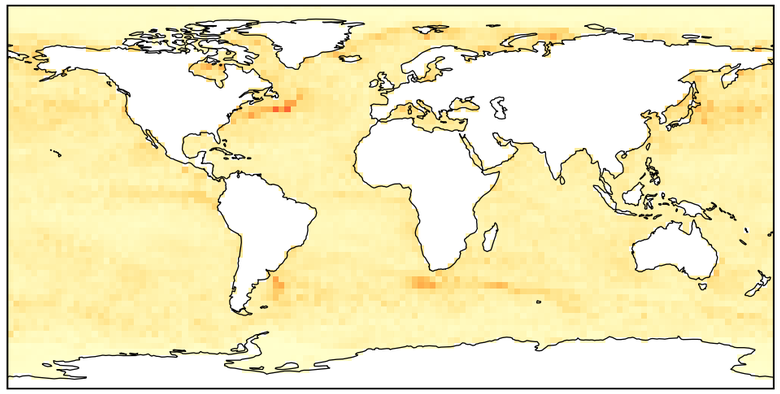
Step 1



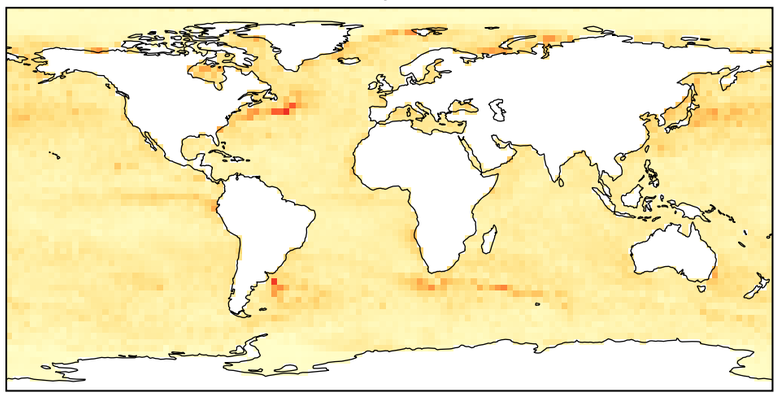
Step 3



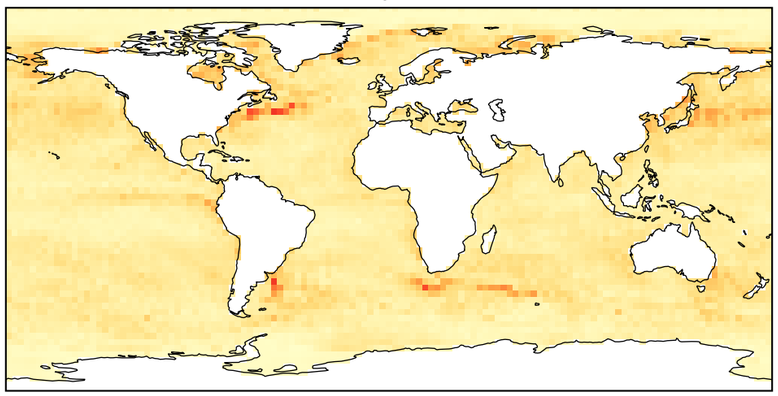
Step 7



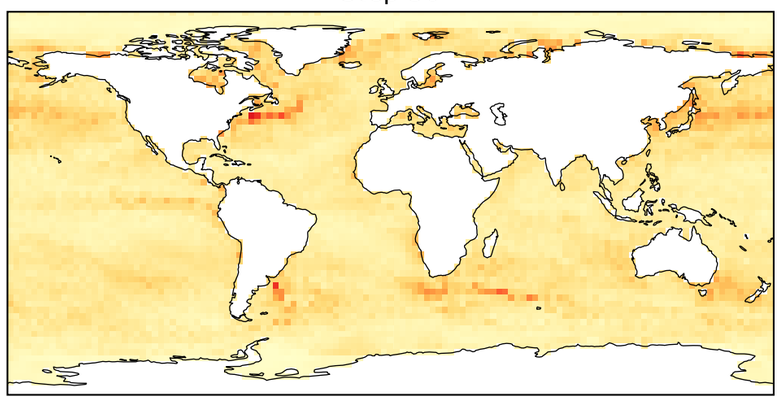
Step 14



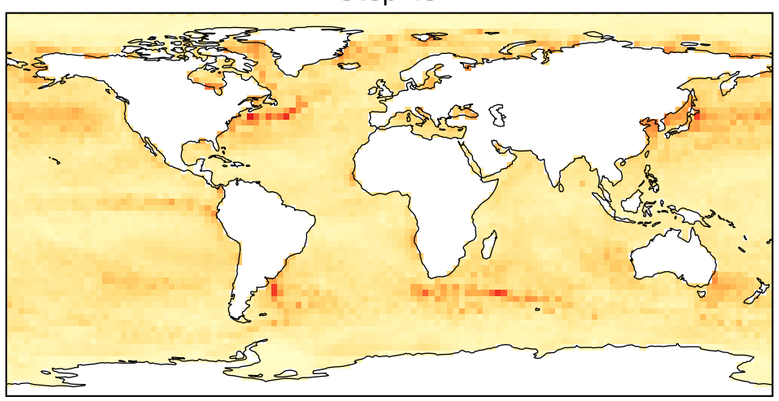
Step 21



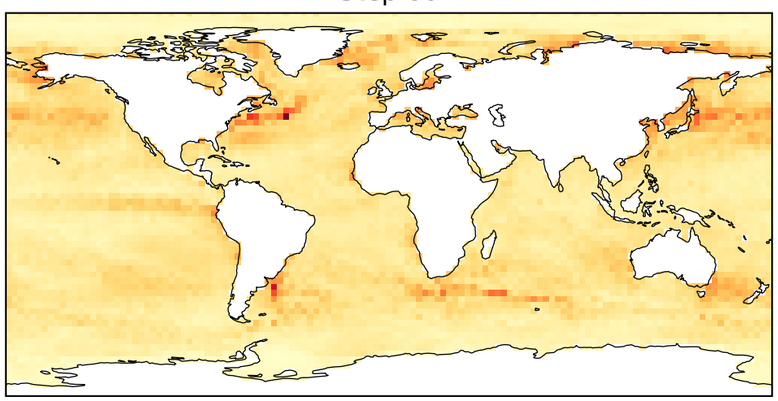
Step 30



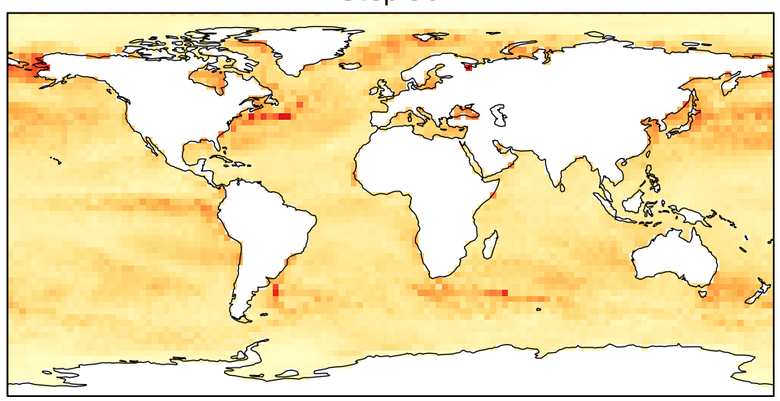
Step 45

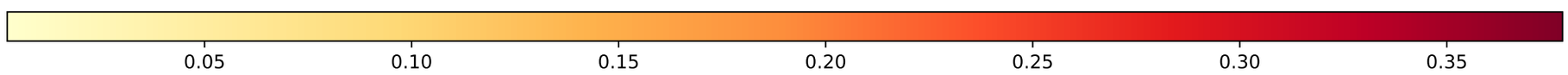


Step 60



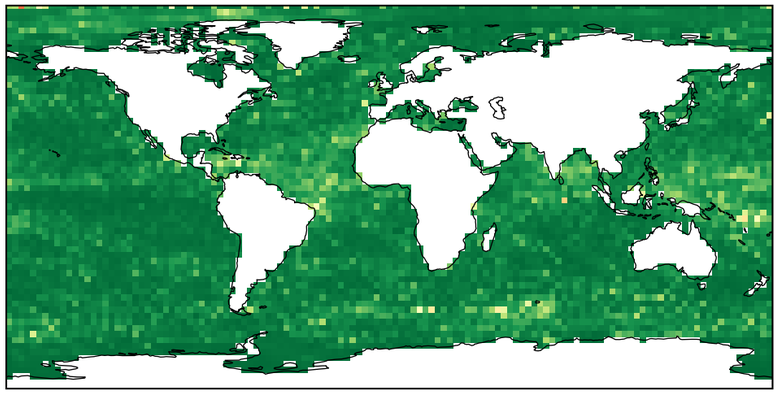
Step 90



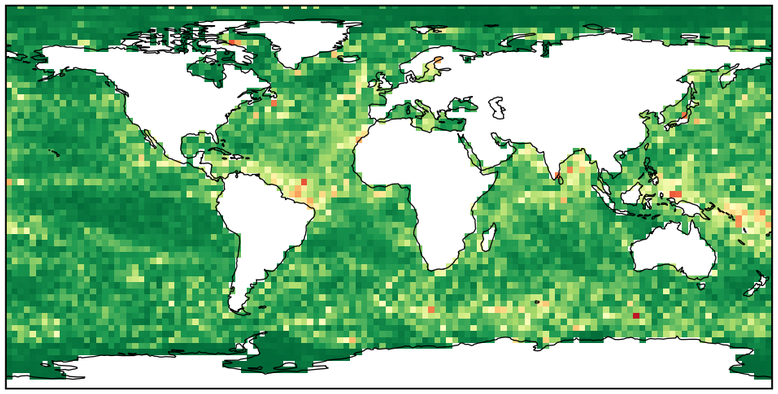


Spatial ACC (ocean only) — Validation

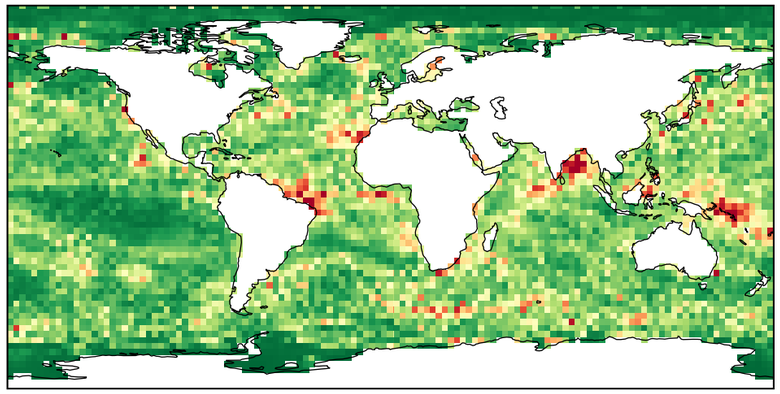
Step 1



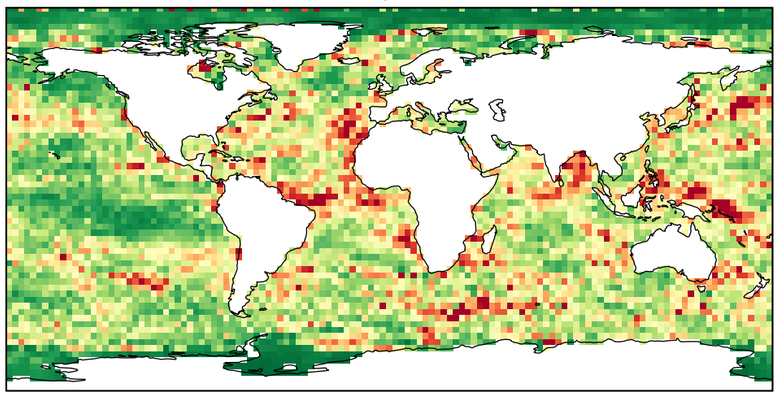
Step 3



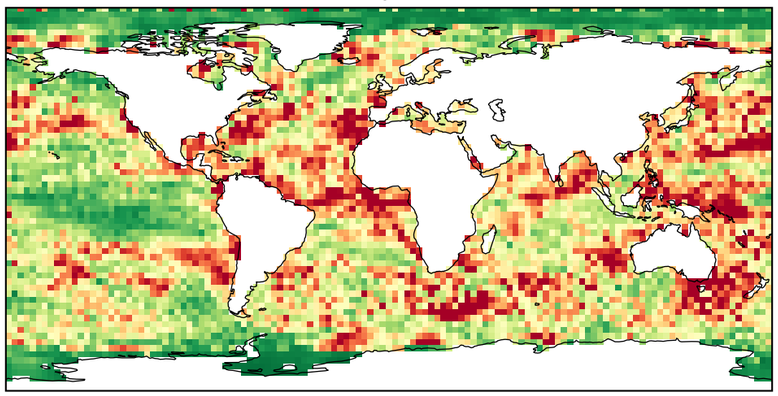
Step 7



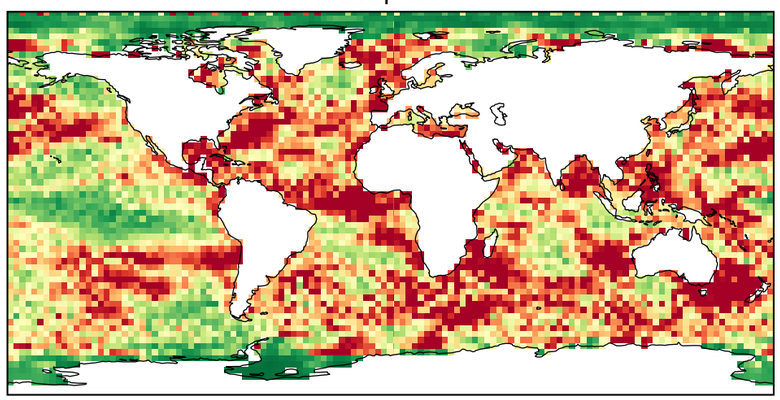
Step 14



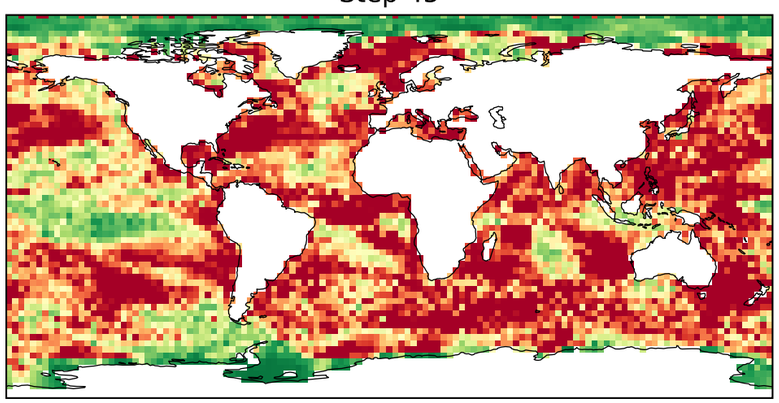
Step 21



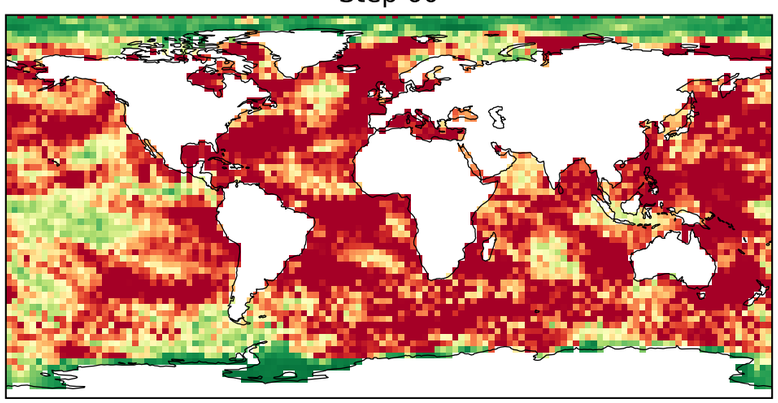
Step 30



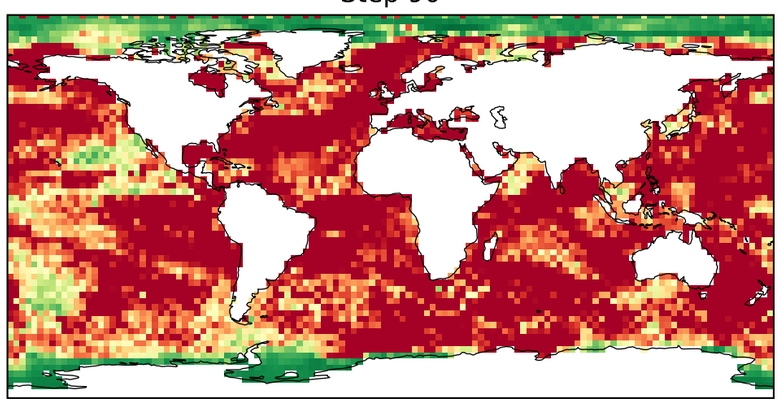
Step 45

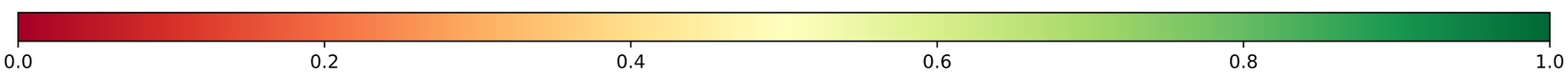


Step 60



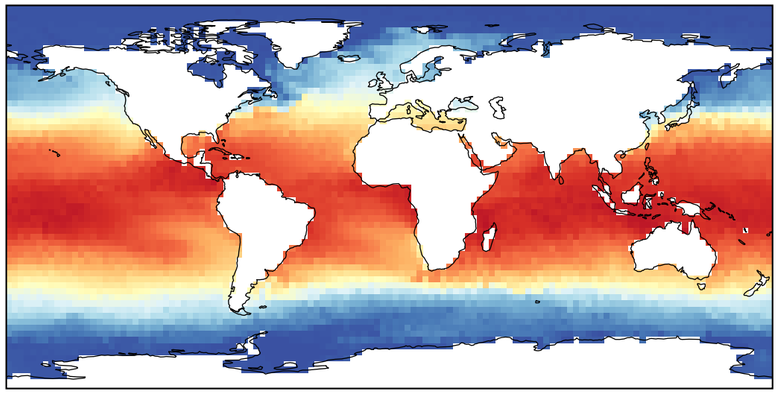
Step 90



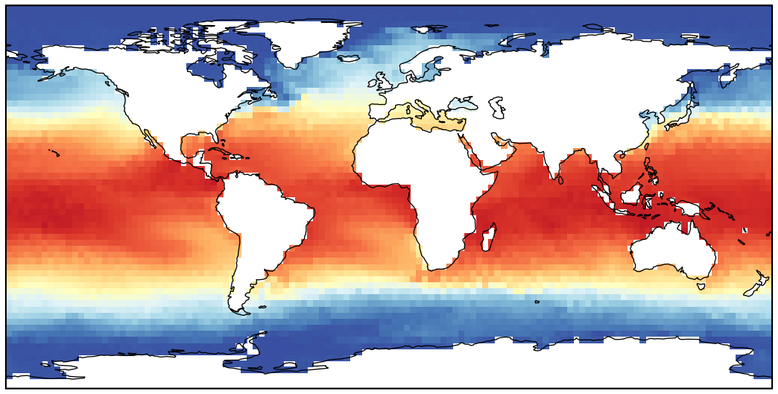


Chain init: 2016-01-11 — Validation (°C)

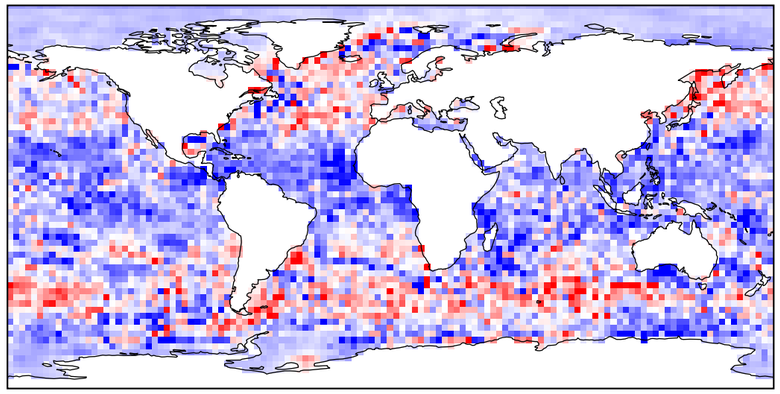
Actual — Step 1 (°C)



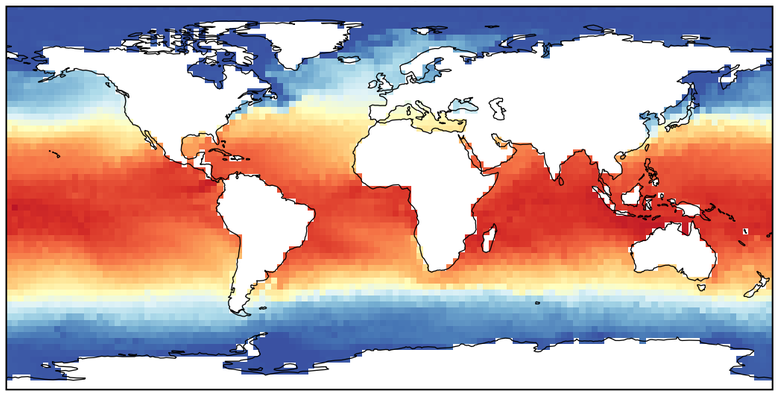
SFNO — Step 1 (°C)



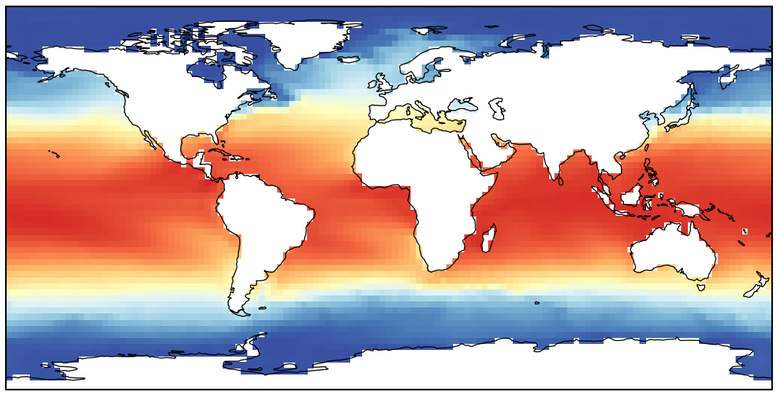
Error — Step 1 (°C)



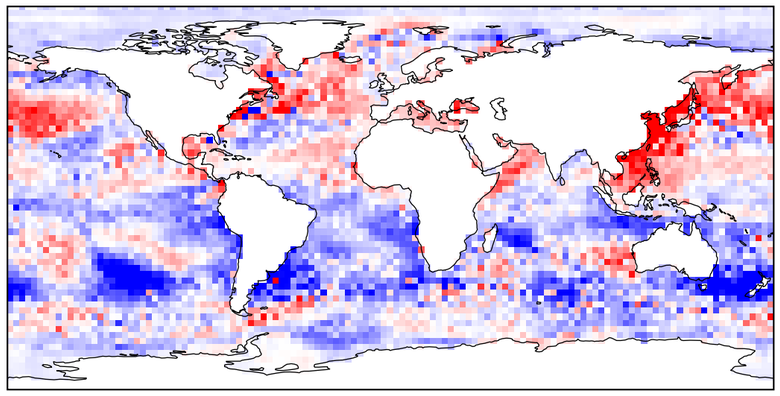
Actual — Step 46 (°C)



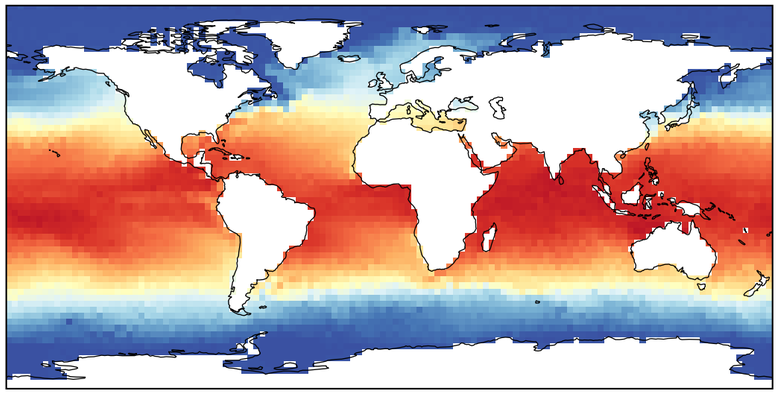
SFNO — Step 46 (°C)



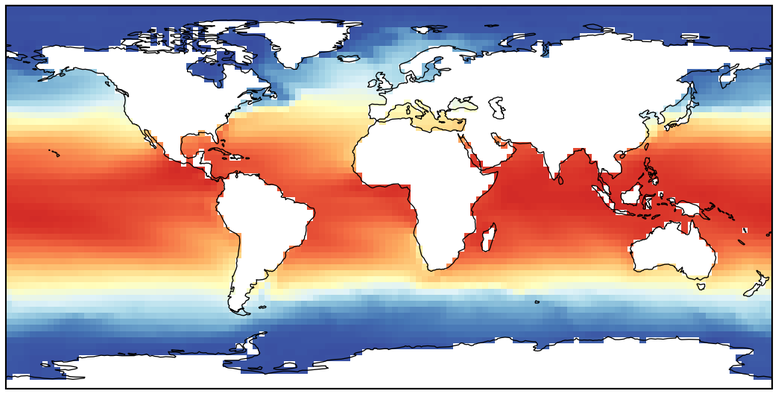
Error — Step 46 (°C)



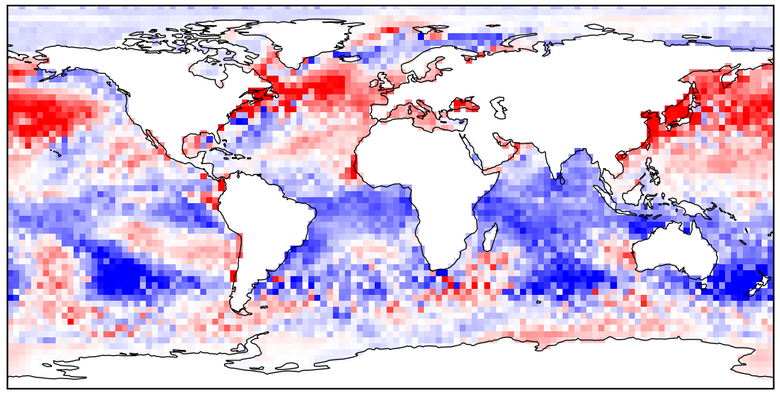
Actual — Step 90 (°C)

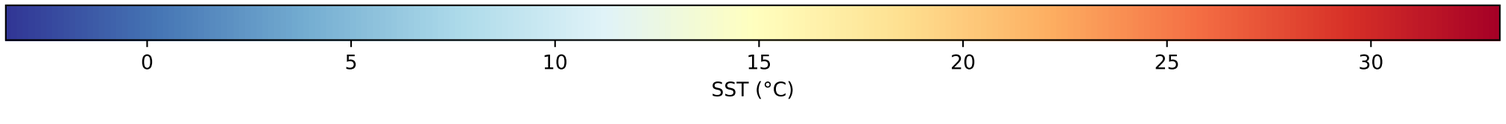


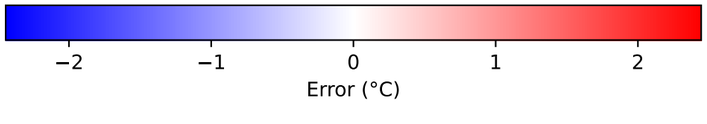
SFNO — Step 90 (°C)



Error — Step 90 (°C)

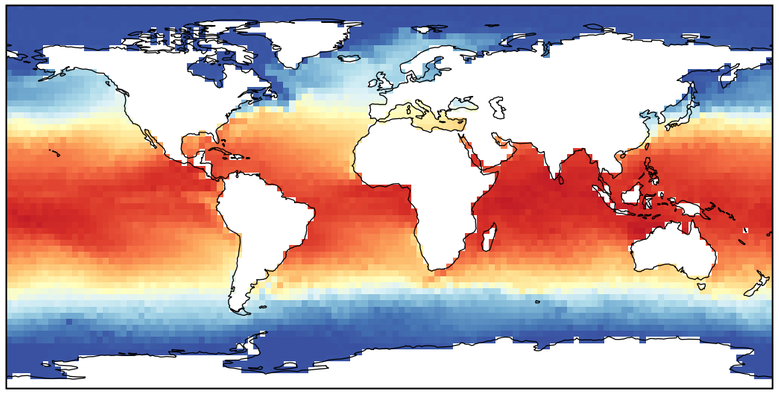




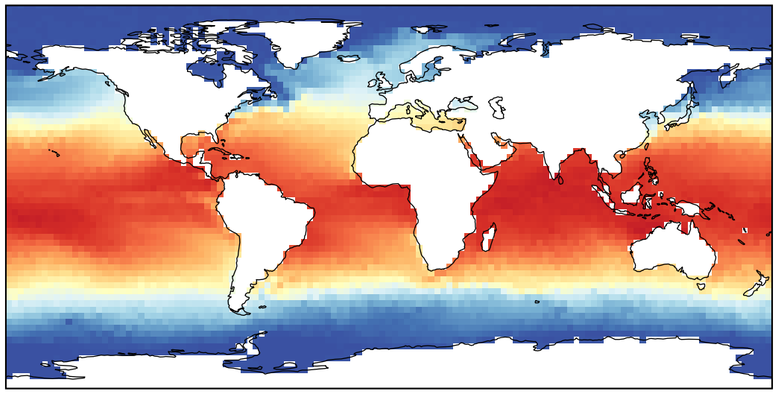


Chain init: 2016-04-10 — Validation (°C)

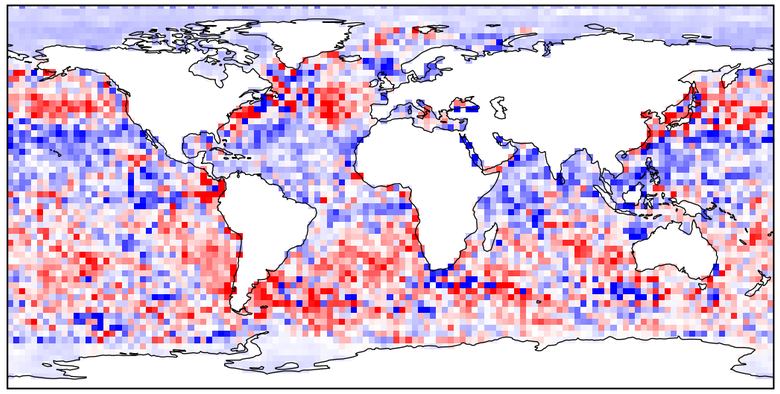
Actual — Step 1 (°C)



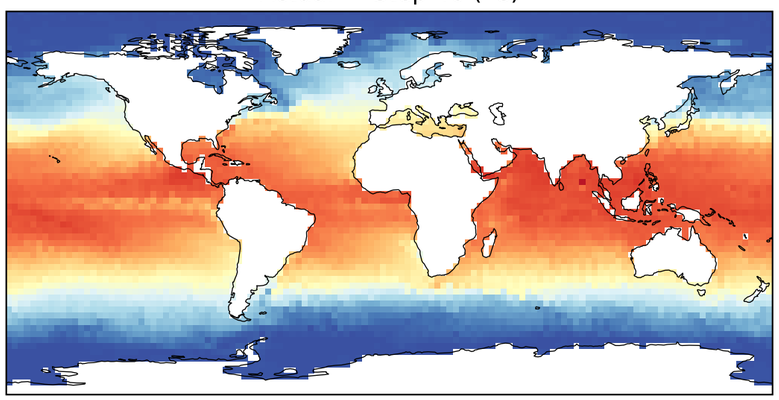
SFNO — Step 1 (°C)



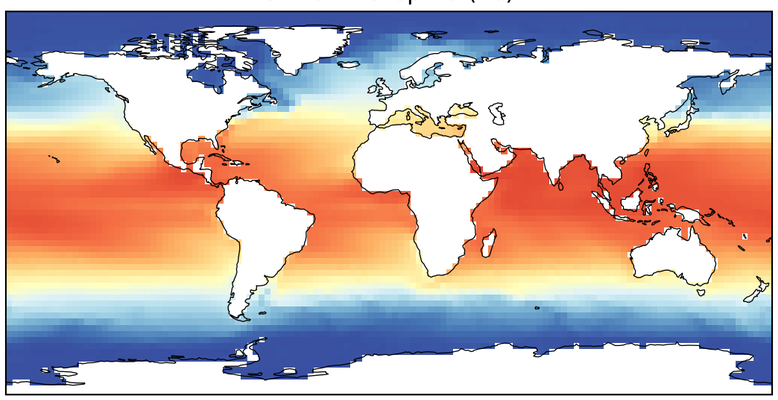
Error — Step 1 (°C)



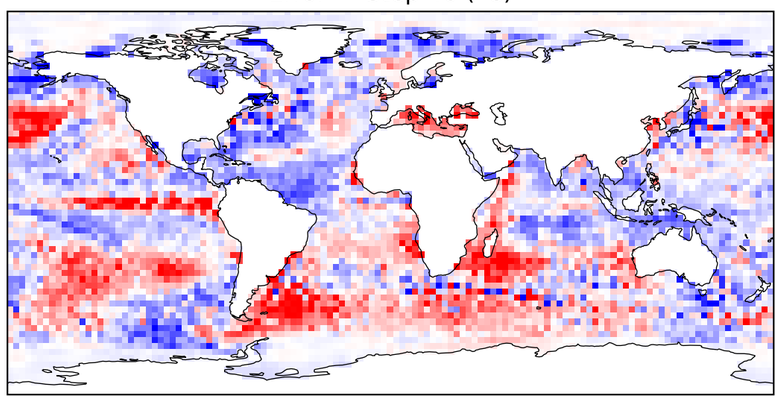
Actual — Step 46 (°C)



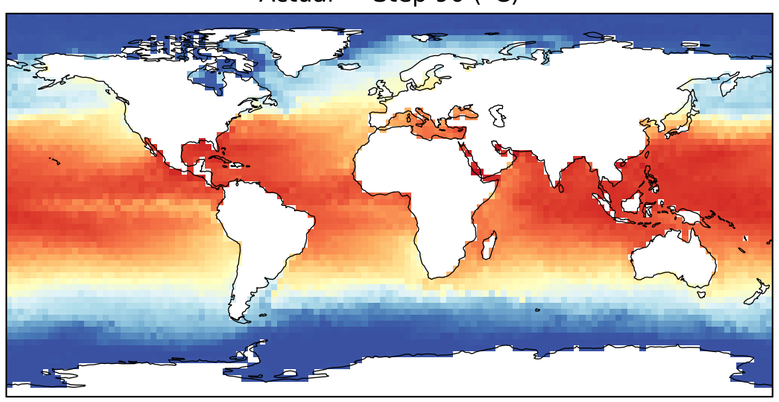
SFNO — Step 46 (°C)



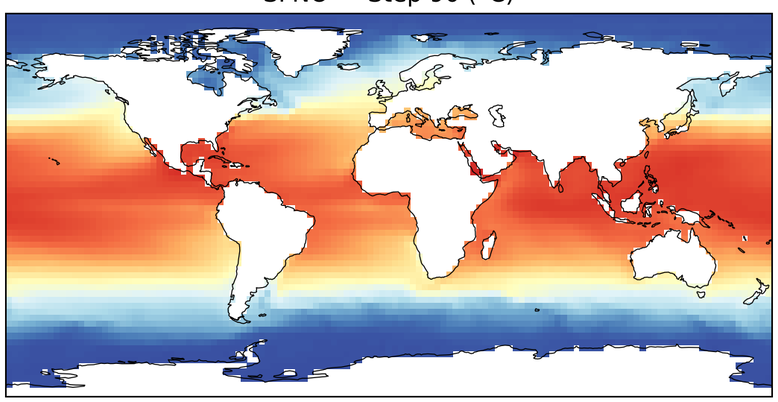
Error — Step 46 (°C)



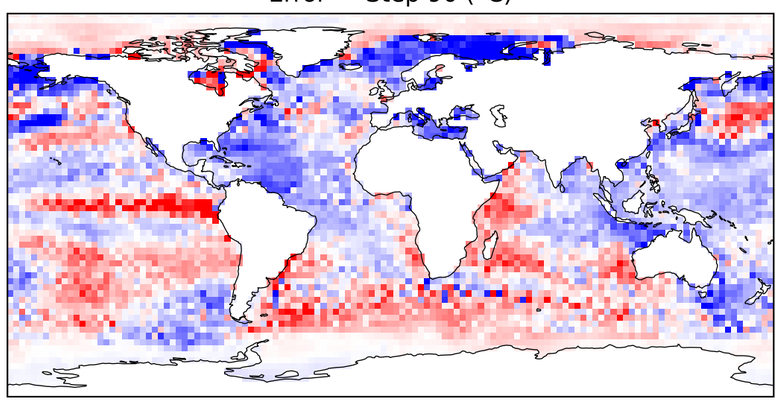
Actual — Step 90 (°C)

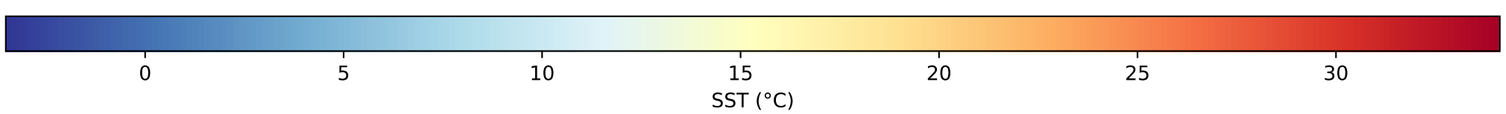


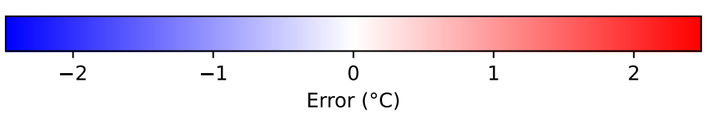
SFNO — Step 90 (°C)



Error — Step 90 (°C)

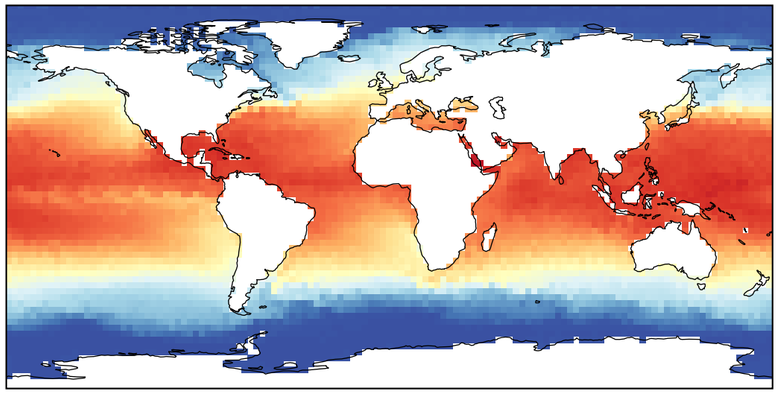




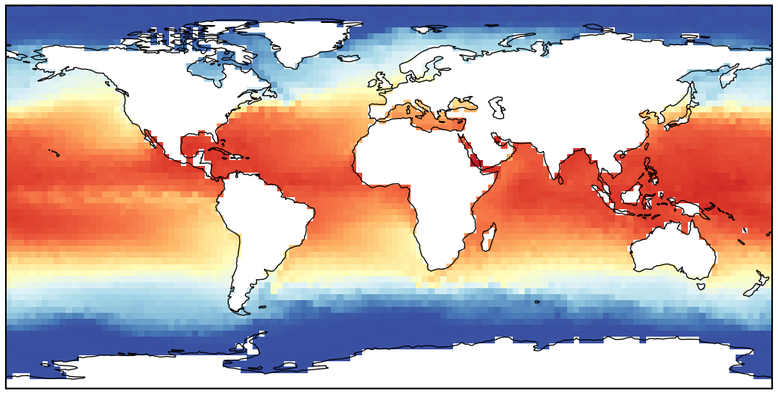


Chain init: 2016-10-07 — Validation (°C)

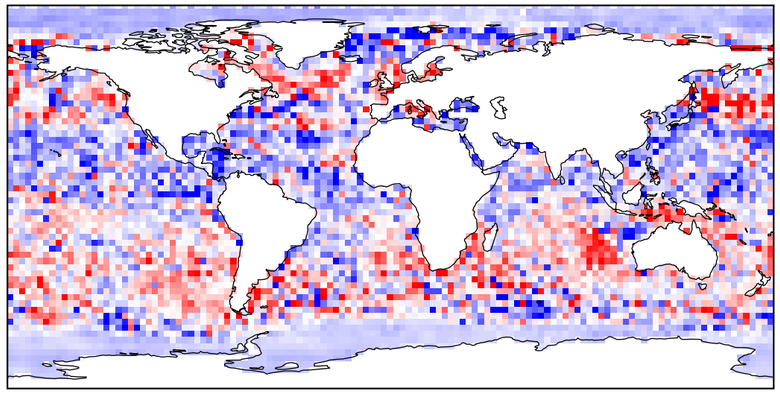
Actual — Step 1 (°C)



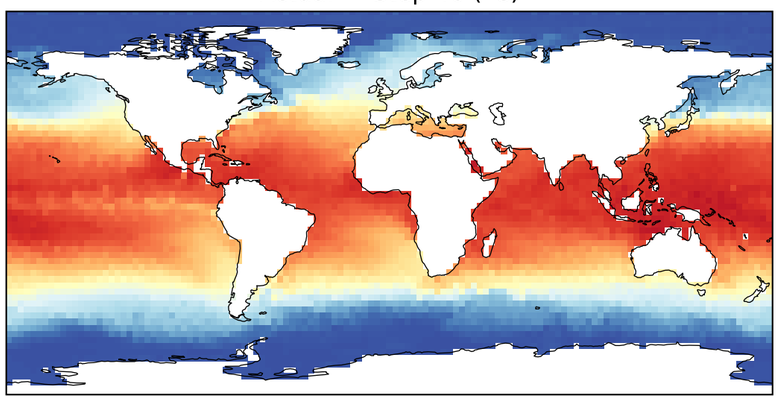
SFNO — Step 1 (°C)



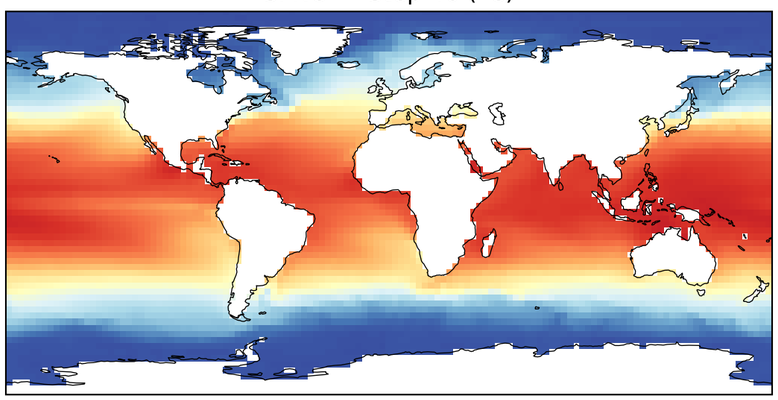
Error — Step 1 (°C)



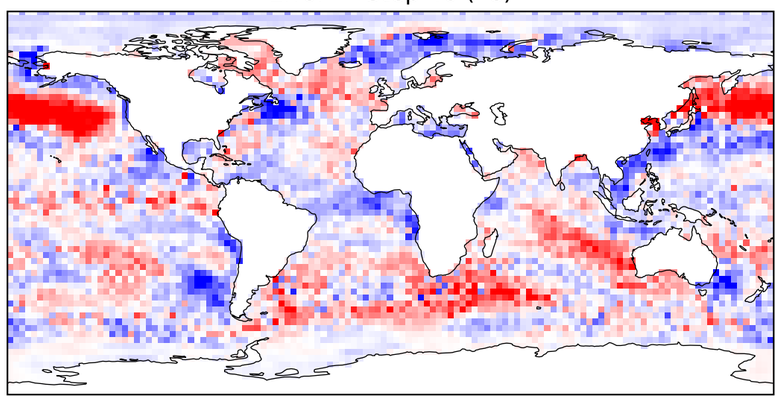
Actual — Step 46 (°C)



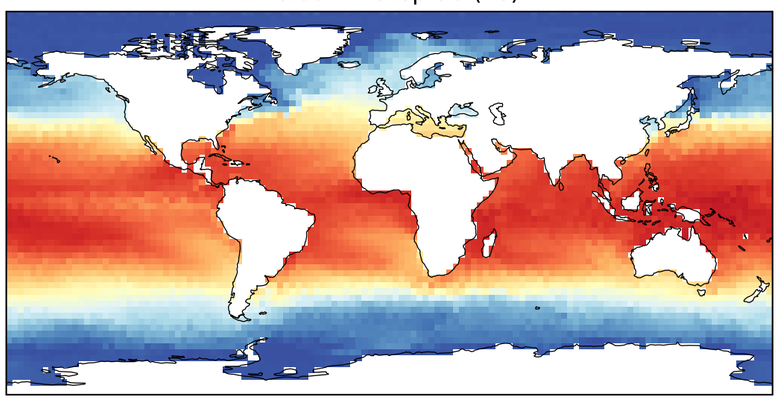
SFNO — Step 46 (°C)



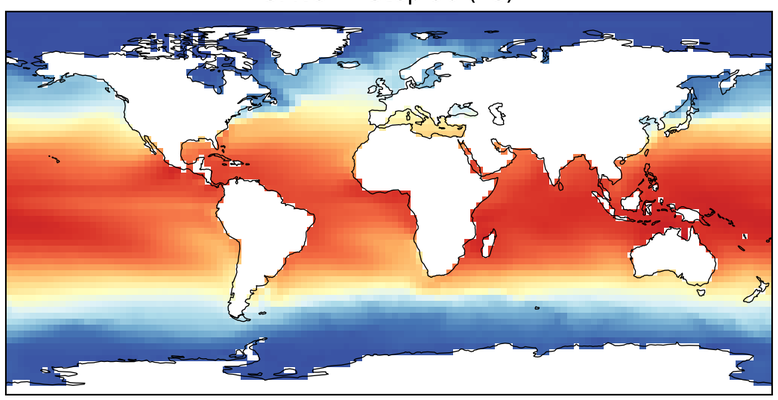
Error — Step 46 (°C)



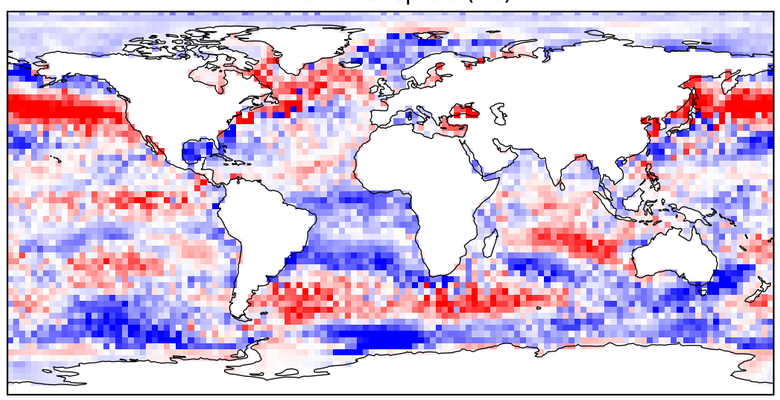
Actual — Step 90 (°C)

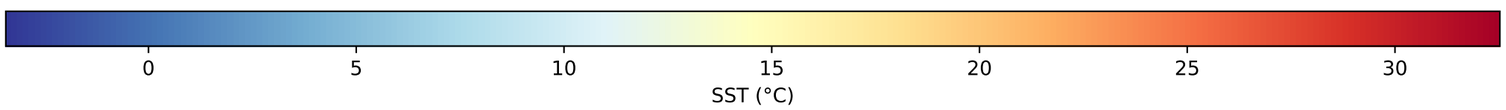


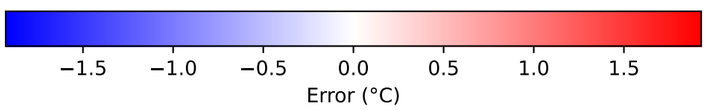
SFNO — Step 90 (°C)



Error — Step 90 (°C)

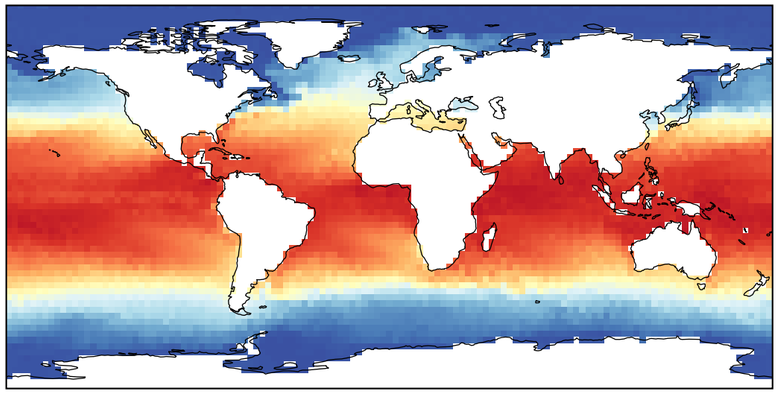




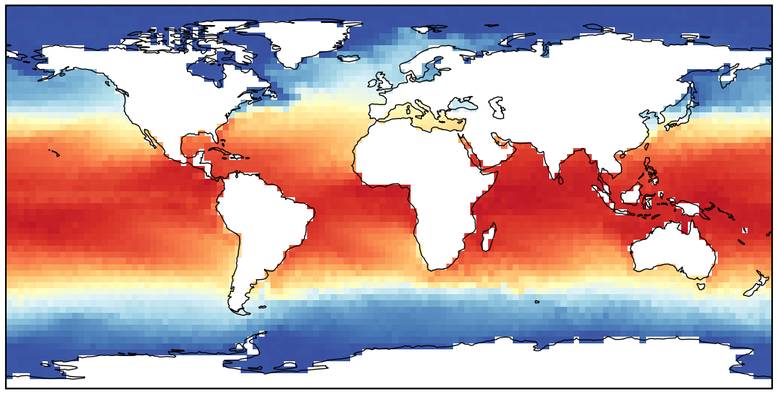


Chain init: 2017-04-05 — Validation (°C)

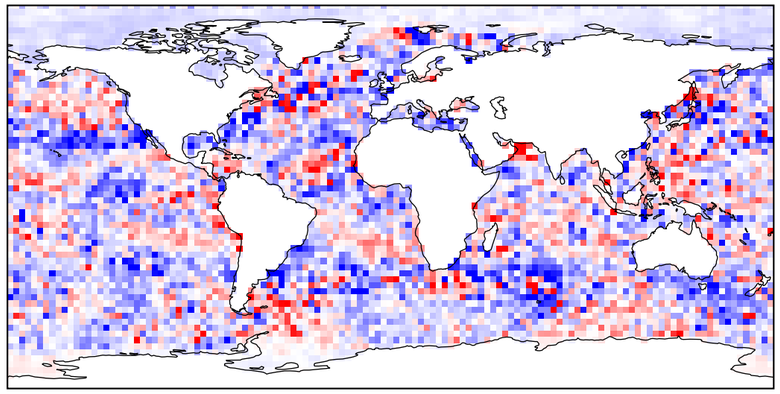
Actual — Step 1 (°C)



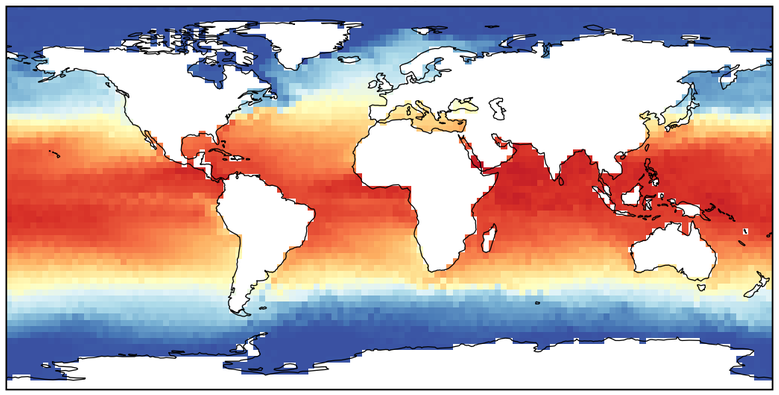
SFNO — Step 1 (°C)



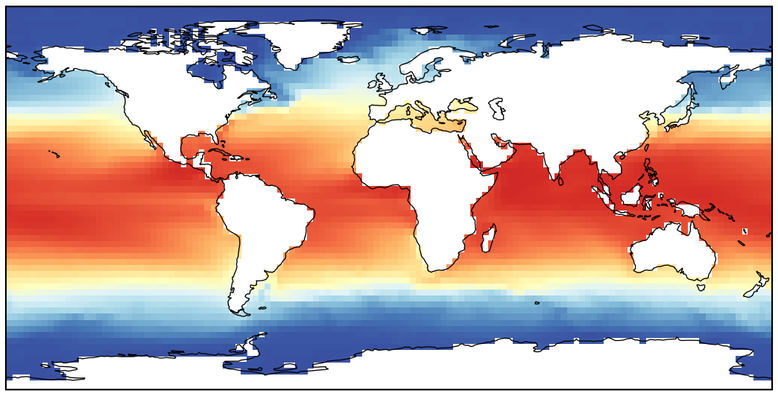
Error — Step 1 (°C)



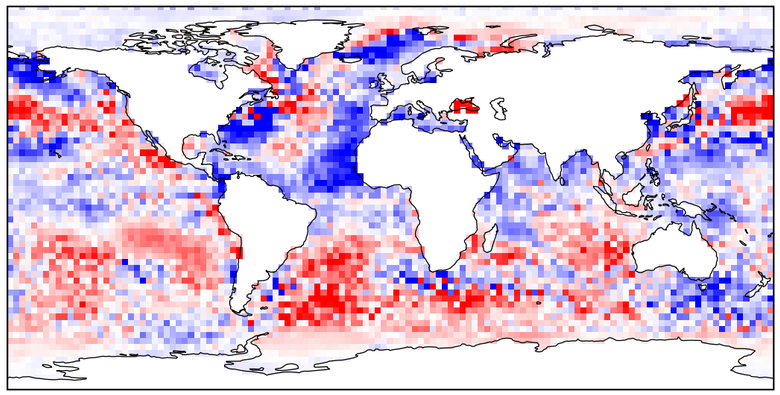
Actual — Step 46 (°C)



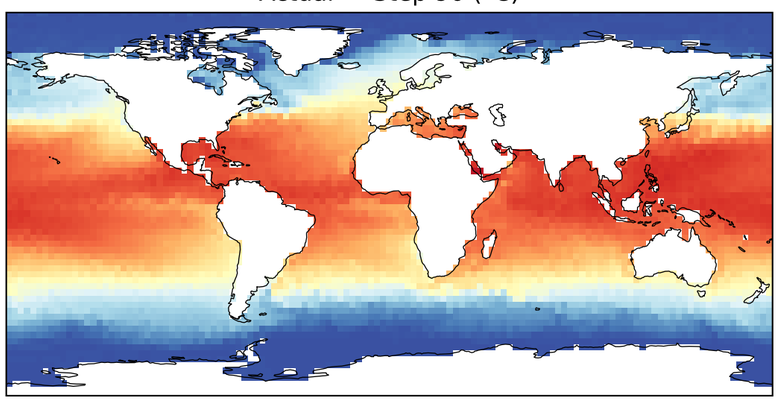
SFNO — Step 46 (°C)



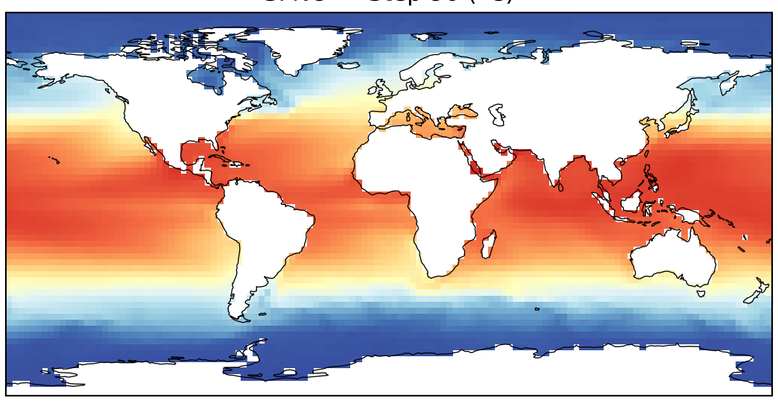
Error — Step 46 (°C)



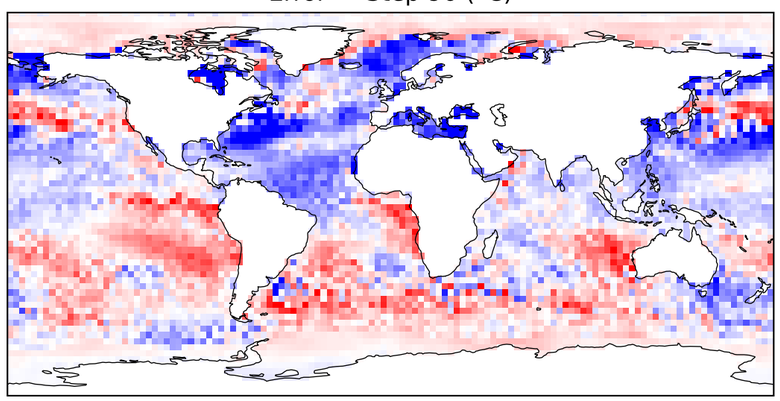
Actual — Step 90 (°C)

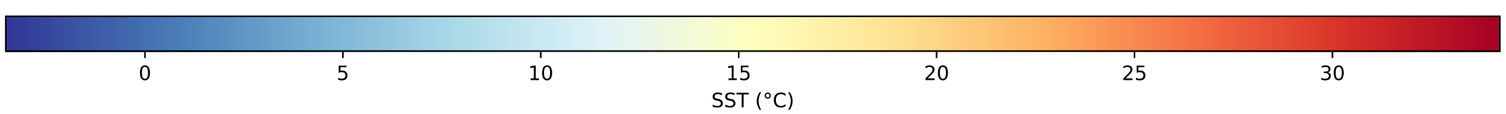


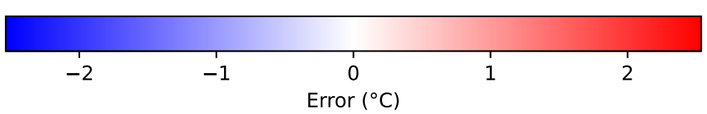
SFNO — Step 90 (°C)



Error — Step 90 (°C)

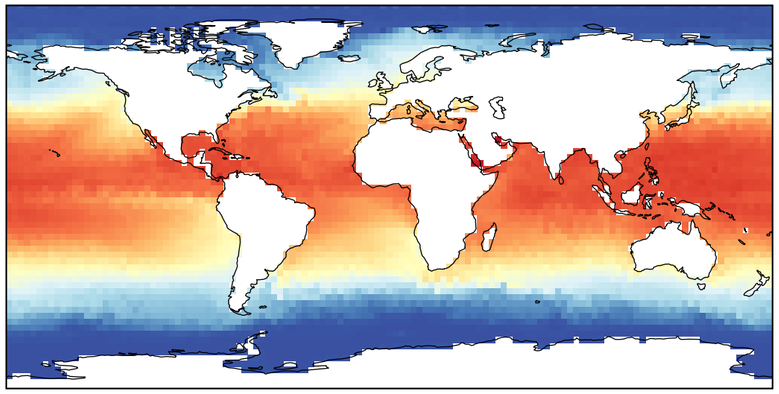




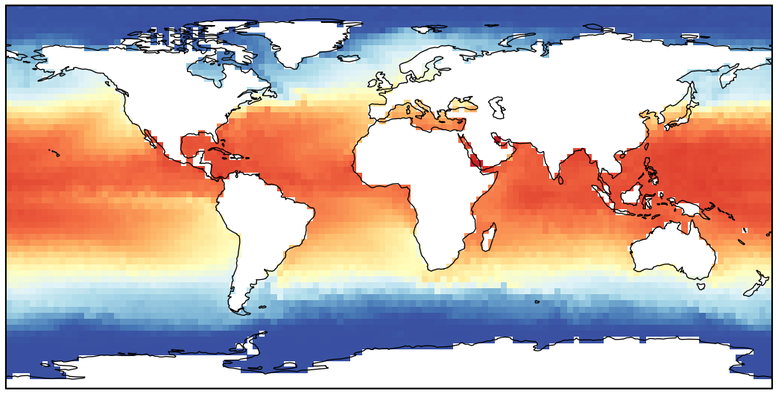


Chain init: 2017-10-02 — Validation (°C)

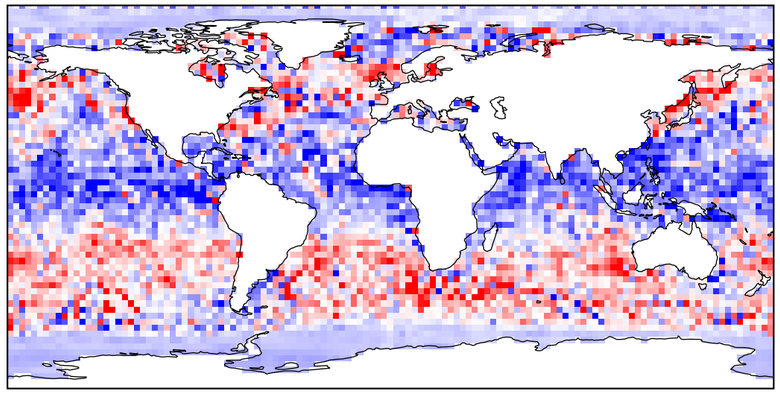
Actual — Step 1 (°C)



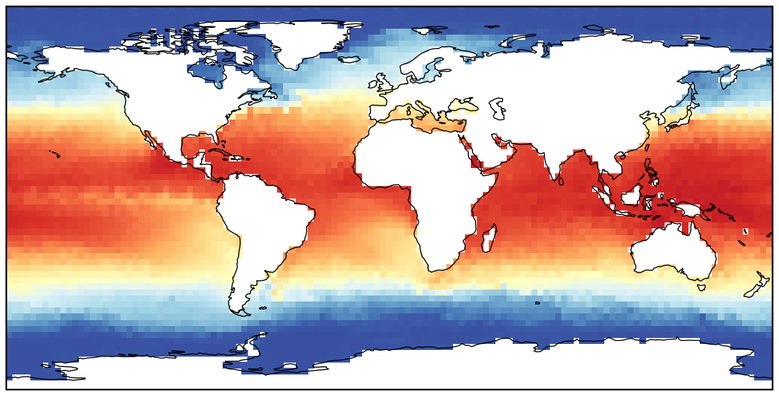
SFNO — Step 1 (°C)



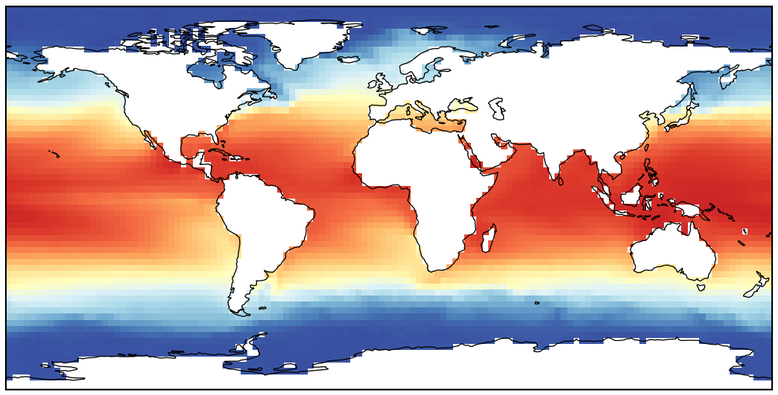
Error — Step 1 (°C)



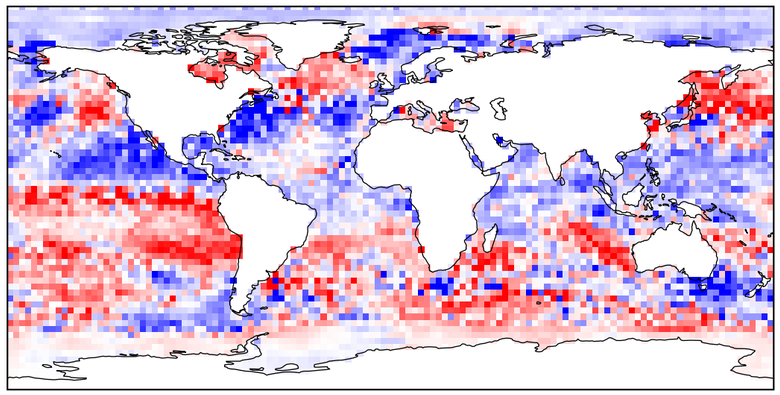
Actual — Step 46 (°C)



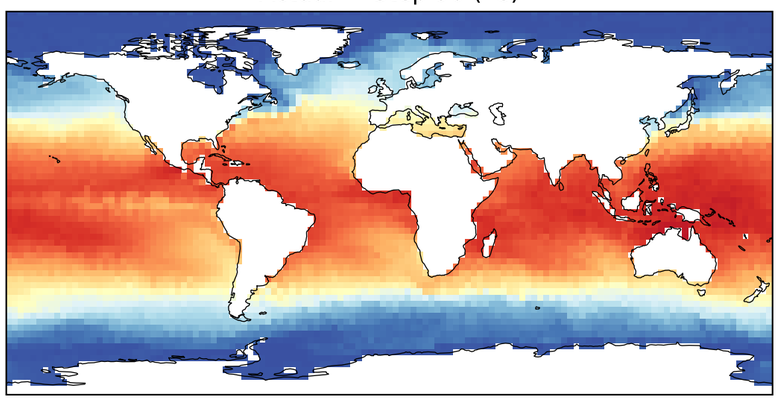
SFNO — Step 46 (°C)



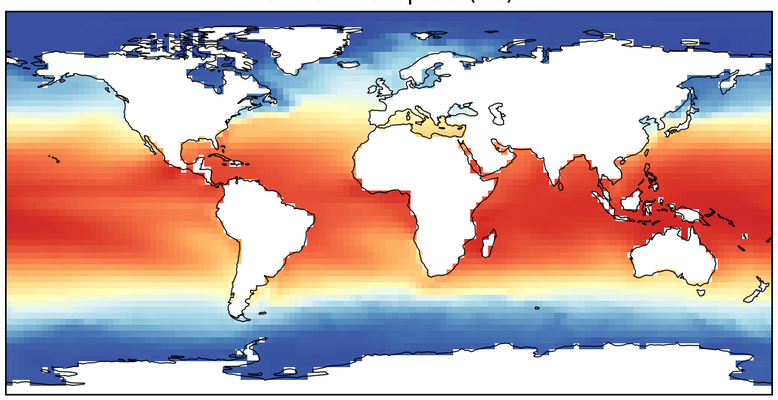
Error — Step 46 (°C)



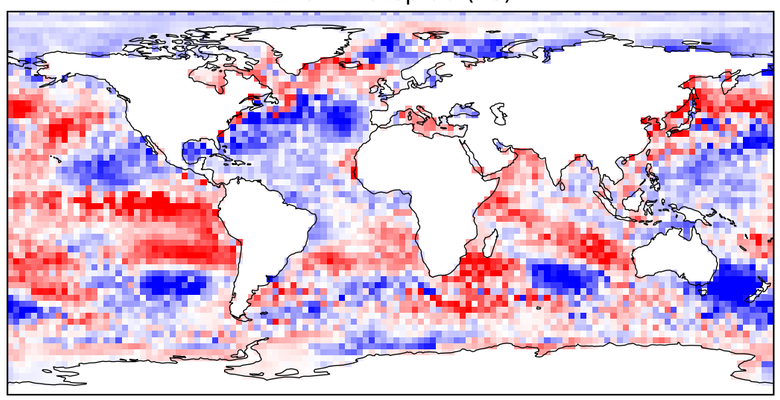
Actual — Step 90 (°C)

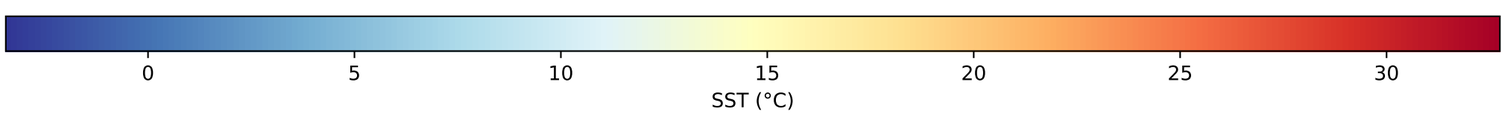


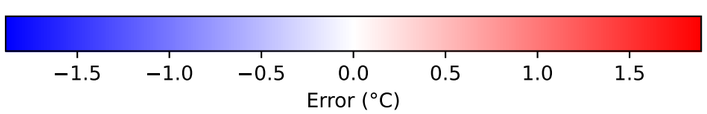
SFNO — Step 90 (°C)



Error — Step 90 (°C)

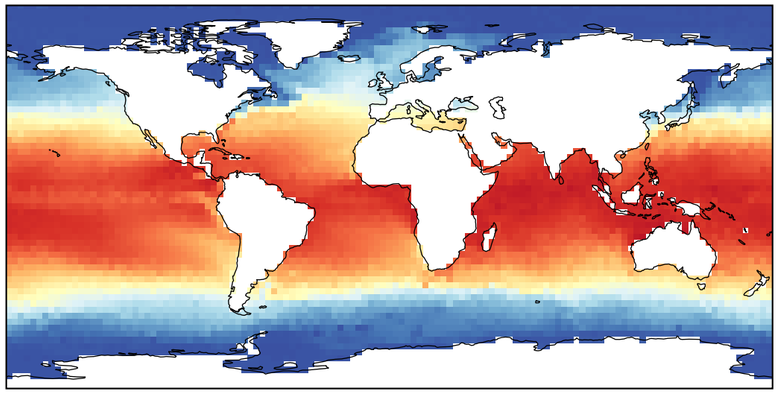




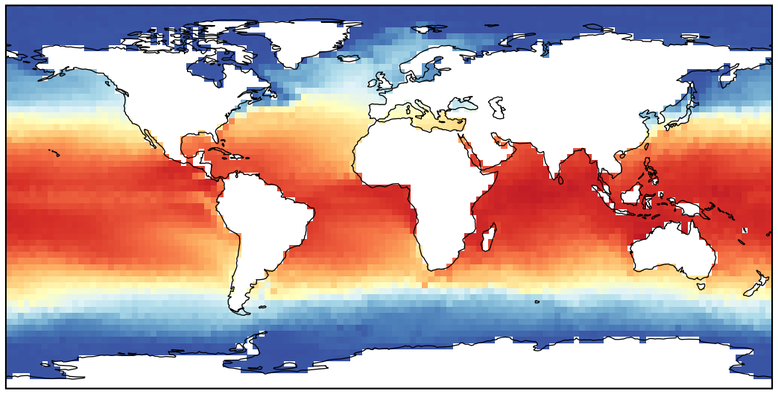


Chain init: 2018-03-31 — Validation (°C)

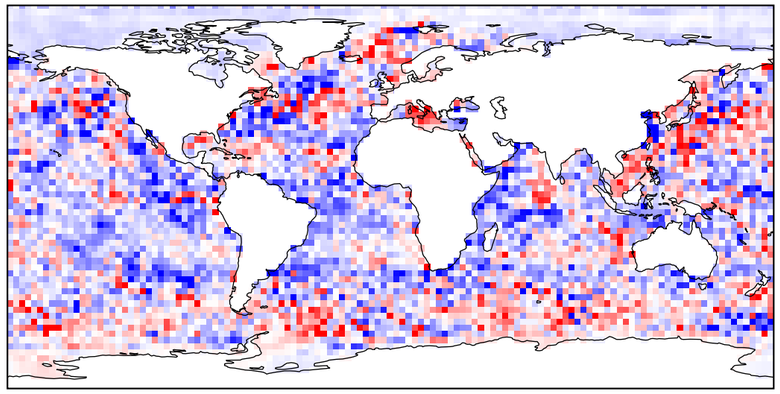
Actual — Step 1 (°C)



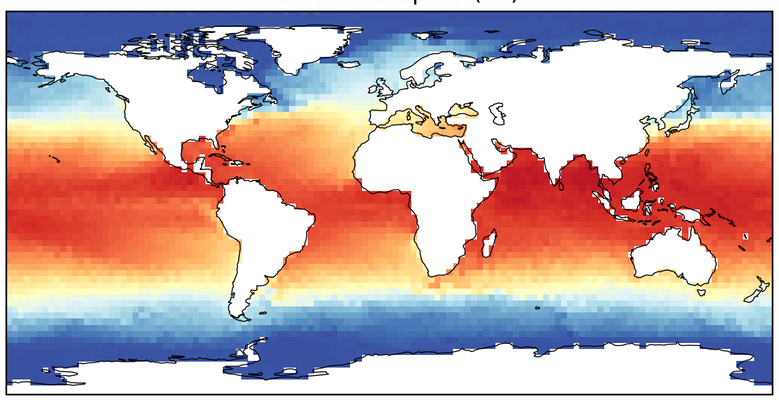
SFNO — Step 1 (°C)



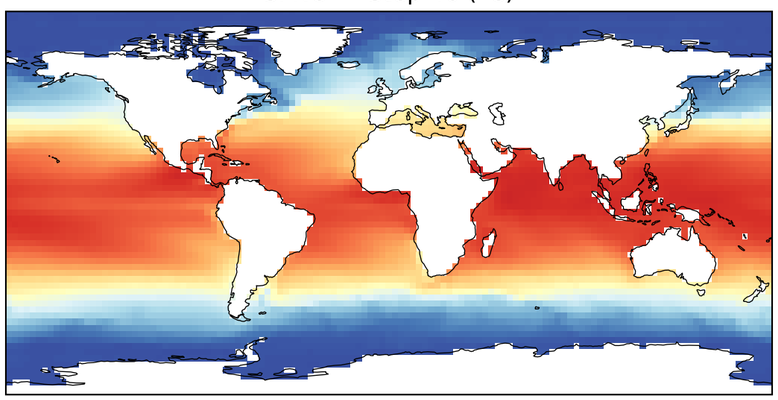
Error — Step 1 (°C)



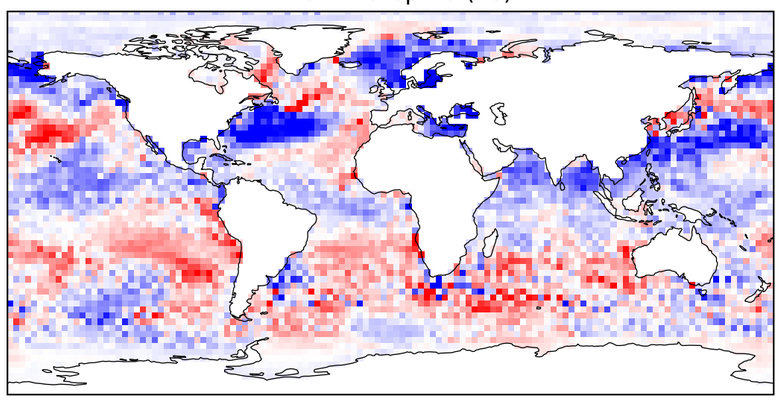
Actual — Step 46 (°C)



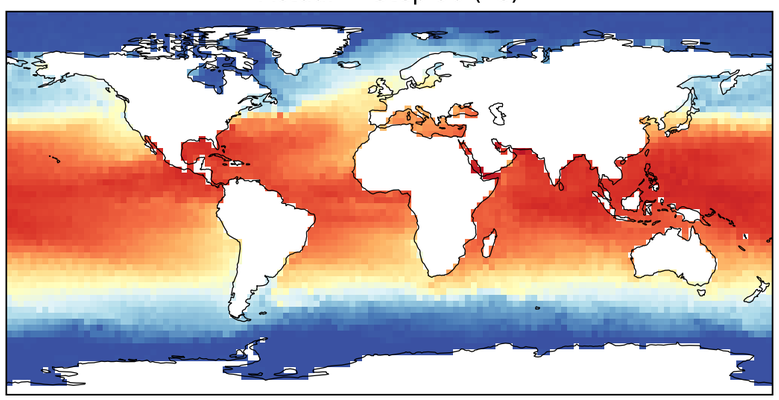
SFNO — Step 46 (°C)



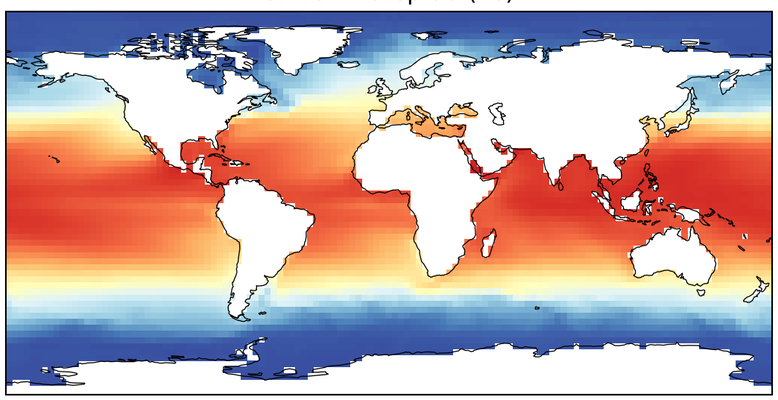
Error — Step 46 (°C)



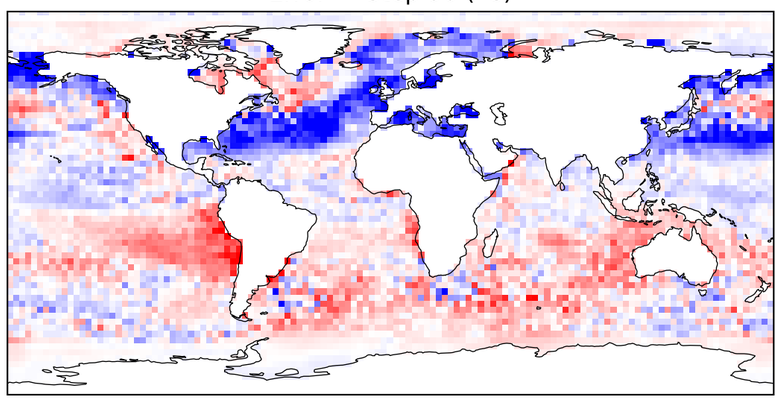
Actual — Step 90 (°C)

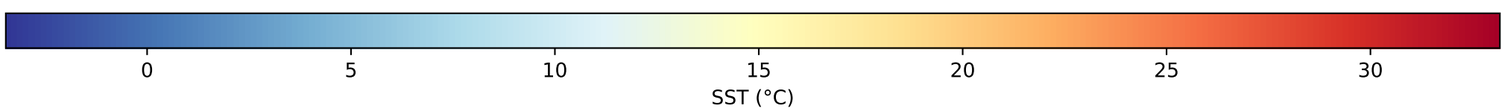


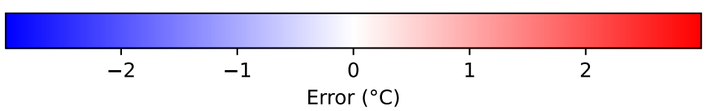
SFNO — Step 90 (°C)



Error — Step 90 (°C)

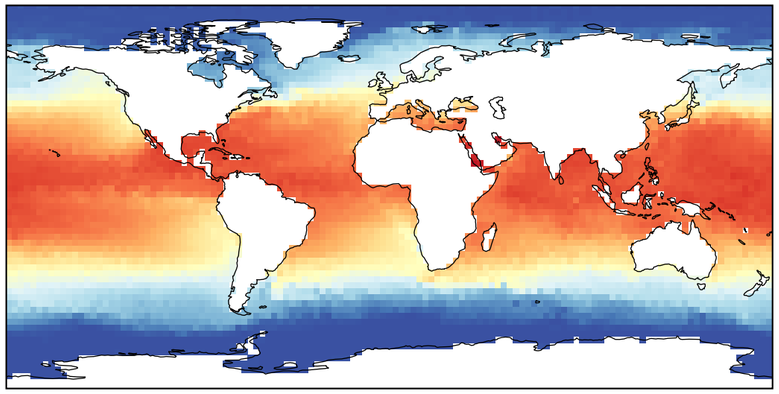




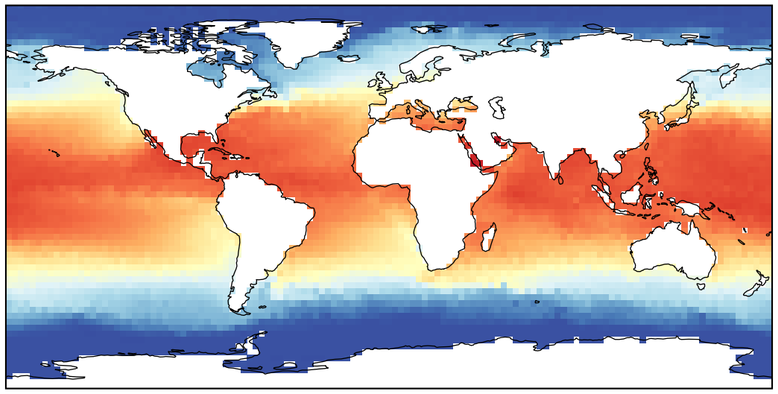


Chain init: 2018-09-27 — Validation (°C)

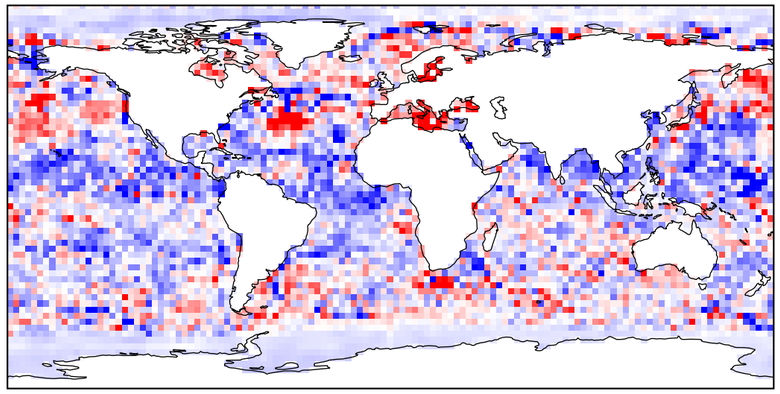
Actual — Step 1 (°C)



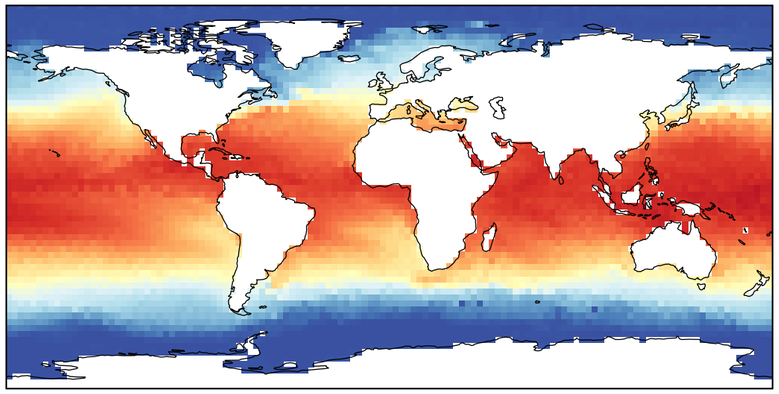
SFNO — Step 1 (°C)



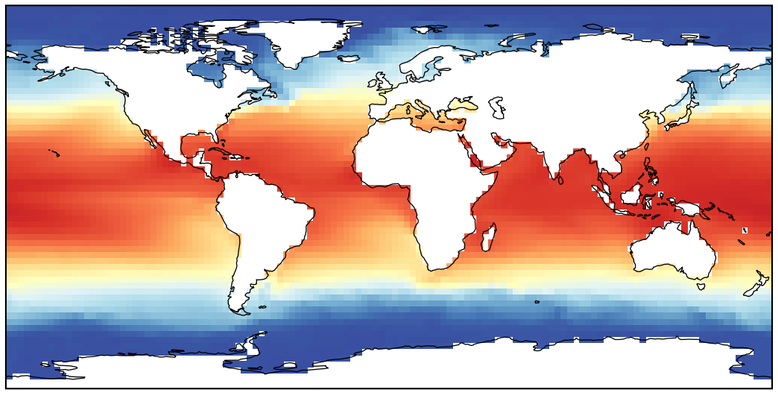
Error — Step 1 (°C)



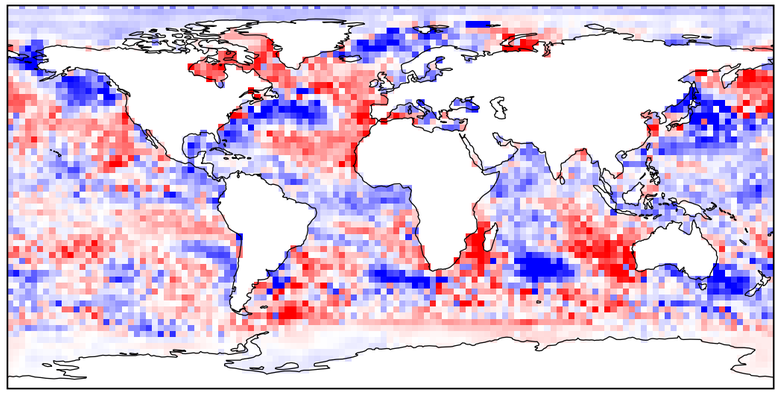
Actual — Step 46 (°C)



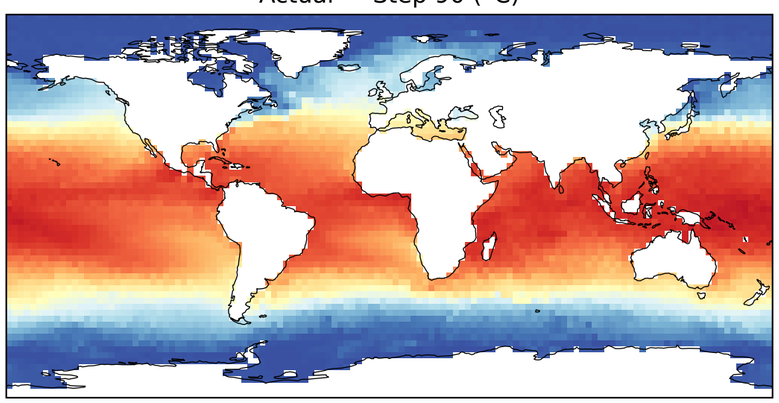
SFNO — Step 46 (°C)



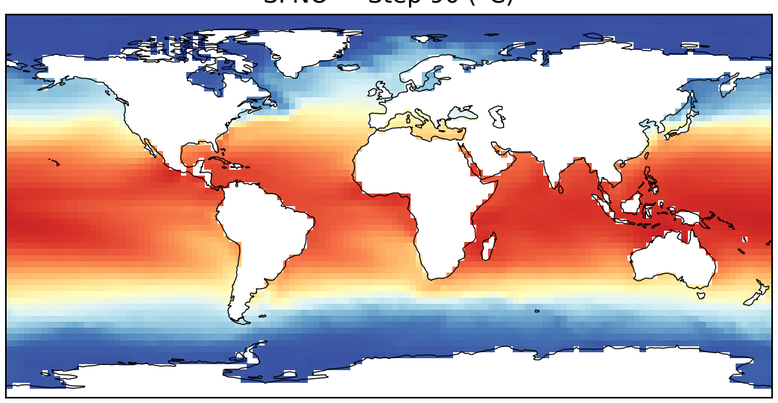
Error — Step 46 (°C)



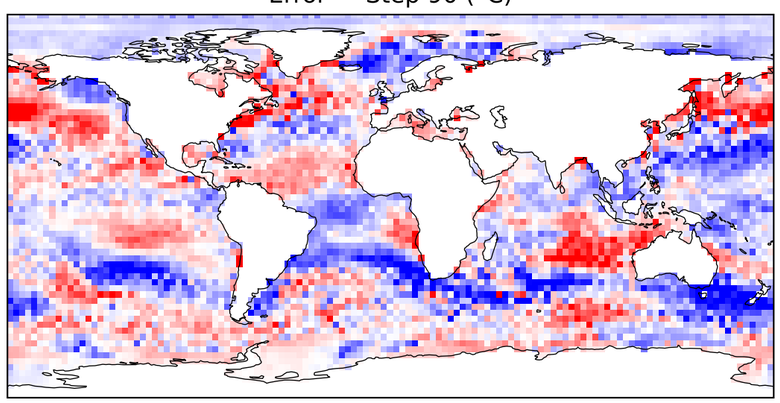
Actual — Step 90 (°C)

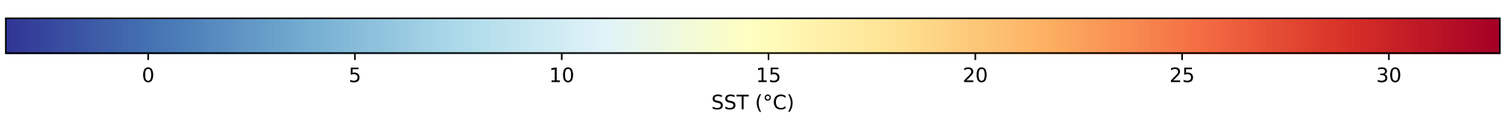


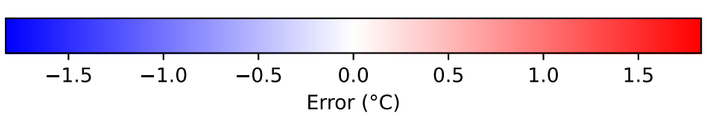
SFNO — Step 90 (°C)



Error — Step 90 (°C)

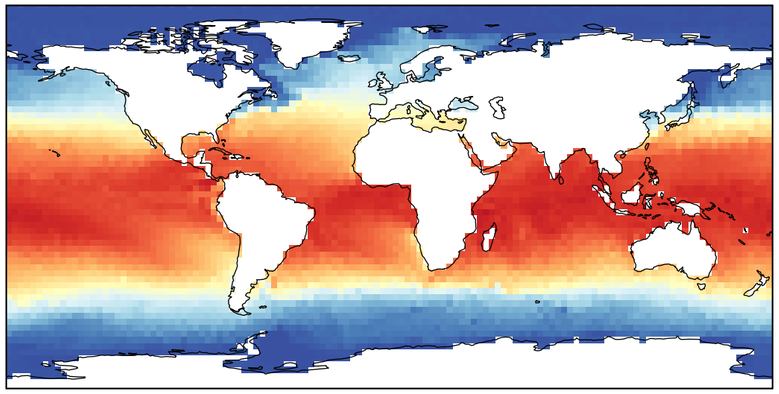




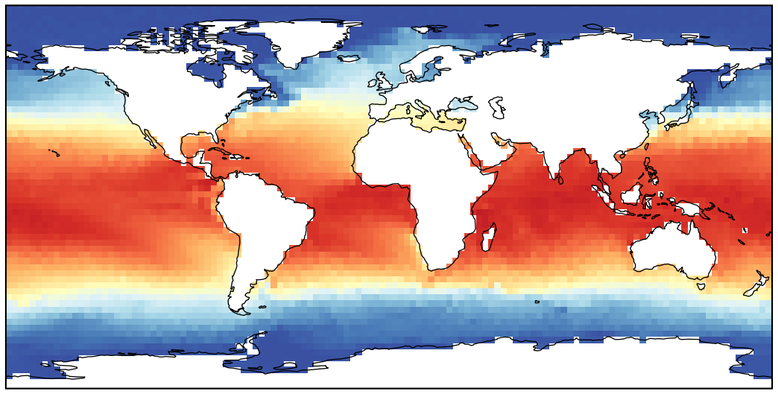


Chain init: 2019-03-26 — Validation (°C)

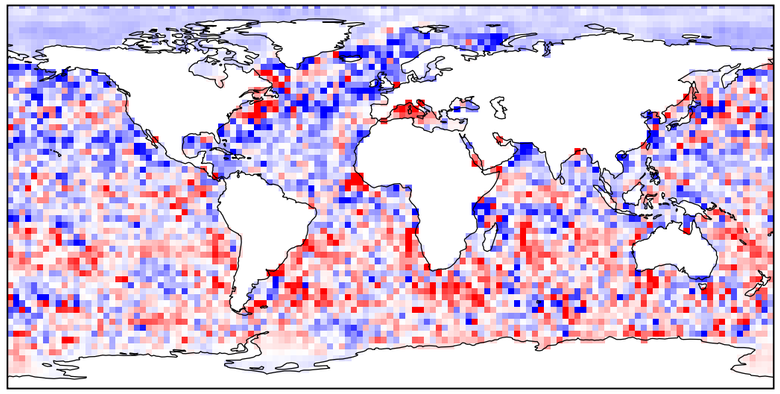
Actual — Step 1 (°C)



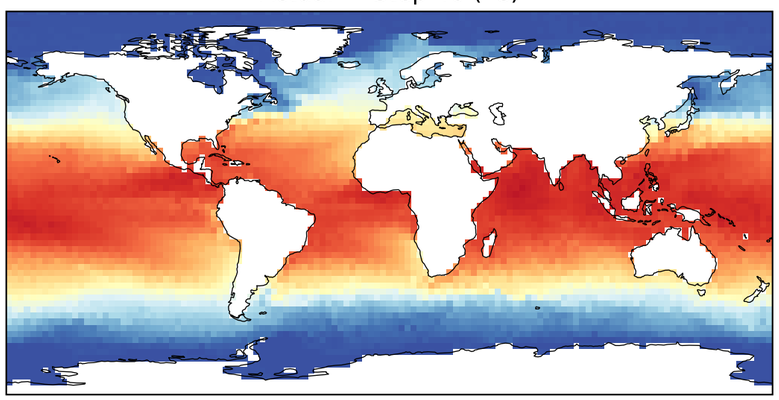
SFNO — Step 1 (°C)



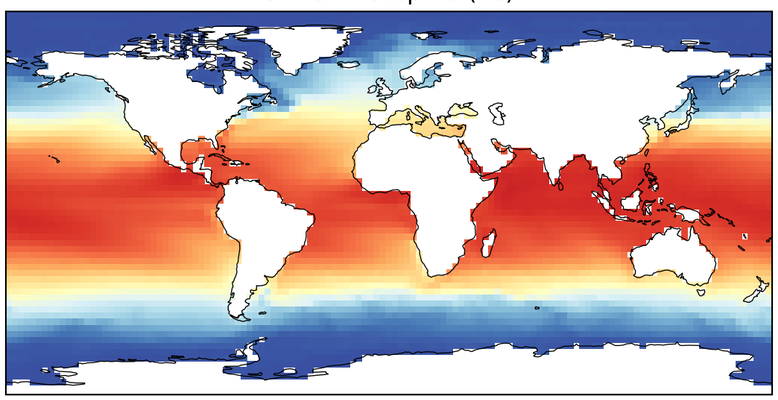
Error — Step 1 (°C)



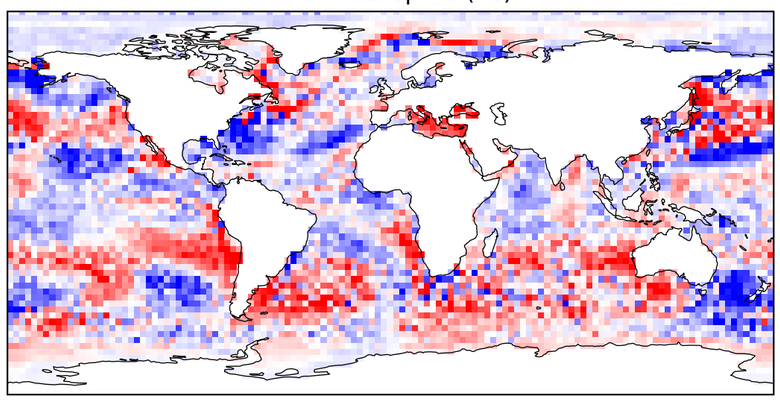
Actual — Step 46 (°C)



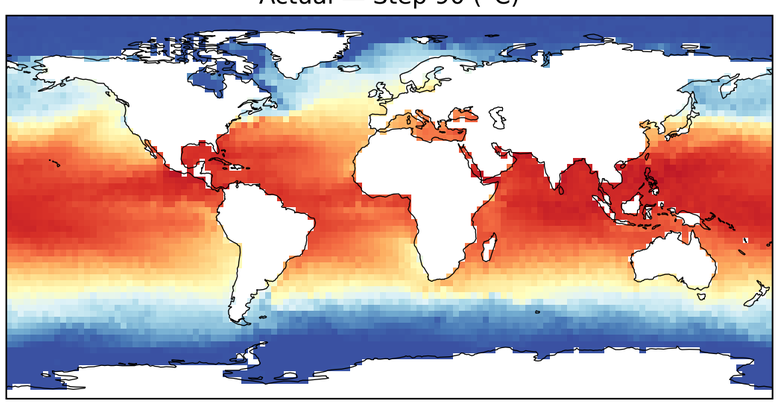
SFNO — Step 46 (°C)



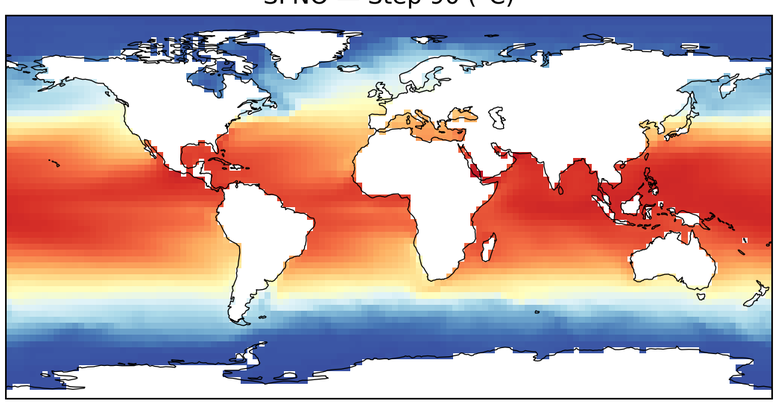
Error — Step 46 (°C)



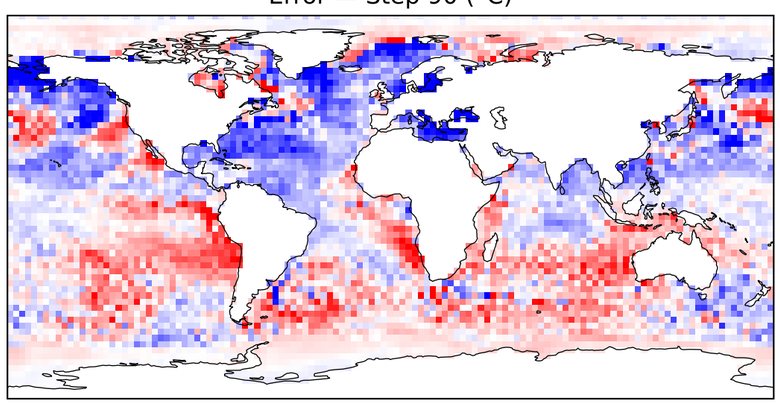
Actual — Step 90 (°C)

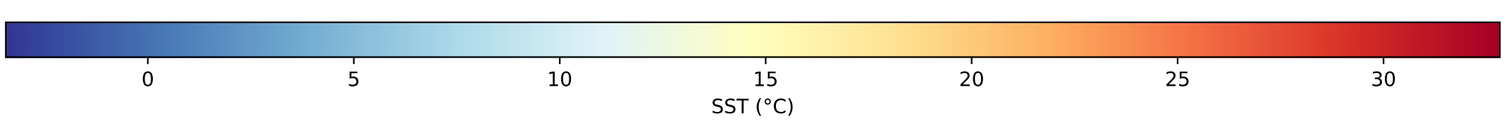


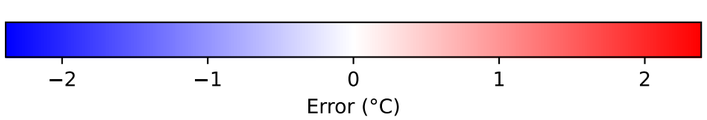
SFNO — Step 90 (°C)



Error — Step 90 (°C)

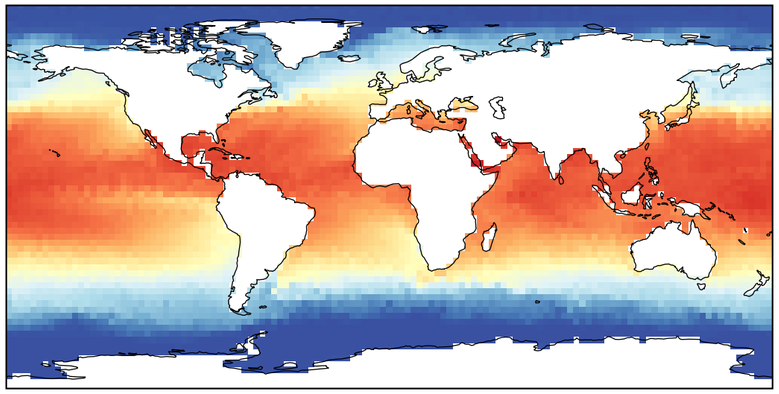




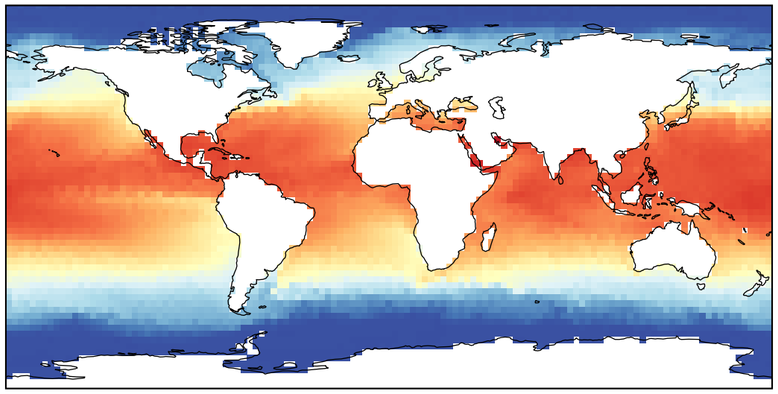


Chain init: 2019-09-22 — Validation (°C)

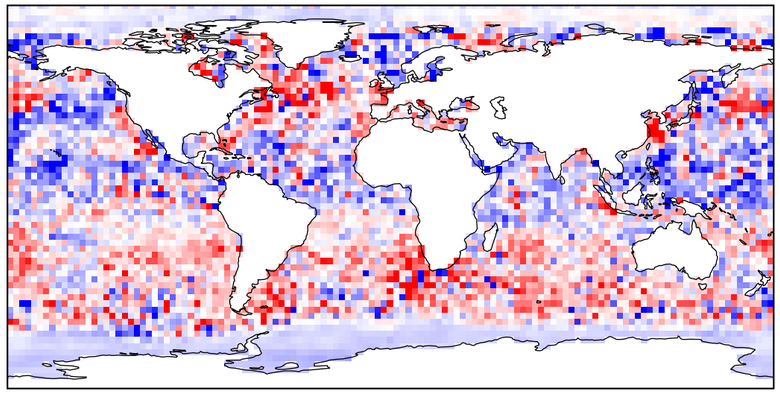
Actual — Step 1 (°C)



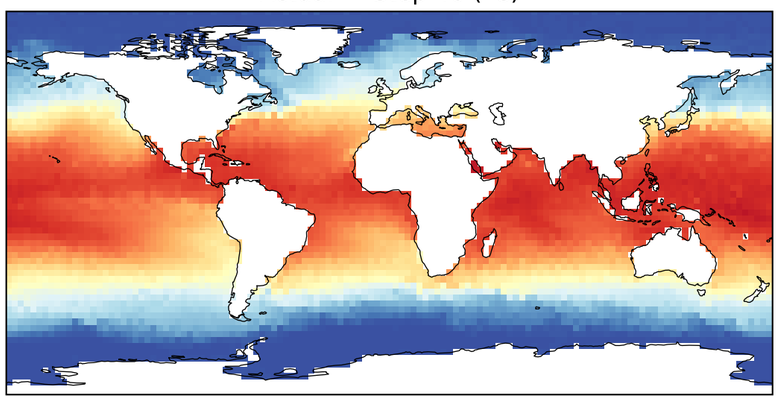
SFNO — Step 1 (°C)



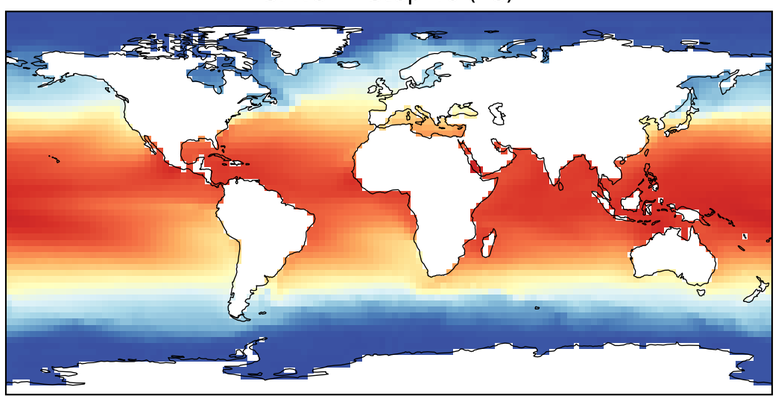
Error — Step 1 (°C)



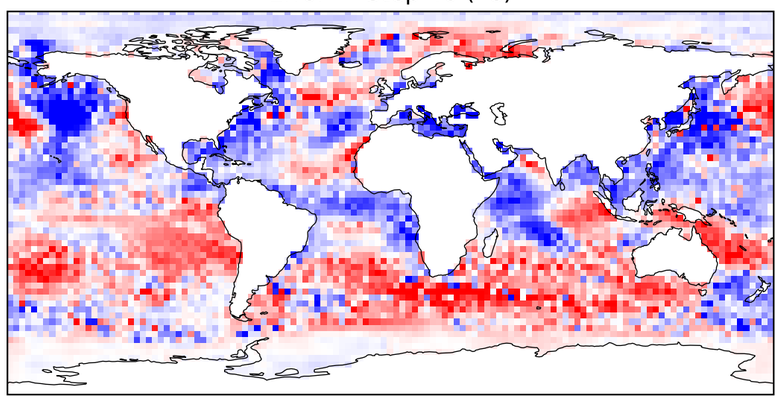
Actual — Step 46 (°C)



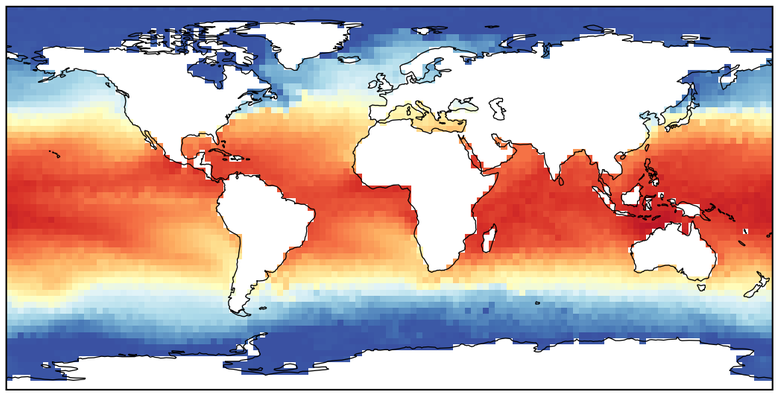
SFNO — Step 46 (°C)



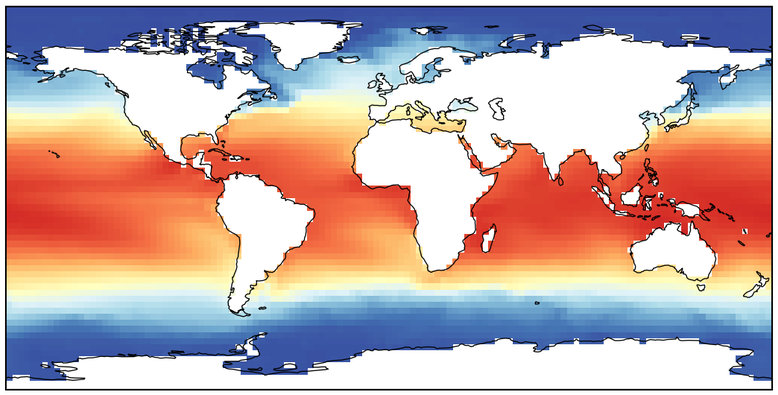
Error — Step 46 (°C)



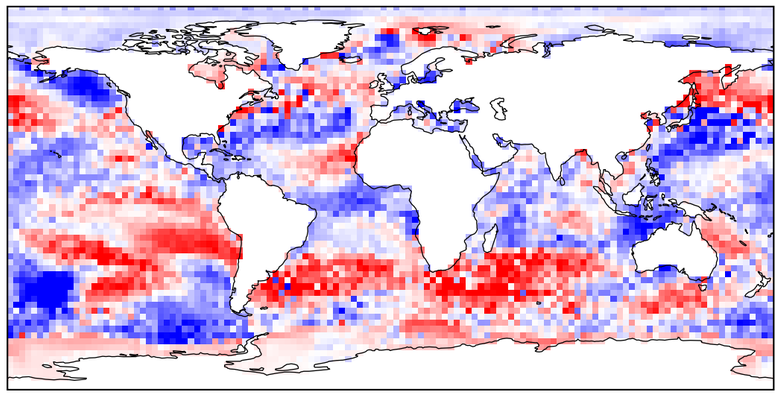
Actual — Step 90 (°C)

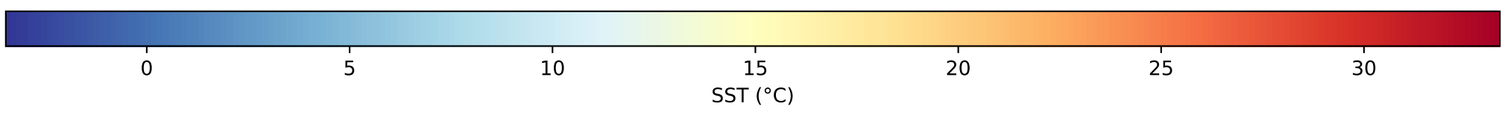


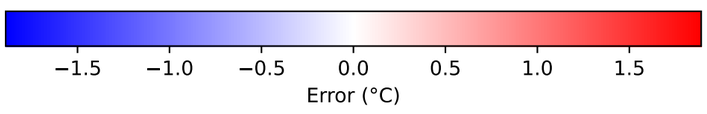
SFNO — Step 90 (°C)



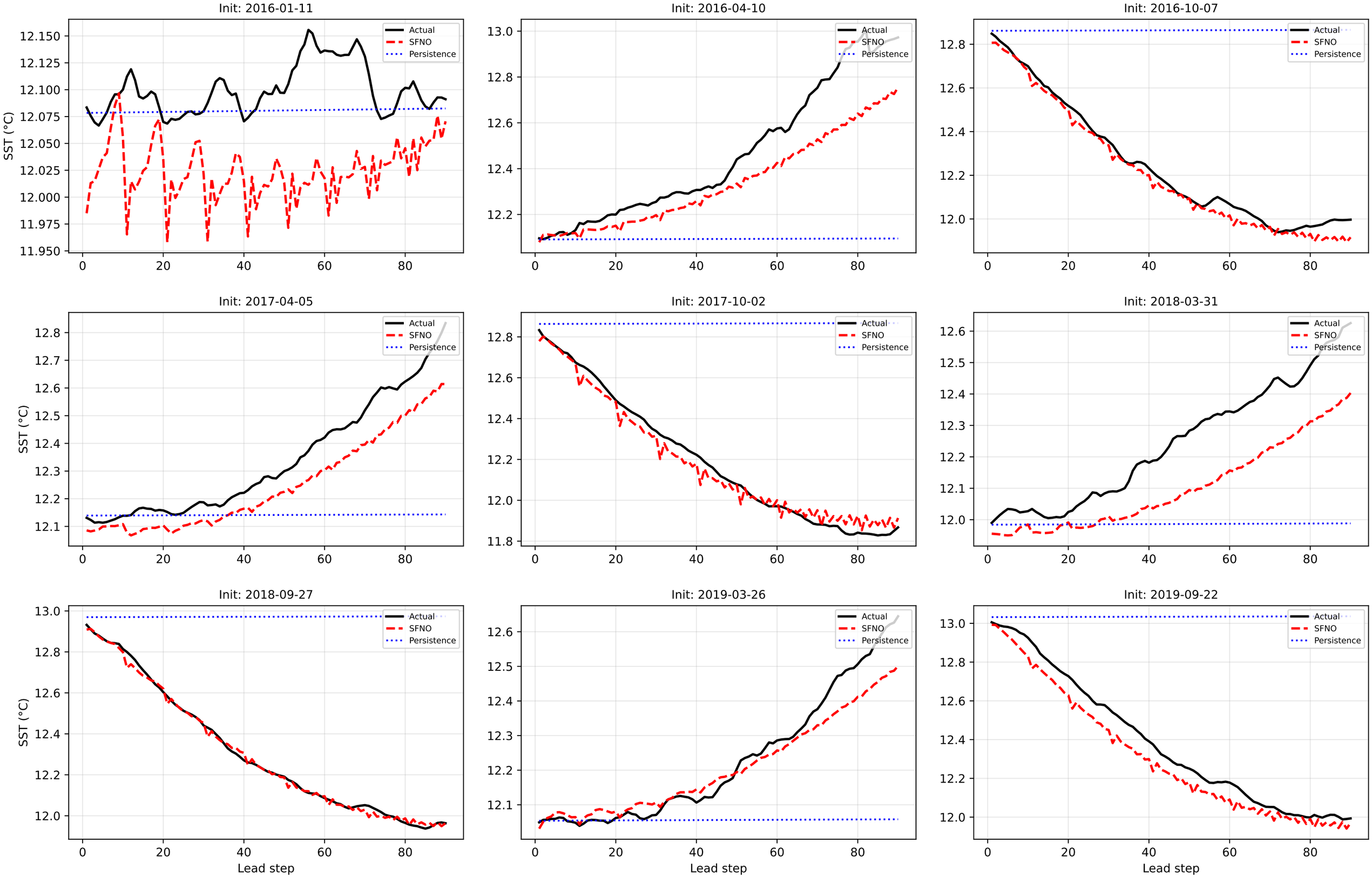
Error — Step 90 (°C)





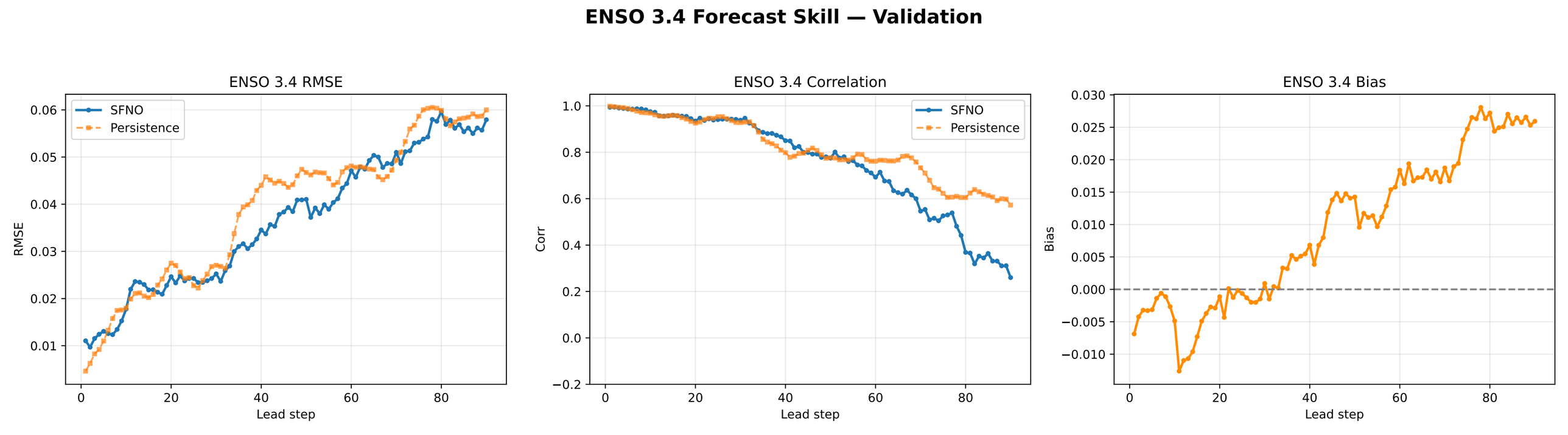


Global-Mean SST — Validation (°C)

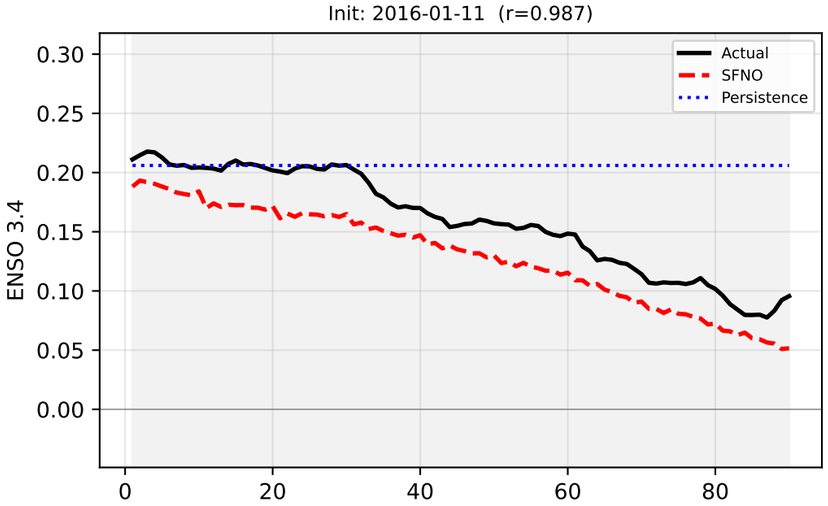


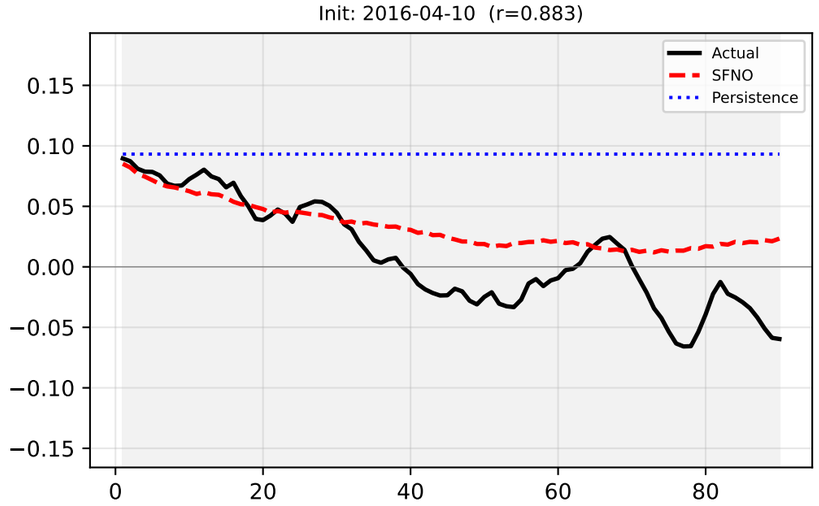
ENSO 3.4 Metrics — Validation

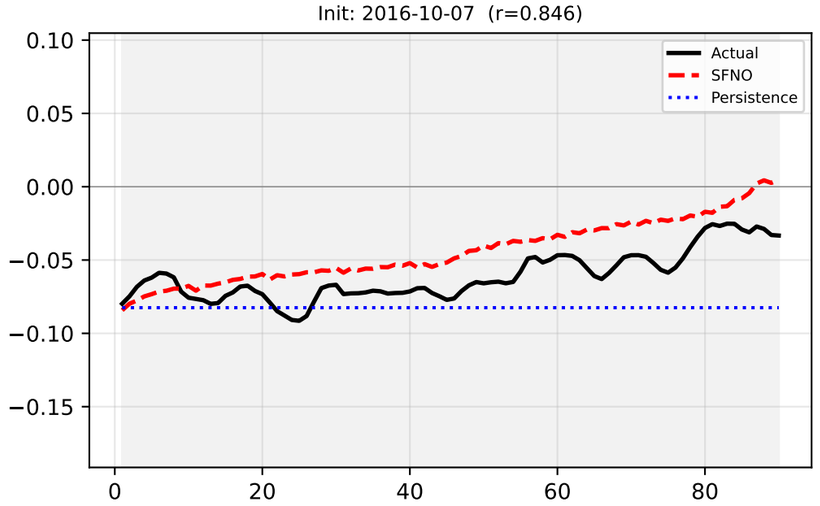
Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0110	0.0046	0.9939	0.9981	-0.0069
2	0.0097	0.0063	0.9942	0.9964	-0.0042
3	0.0115	0.0083	0.9908	0.9938	-0.0032
4	0.0124	0.0092	0.9888	0.9920	-0.0033
5	0.0131	0.0110	0.9861	0.9881	-0.0031
6	0.0126	0.0133	0.9862	0.9833	-0.0014
7	0.0124	0.0158	0.9873	0.9771	-0.0006
8	0.0135	0.0175	0.9867	0.9714	-0.0011
9	0.0153	0.0176	0.9828	0.9705	-0.0027
10	0.0178	0.0180	0.9748	0.9684	-0.0049
11	0.0220	0.0199	0.9724	0.9615	-0.0126
12	0.0236	0.0211	0.9573	0.9562	-0.0110
13	0.0235	0.0212	0.9558	0.9548	-0.0107
14	0.0230	0.0205	0.9567	0.9576	-0.0096
15	0.0218	0.0202	0.9596	0.9592	-0.0073
16	0.0219	0.0209	0.9571	0.9576	-0.0049
17	0.0213	0.0228	0.9557	0.9494	-0.0037
18	0.0209	0.0241	0.9546	0.9431	-0.0027
19	0.0228	0.0261	0.9449	0.9328	-0.0029
20	0.0246	0.0275	0.9344	0.9257	-0.0011
21	0.0233	0.0270	0.9469	0.9303	-0.0043
22	0.0248	0.0256	0.9370	0.9410	0.0001
23	0.0237	0.0243	0.9439	0.9466	-0.0013
24	0.0243	0.0245	0.9389	0.9458	-0.0002
25	0.0243	0.0227	0.9409	0.9531	-0.0006
26	0.0234	0.0222	0.9426	0.9533	-0.0013
27	0.0234	0.0238	0.9425	0.9448	-0.0020
28	0.0238	0.0252	0.9427	0.9366	-0.0020
29	0.0243	0.0268	0.9416	0.9284	-0.0015
30	0.0252	0.0271	0.9380	0.9280	0.0009
31	0.0237	0.0268	0.9469	0.9302	-0.0015
32	0.0259	0.0264	0.9259	0.9312	0.0004
33	0.0269	0.0293	0.9151	0.9144	0.0002
34	0.0300	0.0338	0.8929	0.8869	0.0033
35	0.0310	0.0378	0.8861	0.8557	0.0032
36	0.0316	0.0394	0.8804	0.8439	0.0052
37	0.0306	0.0399	0.8812	0.8370	0.0046
38	0.0314	0.0408	0.8733	0.8272	0.0051
39	0.0326	0.0429	0.8667	0.8091	0.0055
40	0.0345	0.0440	0.8500	0.7981	0.0068
41	0.0337	0.0458	0.8487	0.7782	0.0038
42	0.0357	0.0451	0.8194	0.7835	0.0068
43	0.0353	0.0445	0.8251	0.7945	0.0080
44	0.0378	0.0449	0.7997	0.7965	0.0119
45	0.0383	0.0444	0.7991	0.8085	0.0138
46	0.0393	0.0436	0.7919	0.8178	0.0148
47	0.0384	0.0441	0.7922	0.8079	0.0136
48	0.0409	0.0460	0.7780	0.7888	0.0148
49	0.0409	0.0474	0.7802	0.7738	0.0141
50	0.0410	0.0467	0.7745	0.7763	0.0143
51	0.0372	0.0462	0.8009	0.7751	0.0096
52	0.0392	0.0468	0.7750	0.7672	0.0117
53	0.0380	0.0467	0.7805	0.7680	0.0111
54	0.0399	0.0466	0.7604	0.7665	0.0114
55	0.0389	0.0454	0.7637	0.7763	0.0097
56	0.0404	0.0441	0.7452	0.7919	0.0112
57	0.0412	0.0446	0.7414	0.7902	0.0129
58	0.0434	0.0469	0.7213	0.7685	0.0154
59	0.0444	0.0477	0.7110	0.7616	0.0158
60	0.0471	0.0481	0.6927	0.7614	0.0184
61	0.0457	0.0478	0.7140	0.7659	0.0163
62	0.0479	0.0480	0.6762	0.7649	0.0194
63	0.0474	0.0477	0.6742	0.7628	0.0167
64	0.0493	0.0474	0.6344	0.7625	0.0172
65	0.0504	0.0473	0.6262	0.7666	0.0173
66	0.0500	0.0458	0.6196	0.7823	0.0184
67	0.0478	0.0452	0.6365	0.7845	0.0170
68	0.0487	0.0459	0.6161	0.7758	0.0181
69	0.0485	0.0472	0.5997	0.7581	0.0166
70	0.0510	0.0493	0.5463	0.7330	0.0187
71	0.0486	0.0510	0.5535	0.7100	0.0167
72	0.0511	0.0533	0.5089	0.6791	0.0189
73	0.0513	0.0559	0.5157	0.6473	0.0194
74	0.0530	0.0567	0.5048	0.6412	0.0231
75	0.0531	0.0587	0.5261	0.6229	0.0247
76	0.0538	0.0600	0.5299	0.6055	0.0265
77	0.0542	0.0603	0.5386	0.6065	0.0263
78	0.0580	0.0605	0.4811	0.6097	0.0281
79	0.0576	0.0603	0.4415	0.6050	0.0263
80	0.0596	0.0599	0.3686	0.6048	0.0272
81	0.0569	0.0581	0.3656	0.6243	0.0244
82	0.0578	0.0566	0.3192	0.6391	0.0249
83	0.0561	0.0574	0.3524	0.6298	0.0251
84	0.0569	0.0581	0.3446	0.6190	0.0270
85	0.0554	0.0583	0.3638	0.6136	0.0255
86	0.0562	0.0585	0.3310	0.6072	0.0265
87	0.0550	0.0591	0.3309	0.5917	0.0257
88	0.0562	0.0586	0.3104	0.5994	0.0266
89	0.0557	0.0587	0.3111	0.5977	0.0253
90	0.0579	0.0600	0.2599	0.5725	0.0259

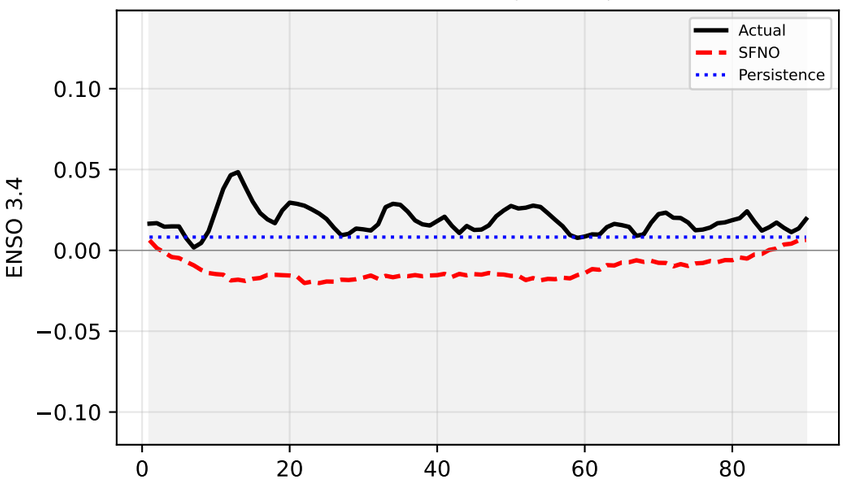

ENSO 3.4 Forecast Skill — Validation
ENSO 3.4 RMSE
SFNO
Persistence
RMSE
Lead step
ENSO 3.4 Correlation
SFNO
Persistence
Corr
Lead step
ENSO 3.4 Bias
Bias
Lead step

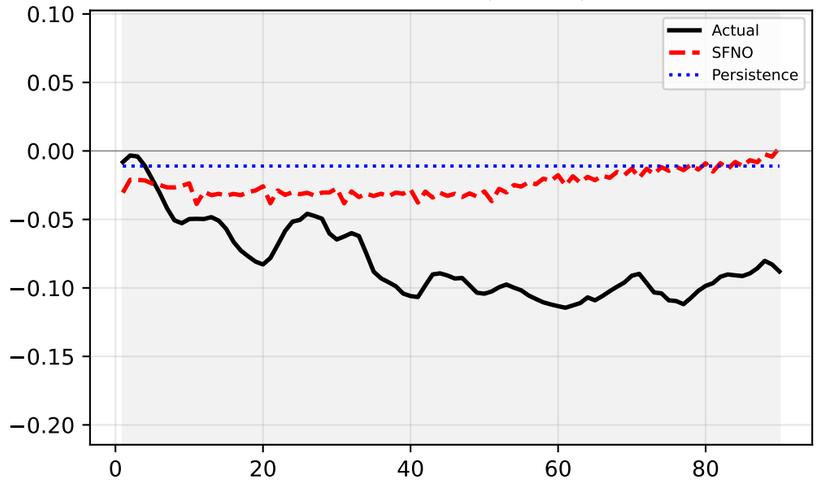
ENSO 3.4 Index — Validation

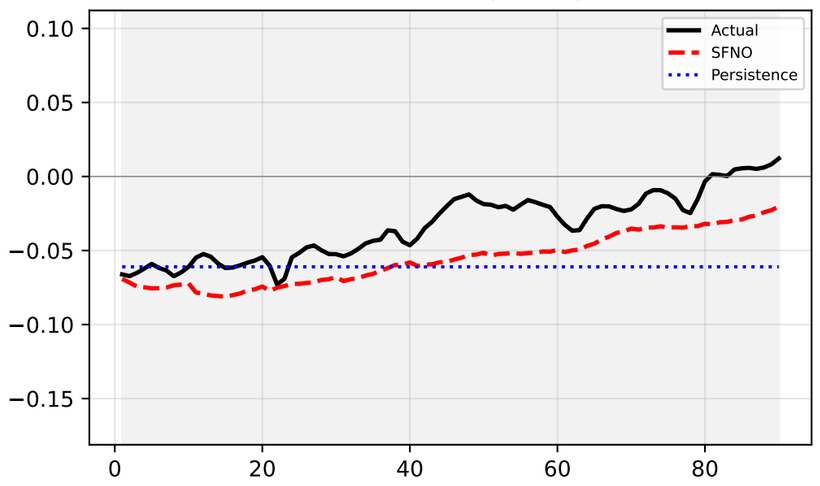


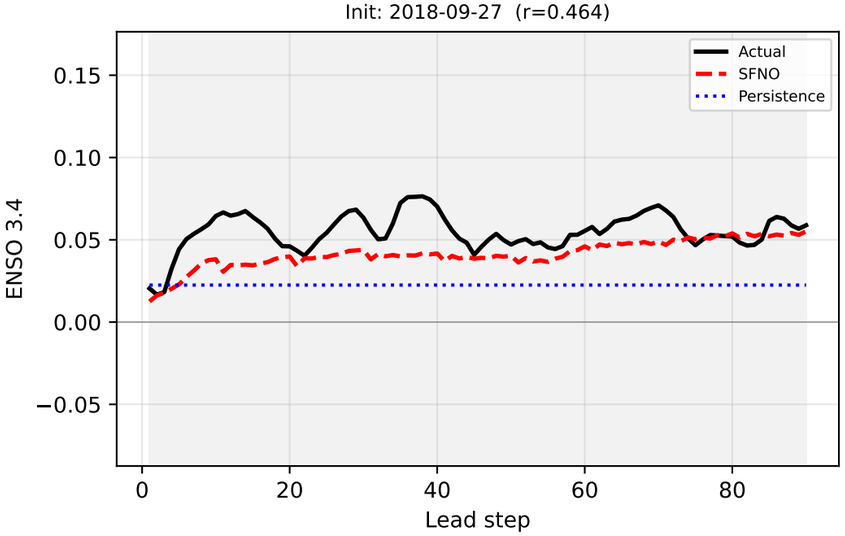


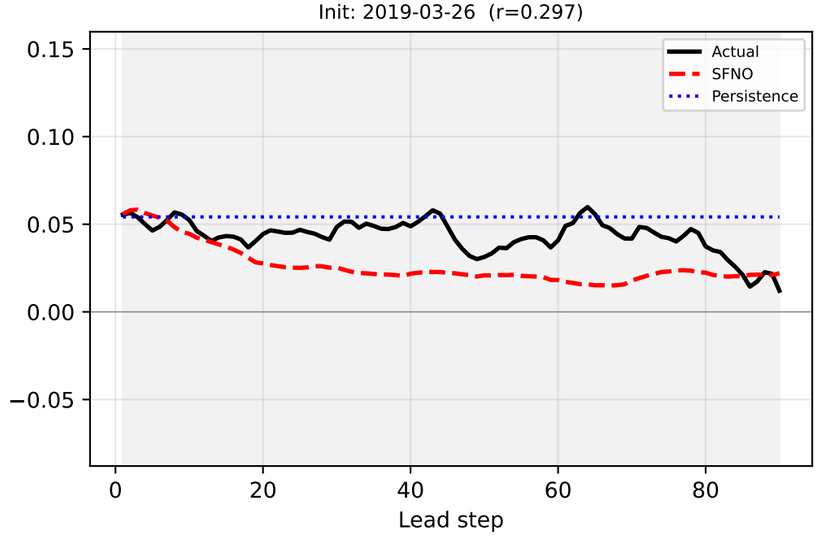


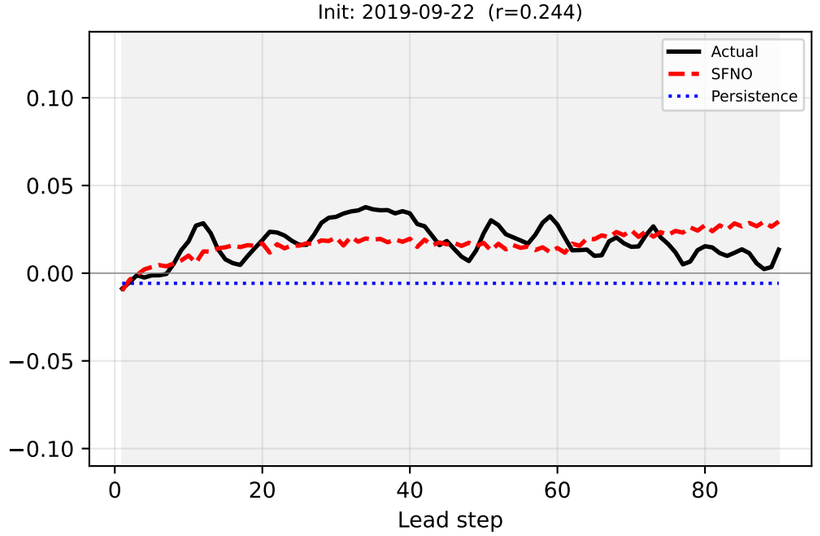




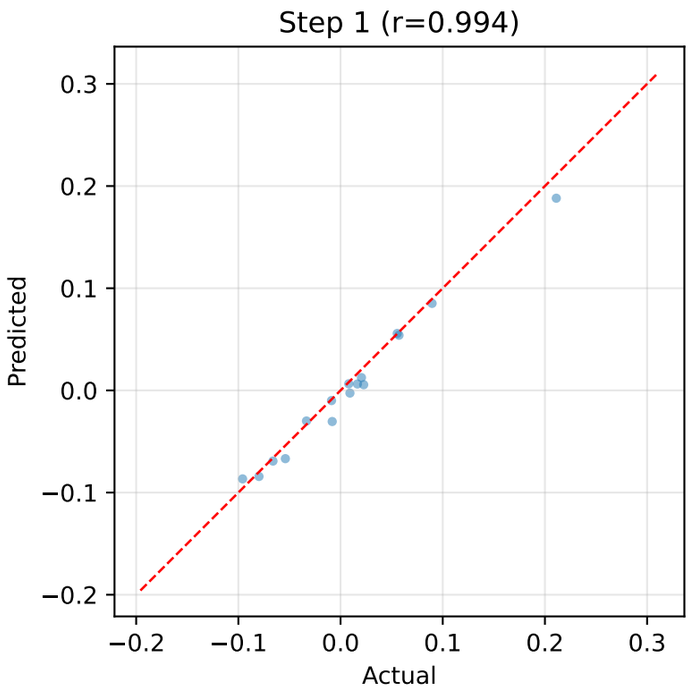


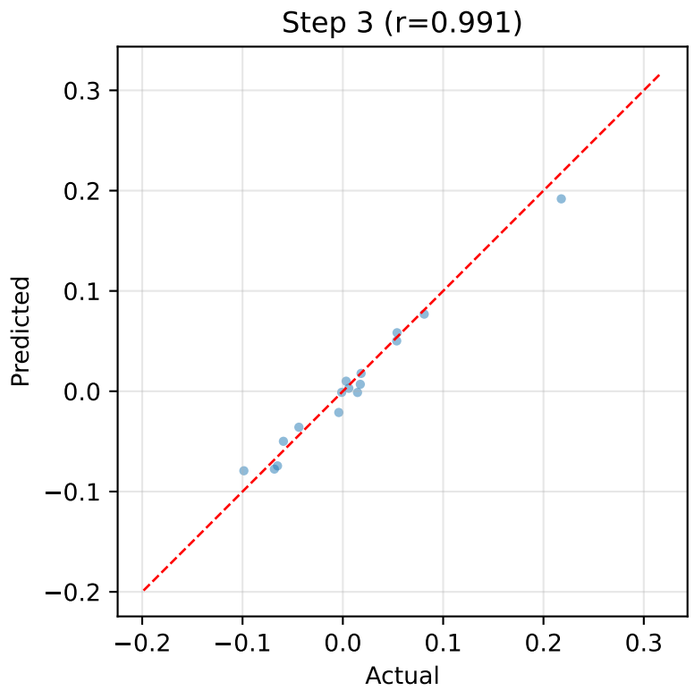


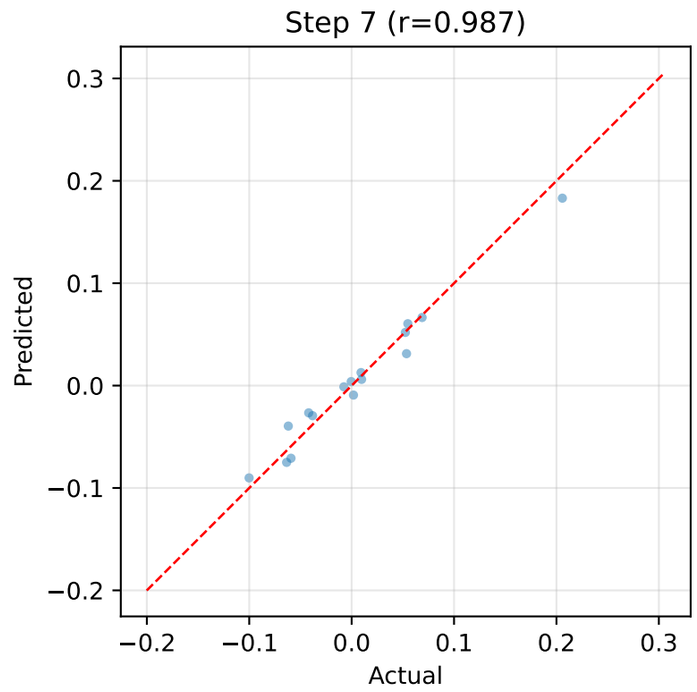


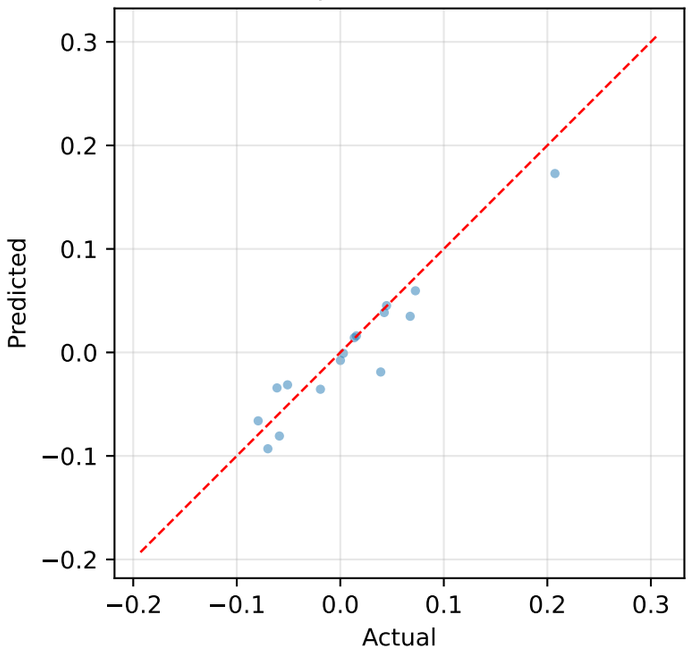


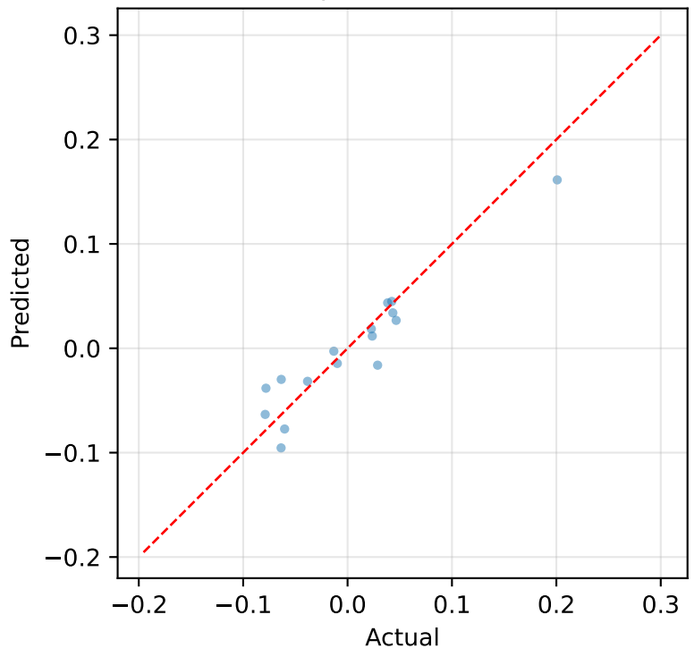
ENSO 3.4 Scatter — Validation

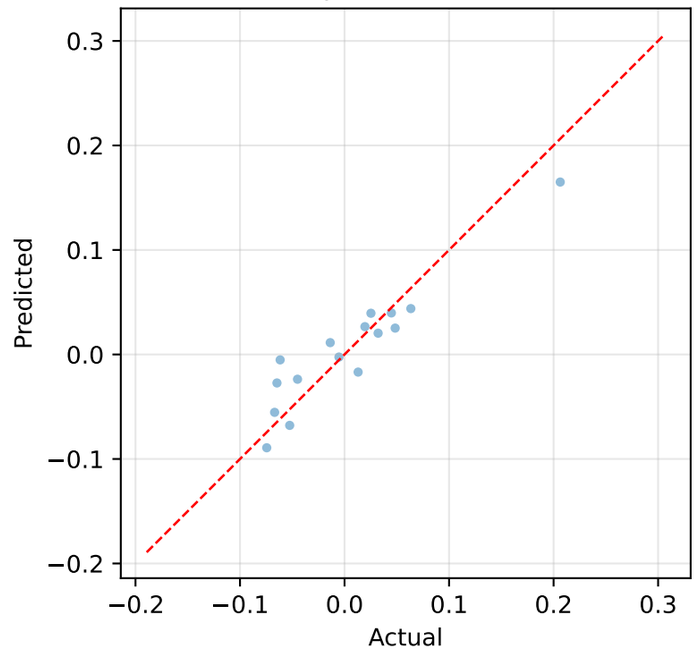


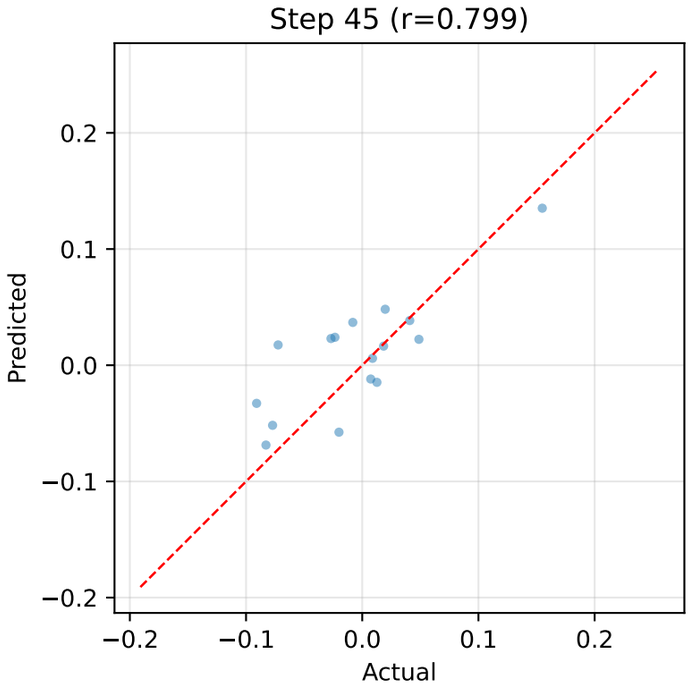


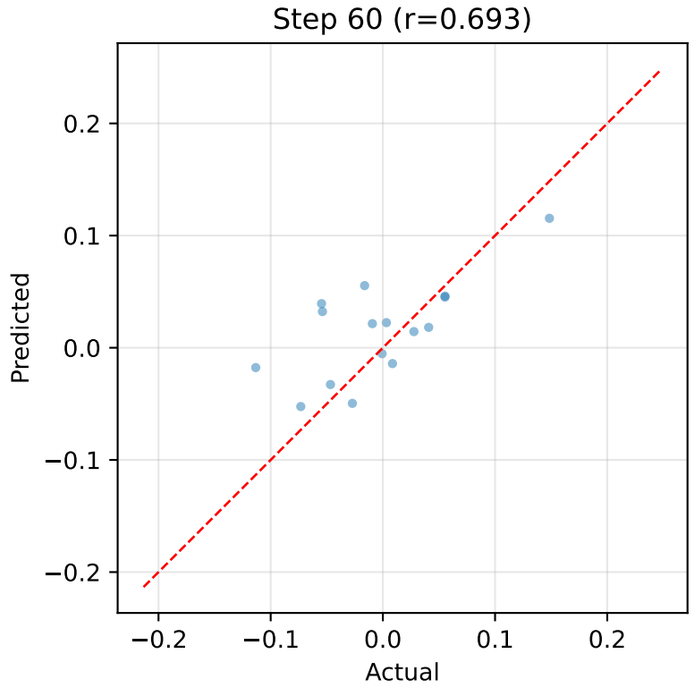


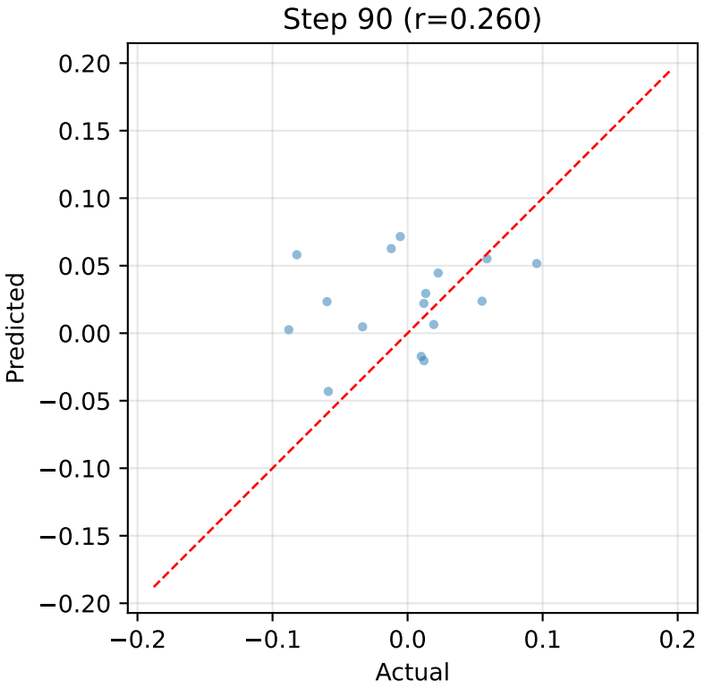








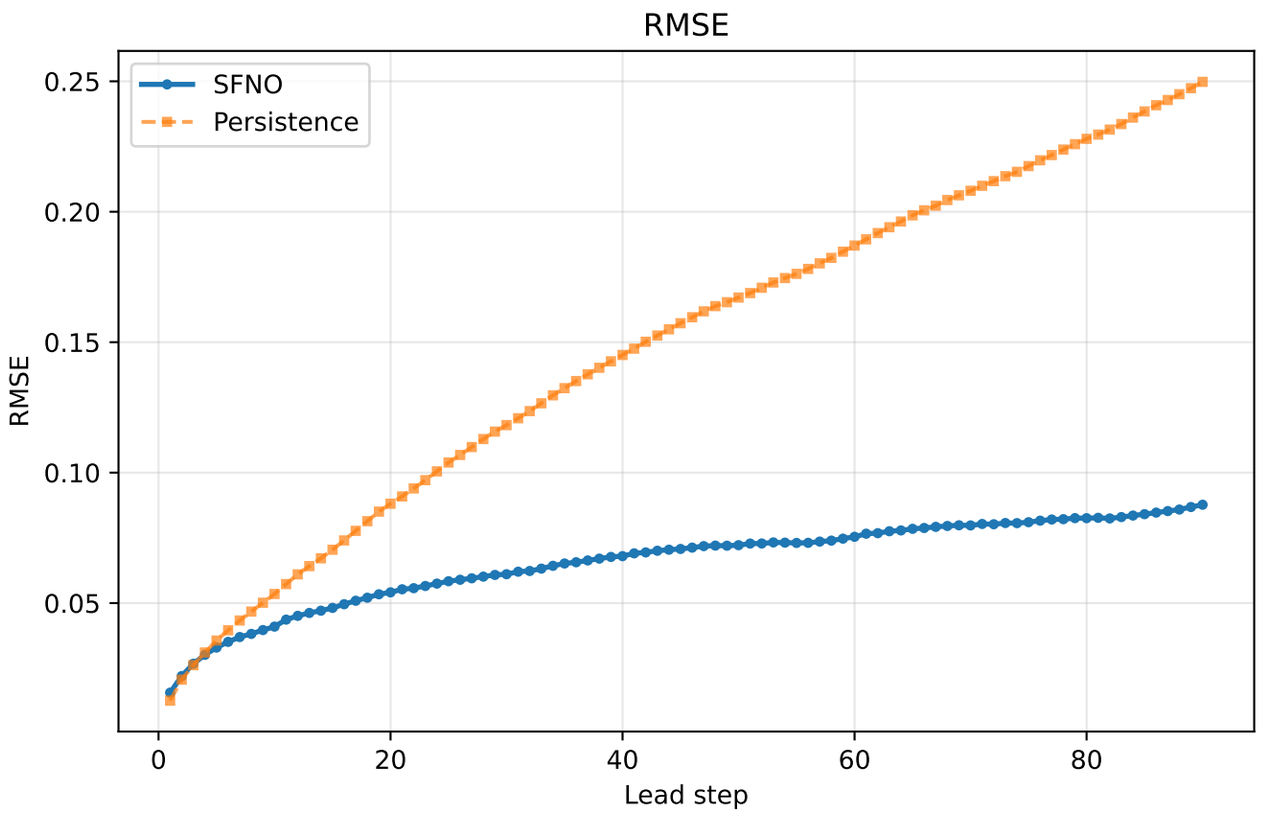


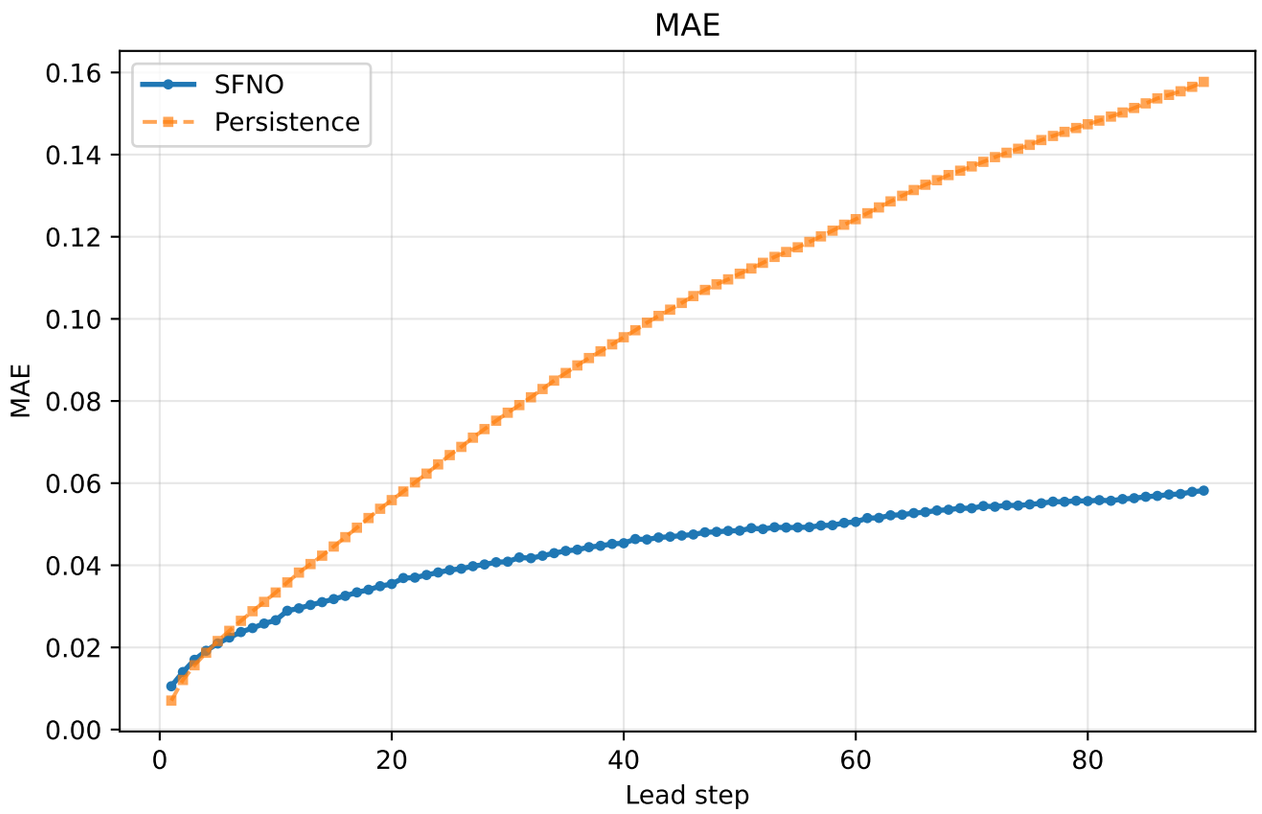


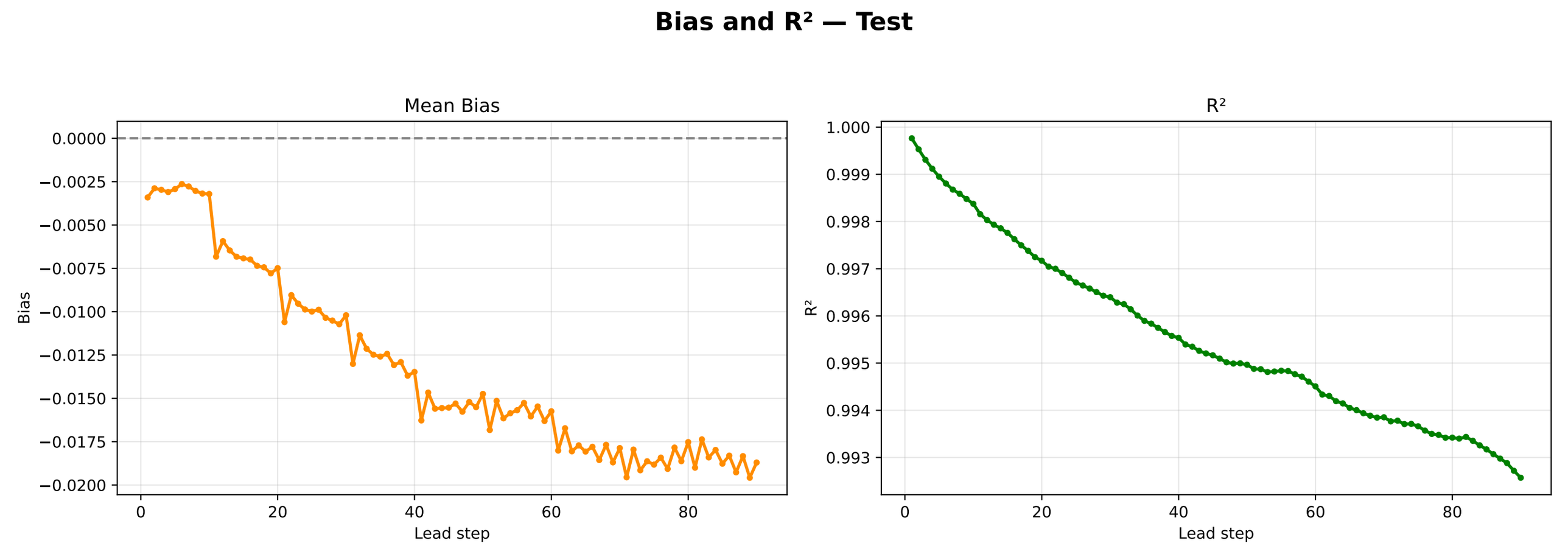
Metrics by Lead Step — Test

Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0157	0.0126	-24.0%	0.0105	-0.0034	0.9713	0.9796	0.9998
2	0.0220	0.0206	-6.7%	0.0140	-0.0029	0.9414	0.9462	0.9995
3	0.0268	0.0262	-2.1%	0.0170	-0.0030	0.9101	0.9132	0.9993
4	0.0302	0.0311	2.8%	0.0192	-0.0031	0.8818	0.8780	0.9991
5	0.0330	0.0357	7.6%	0.0210	-0.0029	0.8554	0.8411	0.9989
6	0.0351	0.0396	11.2%	0.0224	-0.0026	0.8313	0.8072	0.9988
7	0.0370	0.0433	14.6%	0.0237	-0.0028	0.8091	0.7728	0.9987
8	0.0382	0.0467	18.3%	0.0247	-0.0030	0.7904	0.7389	0.9986
9	0.0397	0.0501	20.9%	0.0258	-0.0032	0.7723	0.7063	0.9985
10	0.0410	0.0535	23.4%	0.0266	-0.0032	0.7556	0.6743	0.9984
11	0.0437	0.0573	23.8%	0.0289	-0.0068	0.7232	0.6410	0.9982
12	0.0451	0.0610	26.1%	0.0295	-0.0059	0.7043	0.6068	0.9980
13	0.0462	0.0642	28.0%	0.0303	-0.0065	0.6863	0.5769	0.9979
14	0.0471	0.0672	29.9%	0.0310	-0.0068	0.6722	0.5500	0.9979
15	0.0482	0.0705	31.6%	0.0318	-0.0069	0.6559	0.5219	0.9978
16	0.0496	0.0740	33.0%	0.0325	-0.0070	0.6375	0.4938	0.9976
17	0.0509	0.0777	34.5%	0.0334	-0.0074	0.6199	0.4629	0.9975
18	0.0521	0.0814	36.0%	0.0340	-0.0074	0.6037	0.4320	0.9974
19	0.0534	0.0851	37.3%	0.0349	-0.0078	0.5852	0.4030	0.9972
20	0.0541	0.0881	38.6%	0.0354	-0.0075	0.5710	0.3801	0.9972
21	0.0553	0.0909	39.2%	0.0369	-0.0106	0.5435	0.3592	0.9970
22	0.0557	0.0939	40.7%	0.0370	-0.0090	0.5345	0.3356	0.9970
23	0.0566	0.0971	41.8%	0.0376	-0.0095	0.5206	0.3131	0.9969
24	0.0575	0.1005	42.8%	0.0382	-0.0099	0.5081	0.2889	0.9968
25	0.0584	0.1039	43.8%	0.0388	-0.0100	0.4936	0.2670	0.9967
26	0.0589	0.1068	44.8%	0.0392	-0.0099	0.4796	0.2487	0.9966
27	0.0595	0.1098	45.8%	0.0398	-0.0104	0.4652	0.2287	0.9966
28	0.0601	0.1129	46.7%	0.0402	-0.0105	0.4519	0.2138	0.9965
29	0.0608	0.1157	47.5%	0.0407	-0.0107	0.4360	0.2032	0.9964
30	0.0611	0.1182	48.3%	0.0409	-0.0102	0.4228	0.1947	0.9964
31	0.0620	0.1208	48.7%	0.0419	-0.0130	0.3967	0.1854	0.9963
32	0.0623	0.1236	49.6%	0.0417	-0.0114	0.3877	0.1758	0.9962
33	0.0632	0.1266	50.1%	0.0423	-0.0121	0.3677	0.1635	0.9961
34	0.0643	0.1296	50.4%	0.0430	-0.0125	0.3507	0.1532	0.9960
35	0.0652	0.1324	50.8%	0.0435	-0.0126	0.3325	0.1445	0.9959
36	0.0657	0.1351	51.4%	0.0438	-0.0124	0.3170	0.1347	0.9958
37	0.0663	0.1377	51.8%	0.0444	-0.0131	0.3028	0.1282	0.9957
38	0.0670	0.1402	52.2%	0.0447	-0.0129	0.2896	0.1243	0.9957
39	0.0677	0.1427	52.6%	0.0452	-0.0137	0.2803	0.1209	0.9956
40	0.0680	0.1451	53.1%	0.0454	-0.0135	0.2735	0.1145	0.9955
41	0.0691	0.1475	53.2%	0.0464	-0.0163	0.2503	0.1073	0.9954
42	0.0694	0.1502	53.8%	0.0463	-0.0147	0.2450	0.1002	0.9953
43	0.0701	0.1526	54.1%	0.0468	-0.0156	0.2362	0.0960	0.9953
44	0.0705	0.1549	54.5%	0.0470	-0.0156	0.2293	0.0912	0.9952
45	0.0707	0.1573	55.0%	0.0472	-0.0155	0.2175	0.0831	0.9952
46	0.0713	0.1596	55.3%	0.0475	-0.0153	0.2055	0.0757	0.9951
47	0.0718	0.1618	55.6%	0.0480	-0.0158	0.1955	0.0700	0.9950
48	0.0720	0.1638	56.0%	0.0481	-0.0152	0.1857	0.0685	0.9950
49	0.0720	0.1653	56.5%	0.0484	-0.0155	0.1787	0.0718	0.9950
50	0.0722	0.1671	56.8%	0.0484	-0.0147	0.1699	0.0747	0.9950
51	0.0728	0.1688	56.9%	0.0490	-0.0168	0.1556	0.0768	0.9949
52	0.0729	0.1709	57.4%	0.0488	-0.0151	0.1497	0.0753	0.9949
53	0.0733	0.1729	57.6%	0.0493	-0.0162	0.1423	0.0730	0.9948
54	0.0732	0.1746	58.1%	0.0492	-0.0159	0.1414	0.0736	0.9948
55	0.0731	0.1762	58.5%	0.0492	-0.0157	0.1409	0.0758	0.9948
56	0.0731	0.1781	59.0%	0.0493	-0.0153	0.1370	0.0736	0.9948
57	0.0736	0.1802	59.2%	0.0497	-0.0160	0.1321	0.0694	0.9948
58	0.0739	0.1823	59.4%	0.0498	-0.0155	0.1257	0.0654	0.9947
59	0.0747	0.1847	59.6%	0.0503	-0.0163	0.1177	0.0584	0.9946
60	0.0754	0.1870	59.7%	0.0506	-0.0157	0.1109	0.0535	0.9945
61	0.0766	0.1894	59.6%	0.0515	-0.0180	0.1010	0.0470	0.9943
62	0.0768	0.1918	60.0%	0.0515	-0.0167	0.0959	0.0385	0.9943
63	0.0775	0.1941	60.0%	0.0522	-0.0181	0.0868	0.0316	0.9942
64	0.0778	0.1963	60.3%	0.0523	-0.0177	0.0806	0.0248	0.9941
65	0.0785	0.1986	60.5%	0.0527	-0.0181	0.0759	0.0192	0.9941
66	0.0788	0.2006	60.7%	0.0529	-0.0178	0.0703	0.0171	0.9940
67	0.0792	0.2023	60.8%	0.0533	-0.0186	0.0669	0.0183	0.9939
68	0.0796	0.2045	61.1%	0.0535	-0.0177	0.0599	0.0135	0.9939
69	0.0798	0.2063	61.3%	0.0539	-0.0187	0.0582	0.0125	0.9938
70	0.0798	0.2080	61.7%	0.0539	-0.0179	0.0557	0.0122	0.9939
71	0.0803	0.2099	61.7%	0.0544	-0.0196	0.0501	0.0110	0.9938
72	0.0802	0.2117	62.1%	0.0543	-0.0180	0.0496	0.0120	0.9938
73	0.0807	0.2136	62.2%	0.0546	-0.0191	0.0472	0.0109	0.9937
74	0.0806	0.2153	62.5%	0.0545	-0.0186	0.0476	0.0133	0.9937
75	0.0810	0.2175	62.8%	0.0548	-0.0188	0.0466	0.0086	0.9937
76	0.0816	0.2197	62.9%	0.0551	-0.0184	0.0414	0.0050	0.9936
77	0.0820	0.2217	63.0%	0.0555	-0.0191	0.0396	0.0049	0.9935
78	0.0822	0.2238	63.3%	0.0555	-0.0178	0.0377	0.0037	0.9935
79	0.0826	0.2259	63.5%	0.0557	-0.0186	0.0353	0.0019	0.9934
80	0.0825	0.2279	63.8%	0.0556	-0.0175	0.0332	-0.0015	0.9934
81	0.0827	0.2296	64.0%	0.0559	-0.0190	0.0344	-0.0020	0.9934
82	0.0824	0.2315	64.4%	0.0557	-0.0174	0.0344	-0.0056	0.9934
83	0.0830	0.2336	64.5%	0.0561	-0.0184	0.0344	-0.0076	0.9934
84	0.0836	0.2361	64.6%	0.0563	-0.0180	0.0345	-0.0090	0.9933
85	0.0841	0.2385	64.7%	0.0567	-0.0188	0.0351	-0.0119	0.9932
86	0.0847	0.2408	64.8%	0.0569	-0.0183	0.0319	-0.0147	0.9931
87	0.0853	0.2428	64.9%	0.0572	-0.0193	0.0318	-0.0142	0.9930
88	0.0859	0.2451	65.0%	0.0574	-0.0183	0.0289	-0.0133	0.9929
89	0.0868	0.2474	64.9%	0.0579	-0.0196	0.0275	-0.0151	0.9927
90	0.0877	0.2498	64.9%	0.0582	-0.0187	0.0230	-0.0180	0.9926

Forecast Skill vs Lead Step — Test

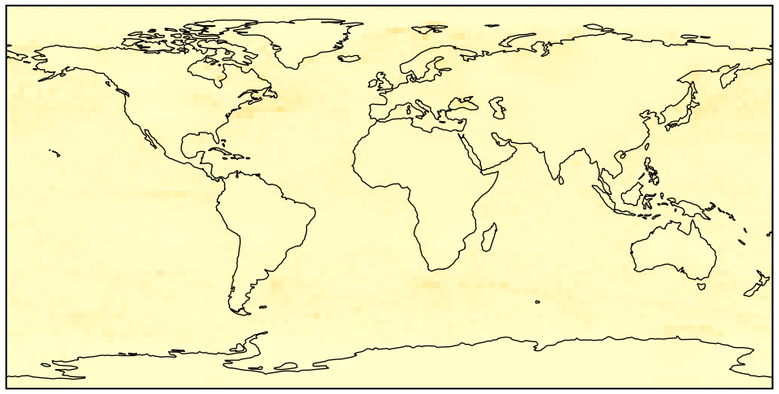




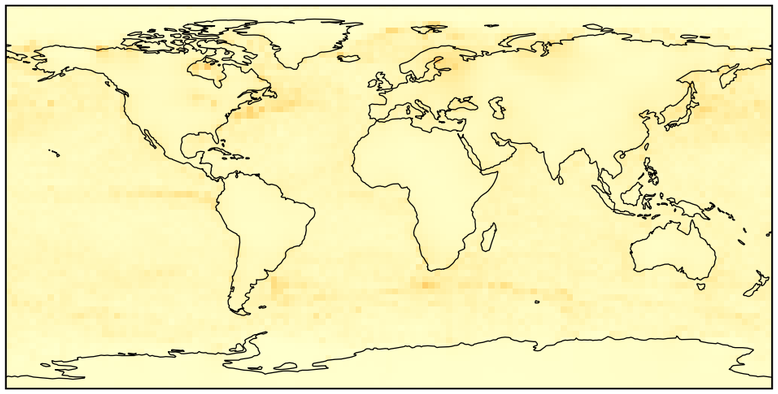

Bias and R² — Test
Mean Bias
Bias
0.0000
−0.0025
−0.0050
−0.0075
−0.0100
−0.0125
−0.0150
−0.0175
−0.0200
0
20
40
60
80
Lead step
R²
R²
1.000
0.999
0.998
0.997
0.996
0.995
0.994
0.993
0
20
40
60
80
Lead step

Spatial RMSE (all grid) — Test

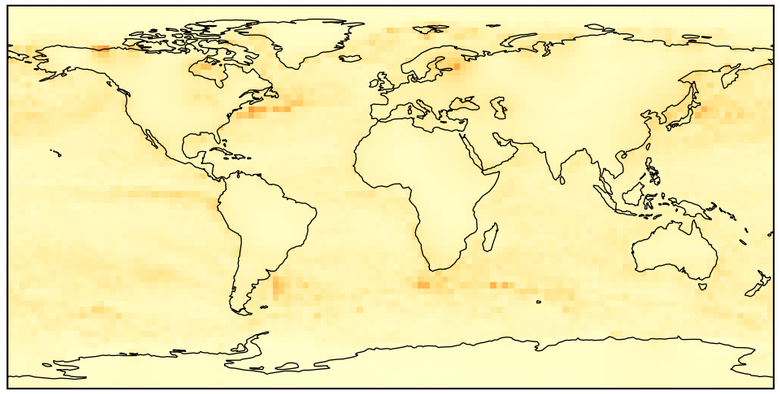
Step 1



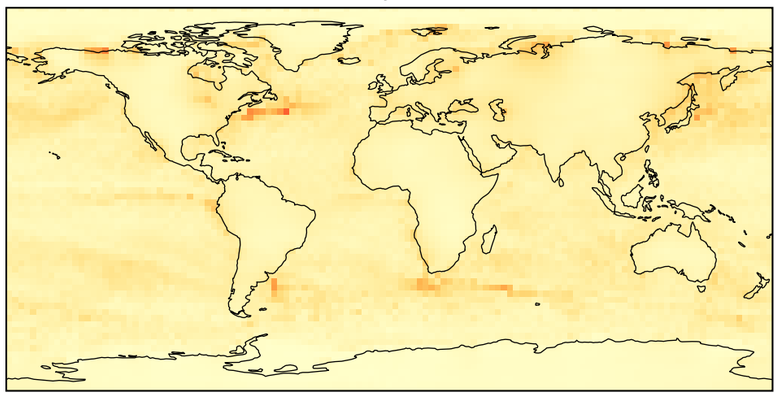
Step 3



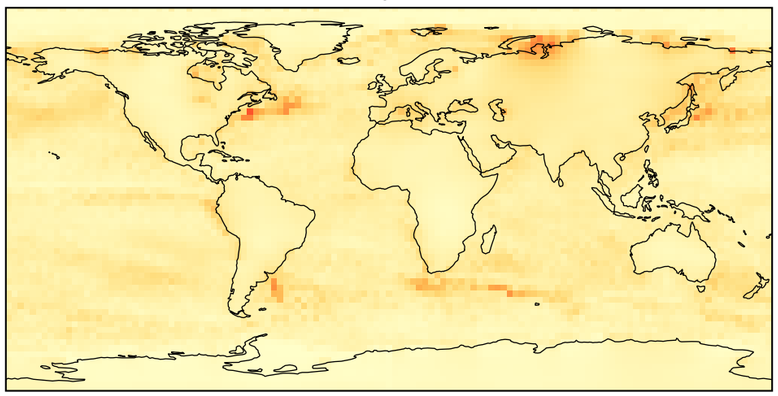
Step 7



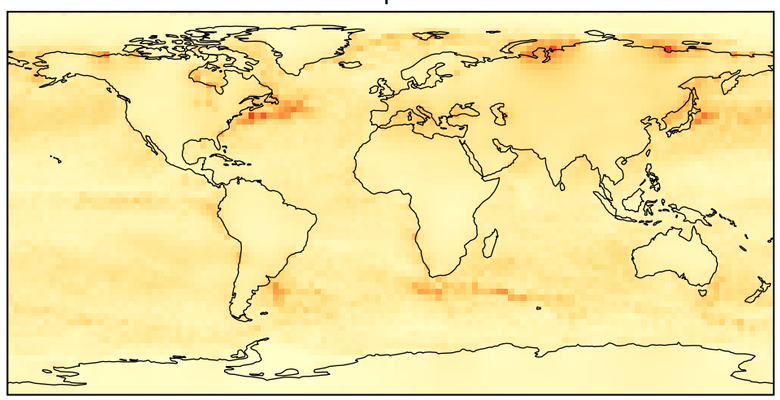
Step 14



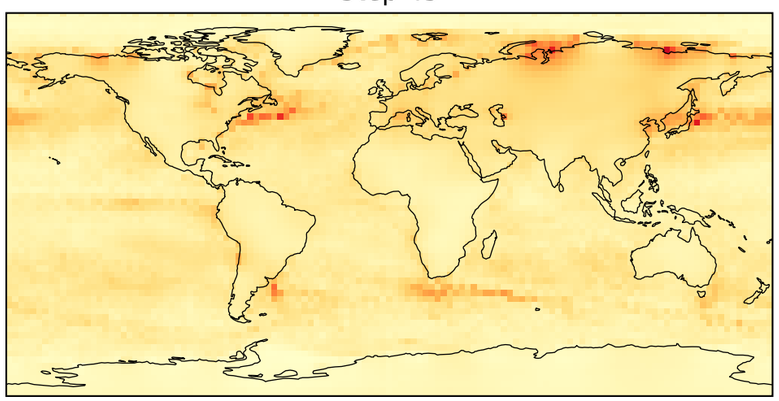
Step 21



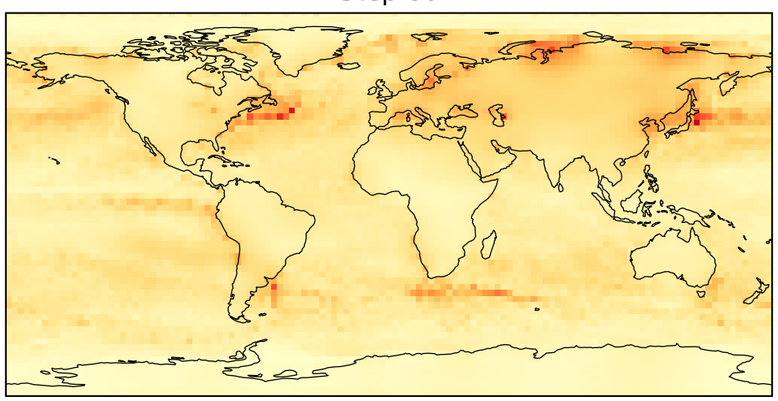
Step 30



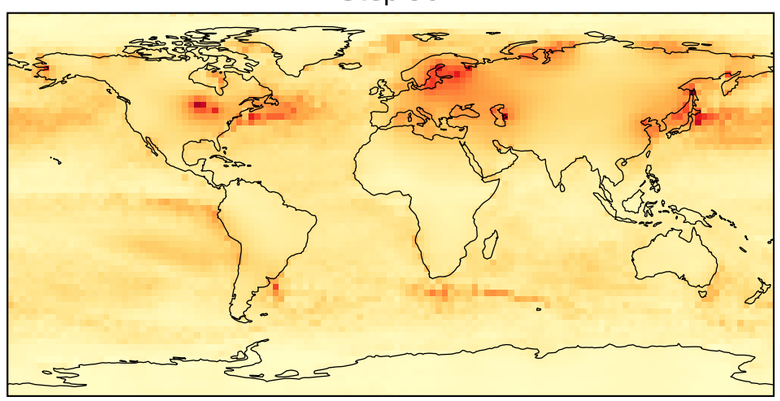
Step 45



Step 60

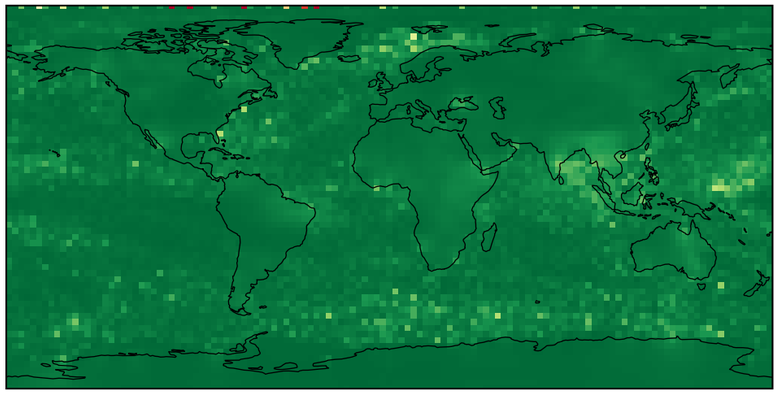


Step 90

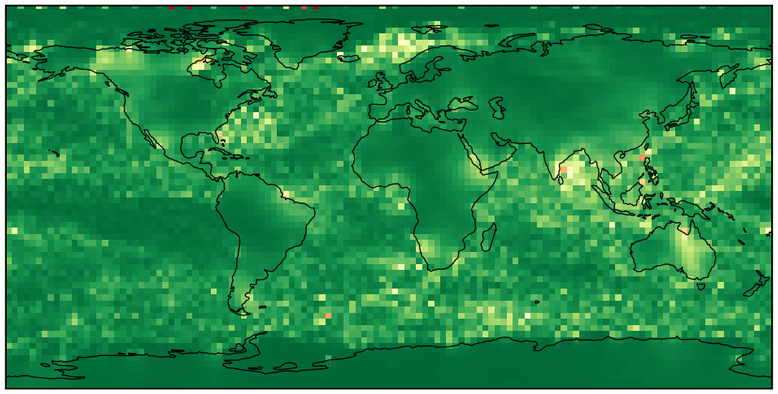


Spatial ACC (all grid) — Test

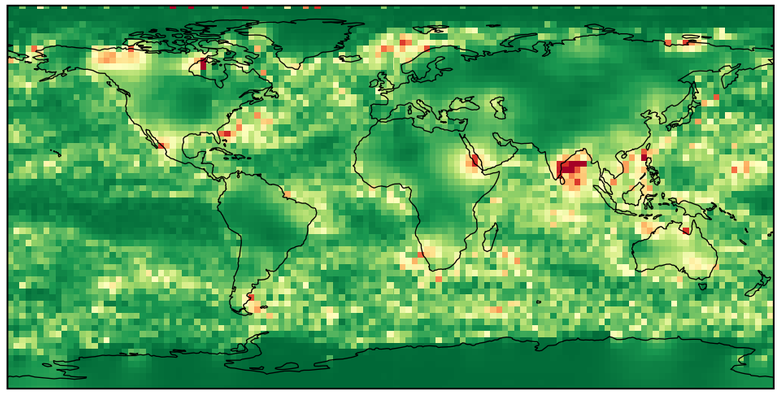
Step 1



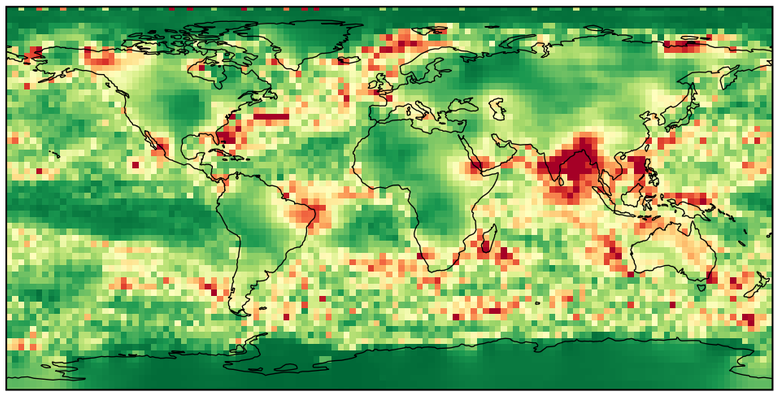
Step 3



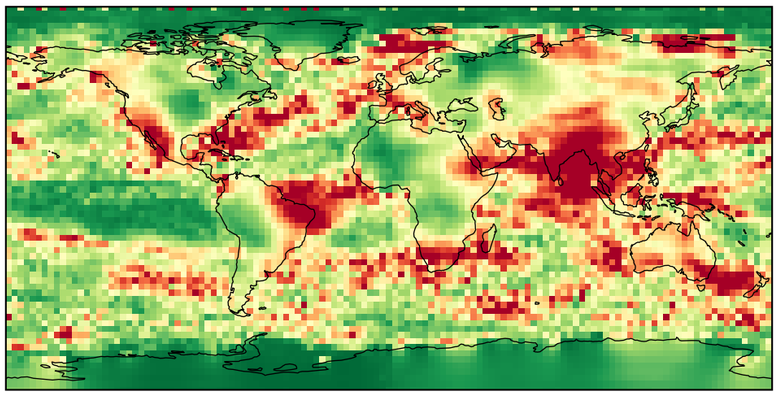
Step 7



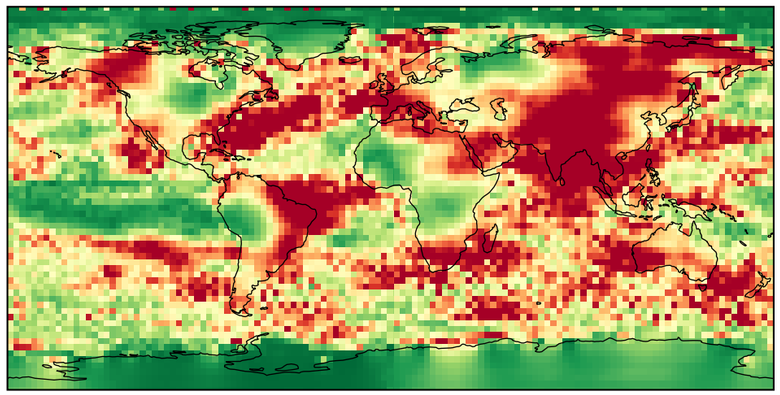
Step 14



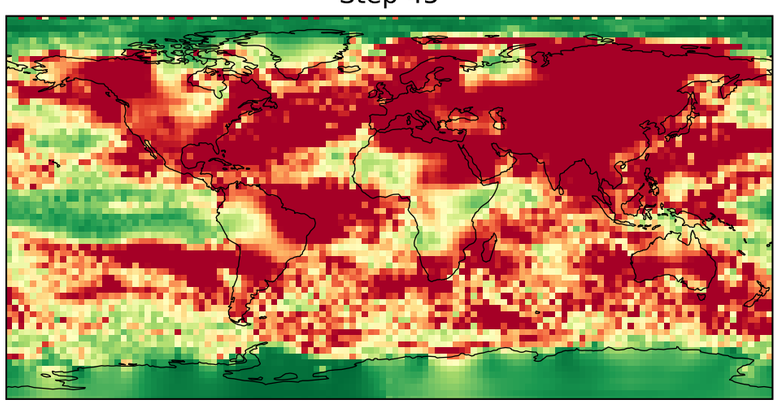
Step 21



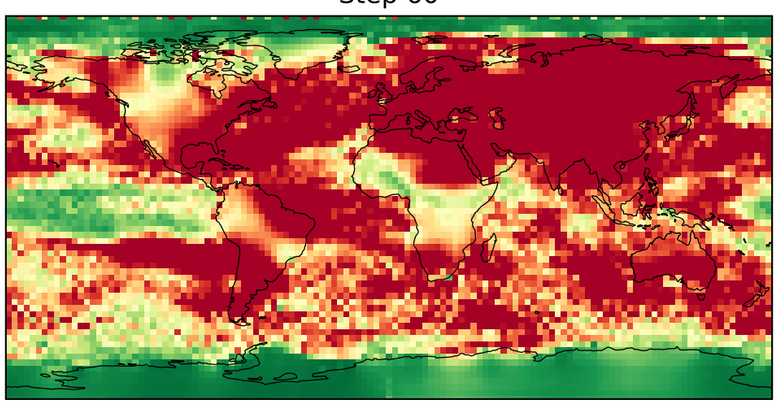
Step 30



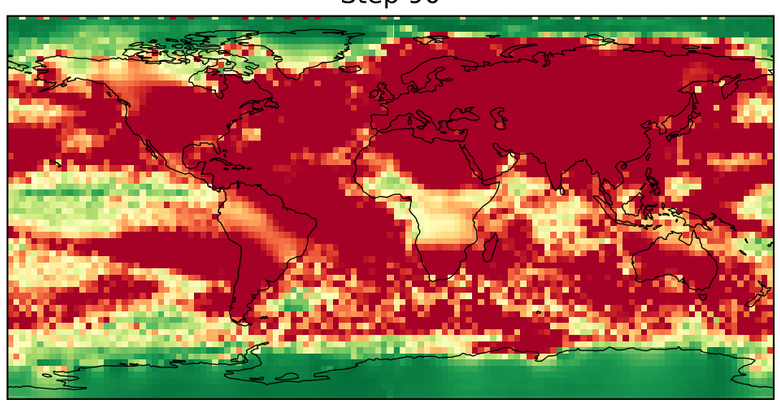
Step 45



Step 60

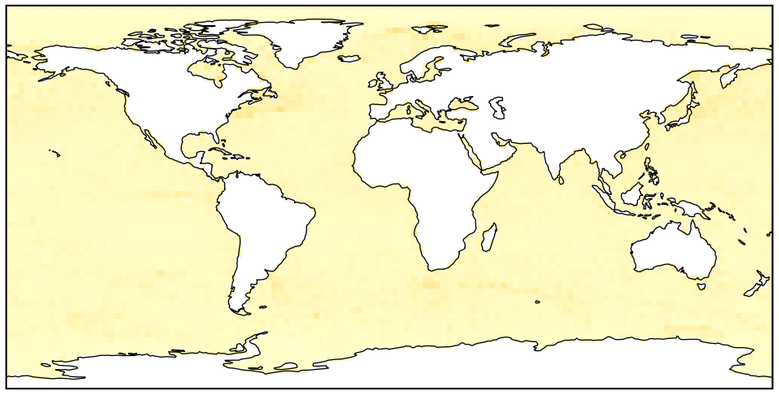


Step 90

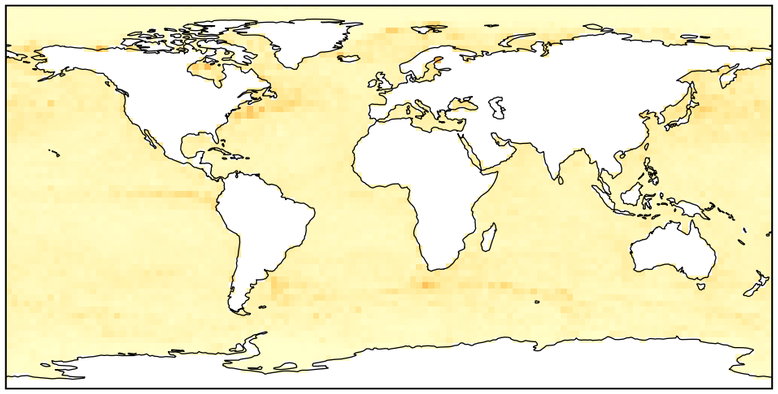


Spatial RMSE (ocean only) — Test

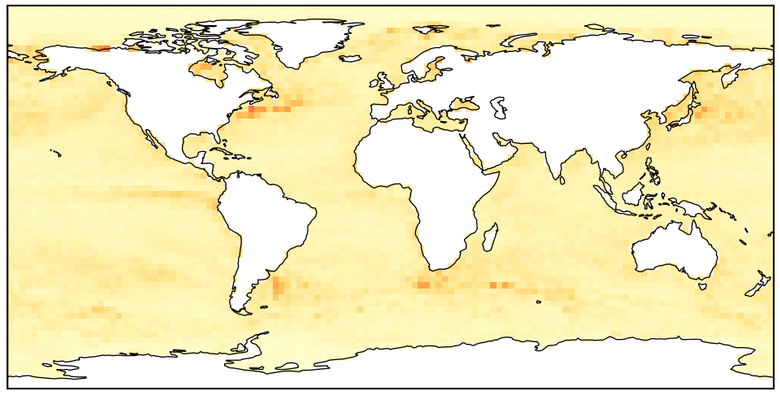
Step 1



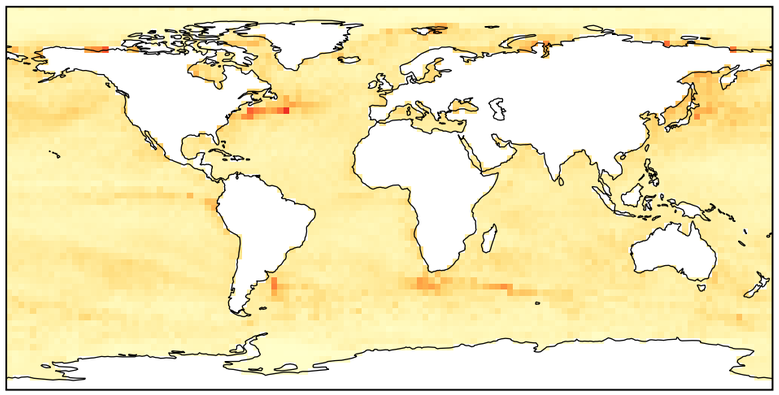
Step 3



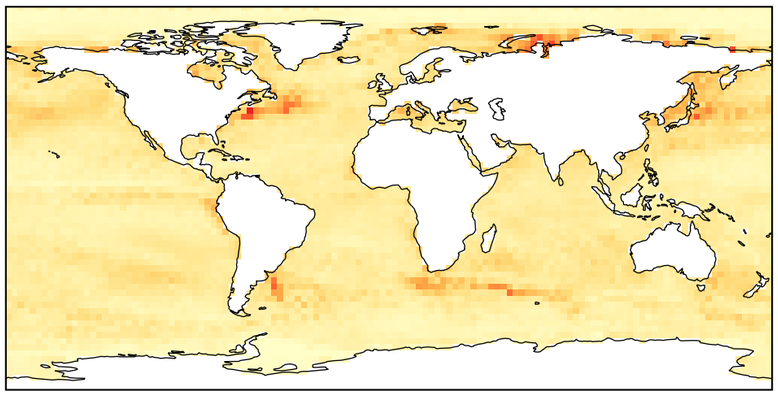
Step 7



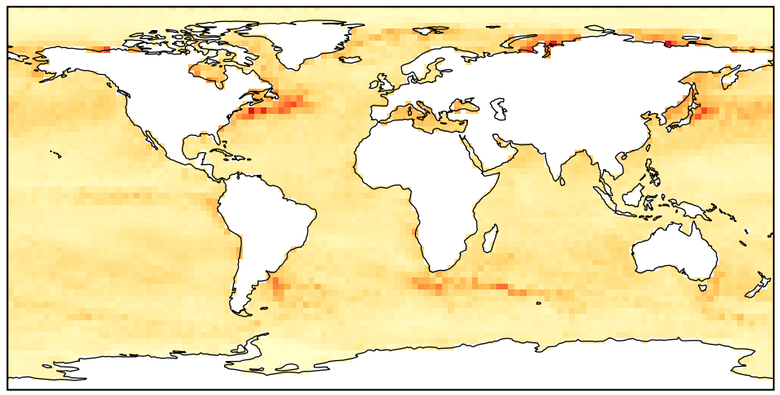
Step 14



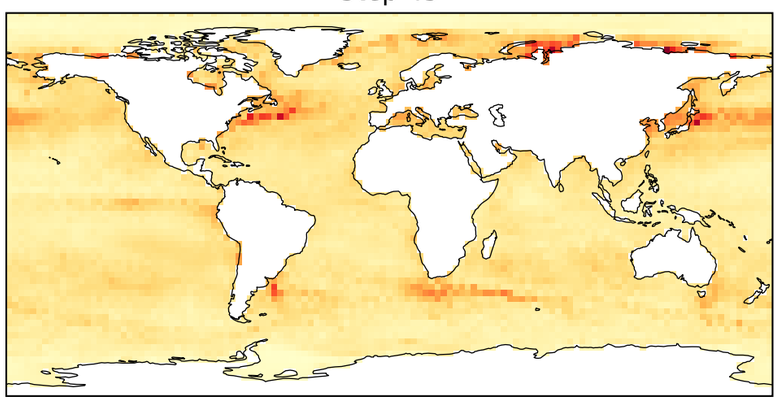
Step 21



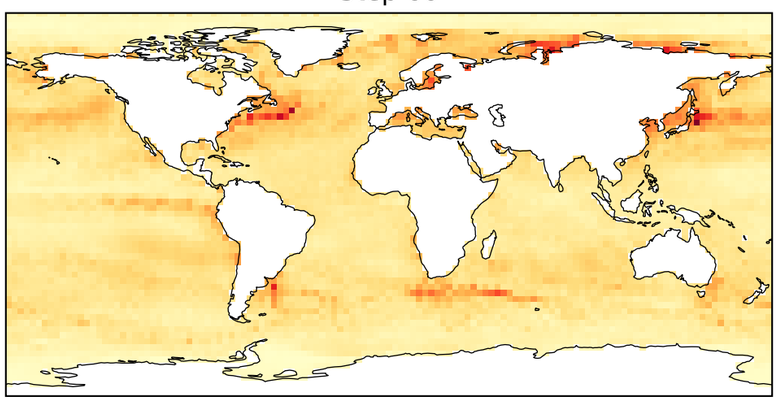
Step 30



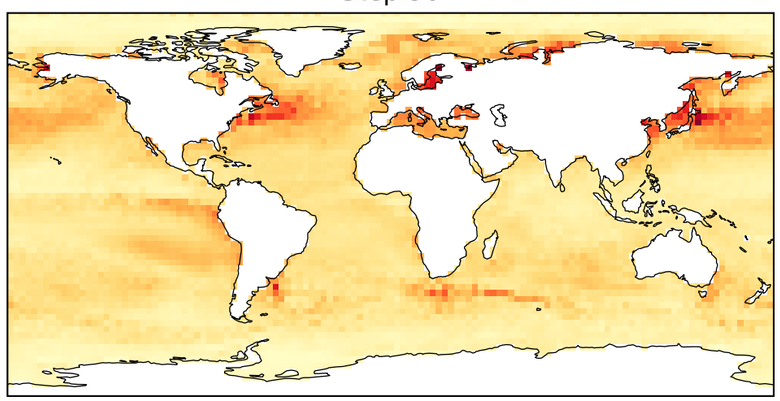
Step 45

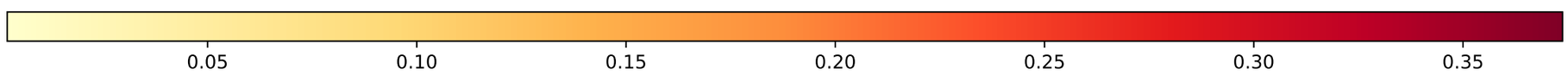


Step 60



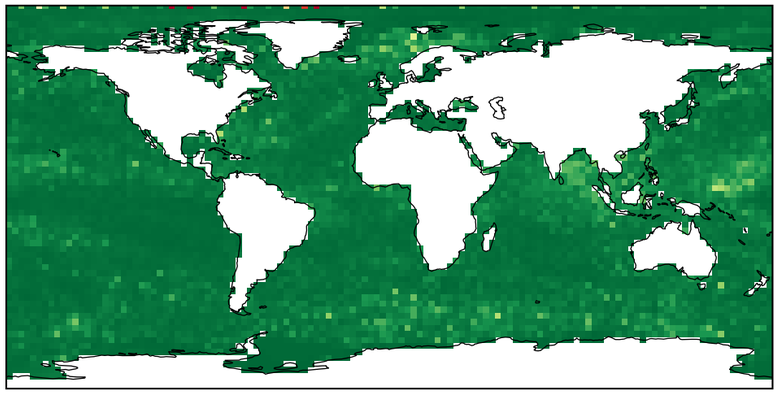
Step 90



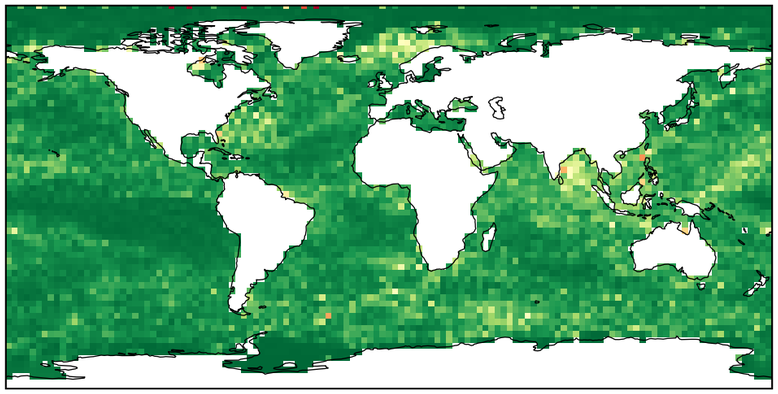


Spatial ACC (ocean only) — Test

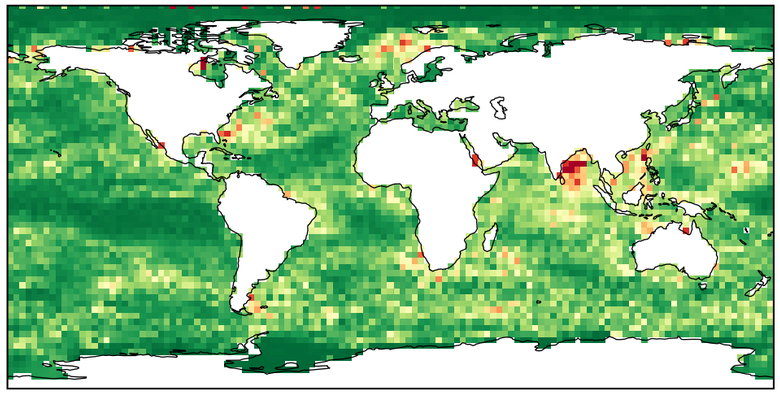
Step 1



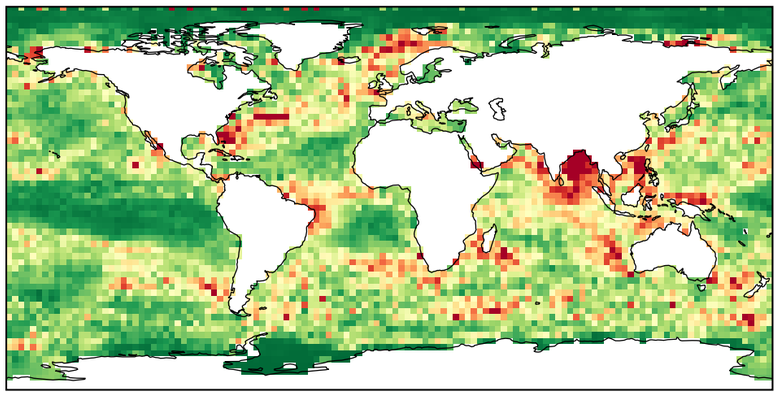
Step 3



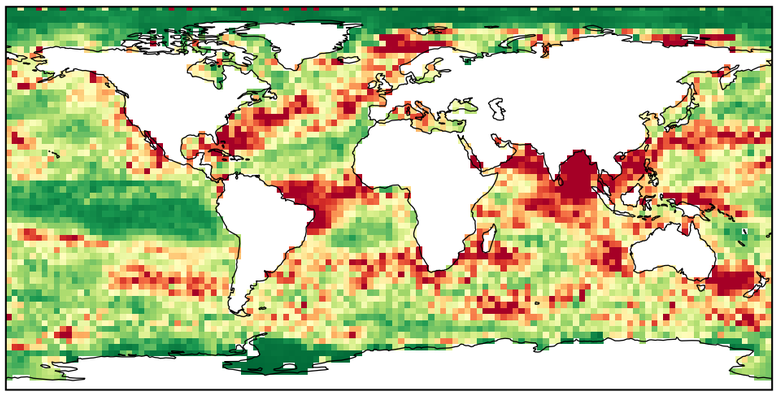
Step 7



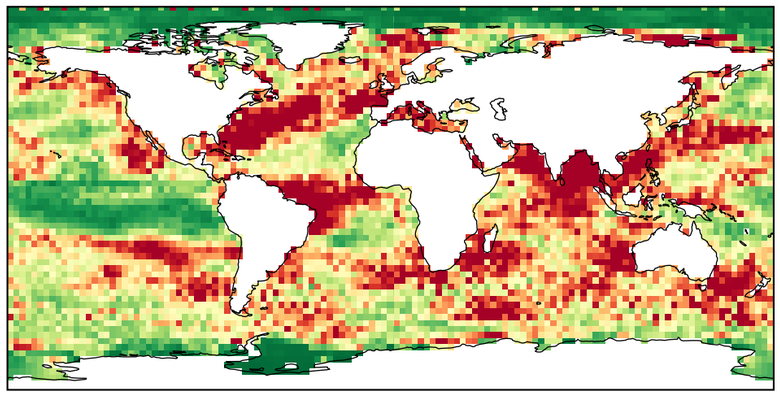
Step 14



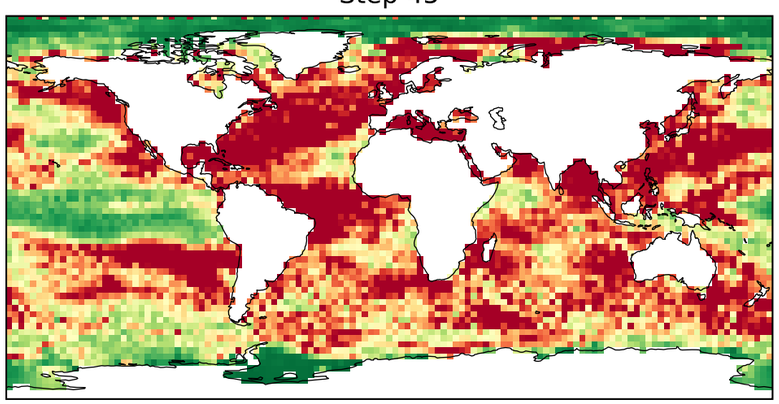
Step 21



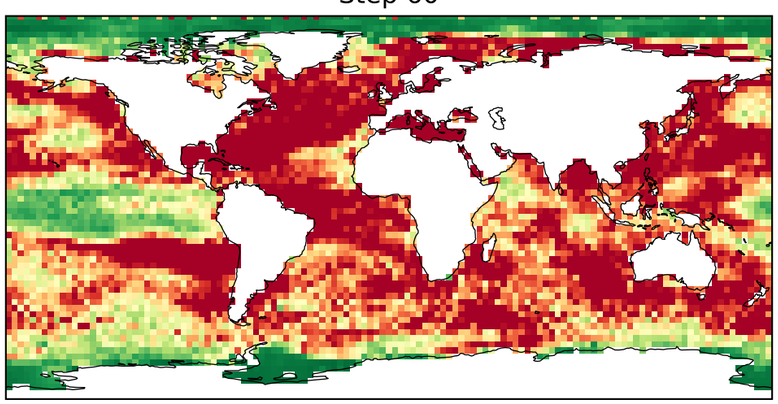
Step 30



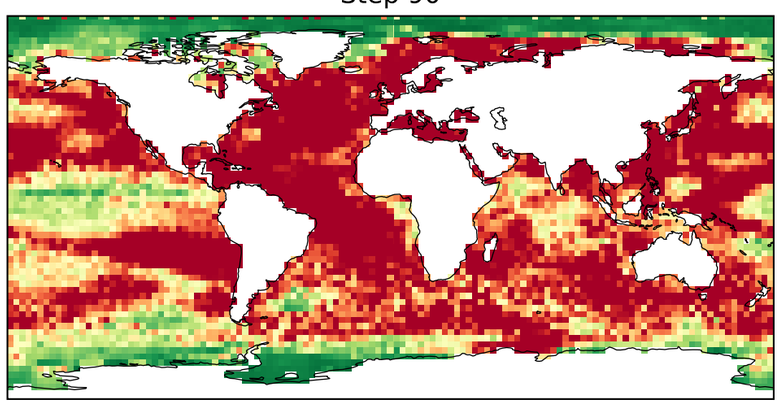
Step 45



Step 60

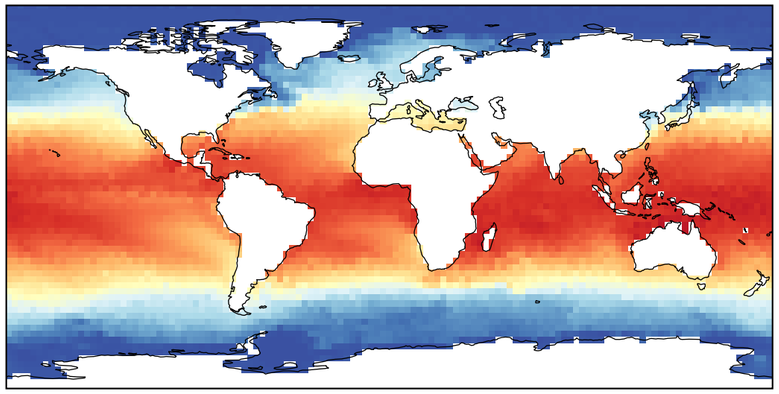


Step 90

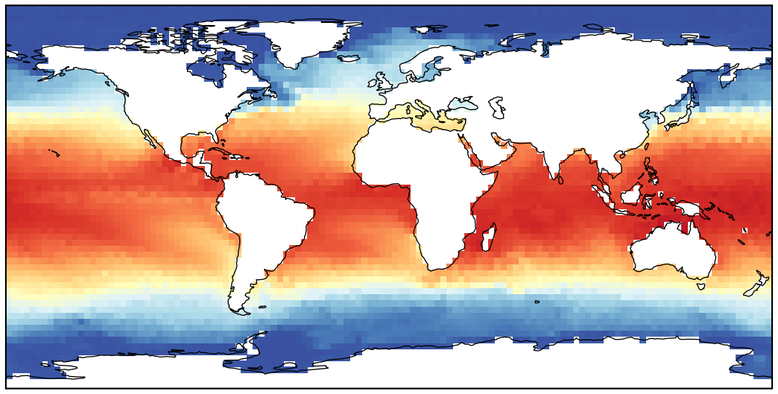


Chain init: 2020-01-11 — Test (°C)

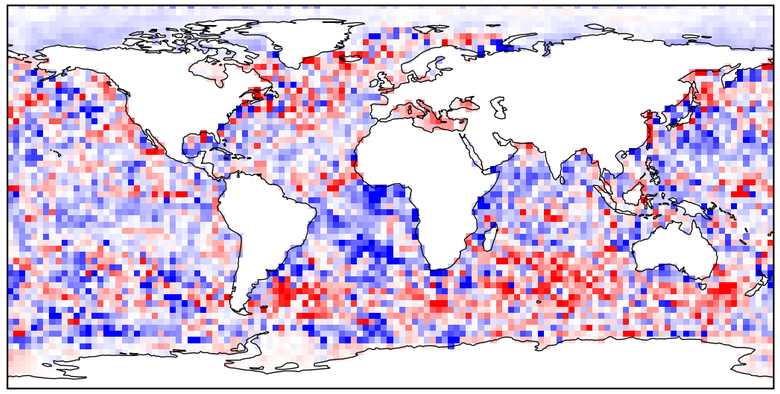
Actual — Step 1 (°C)



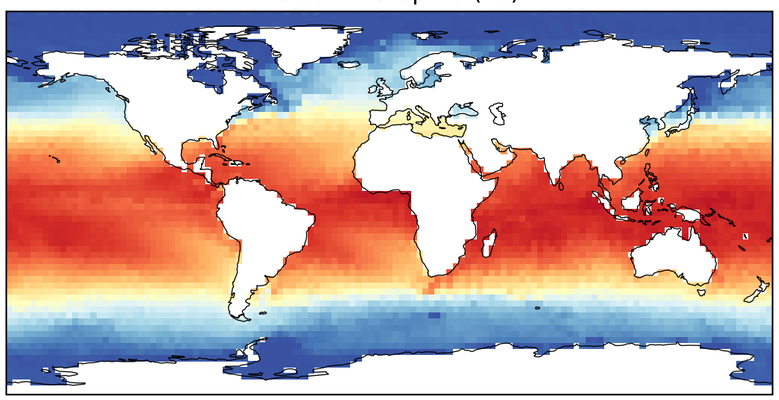
SFNO — Step 1 (°C)



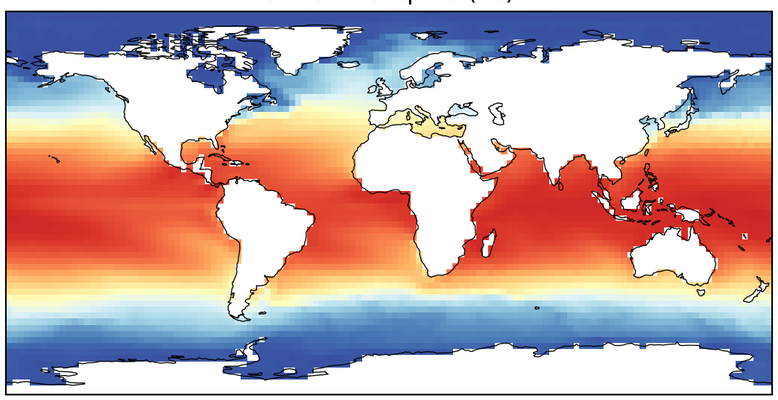
Error — Step 1 (°C)



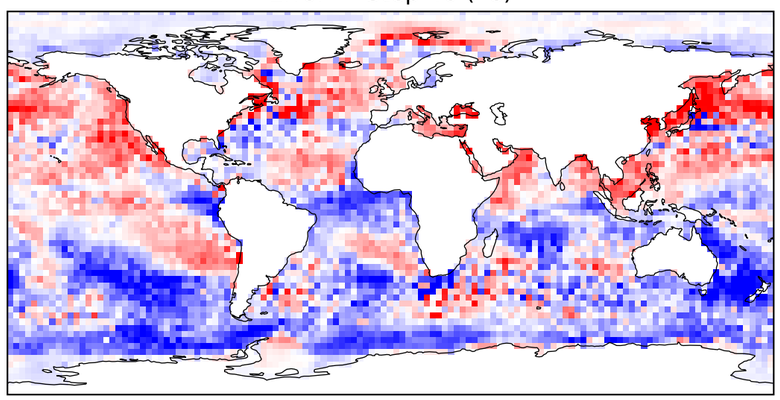
Actual — Step 46 (°C)



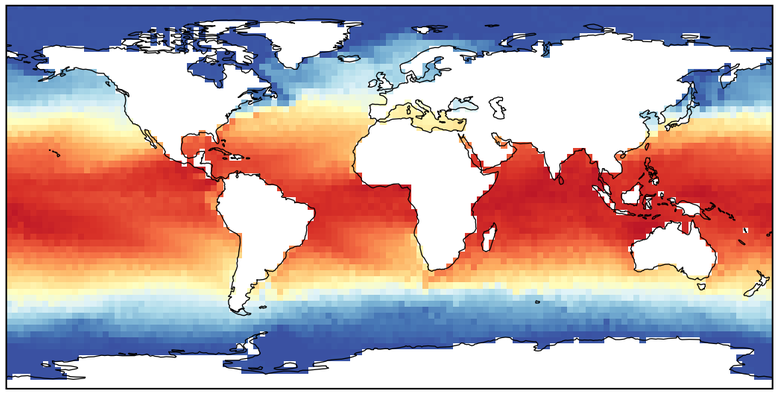
SFNO — Step 46 (°C)



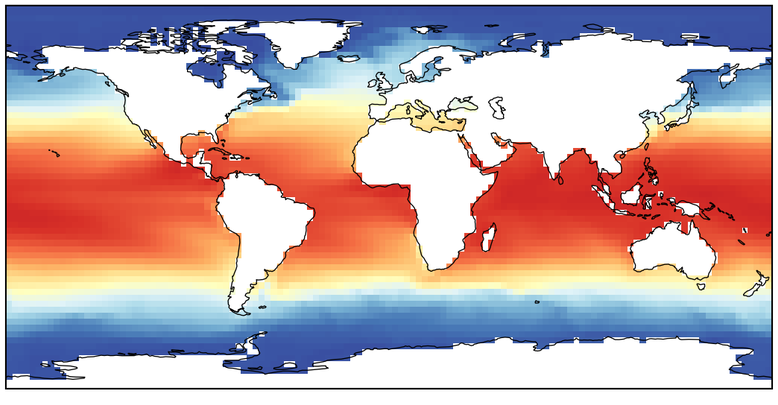
Error — Step 46 (°C)



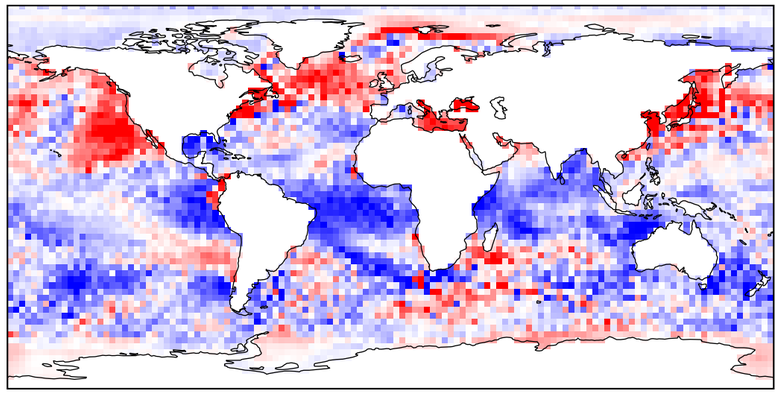
Actual — Step 90 (°C)

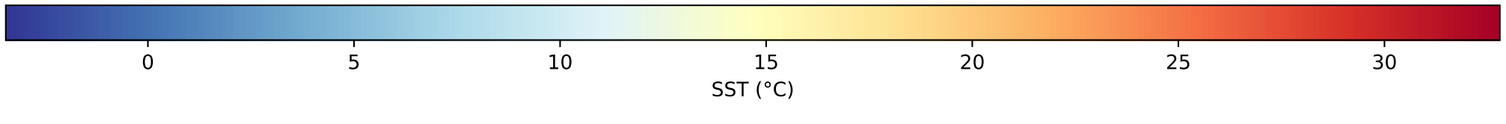


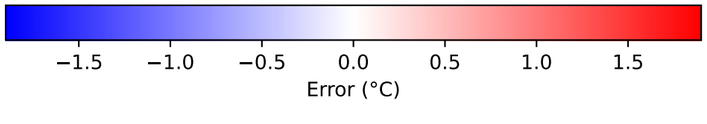
SFNO — Step 90 (°C)



Error — Step 90 (°C)

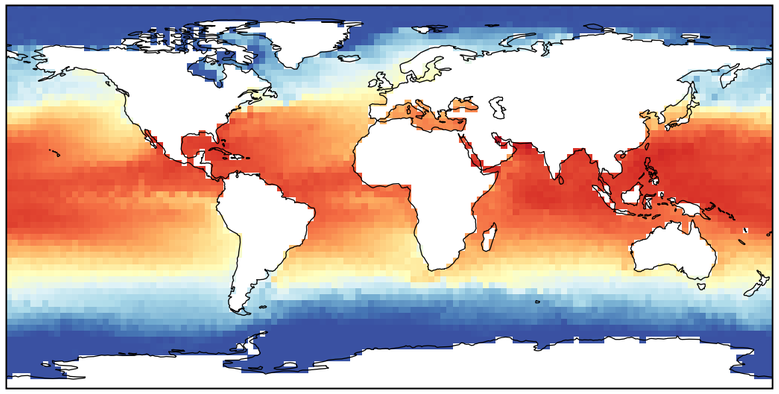




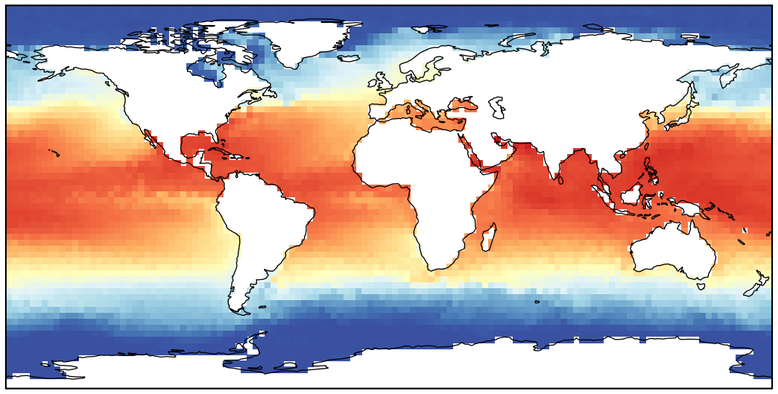


Chain init: 2020-07-09 — Test (°C)

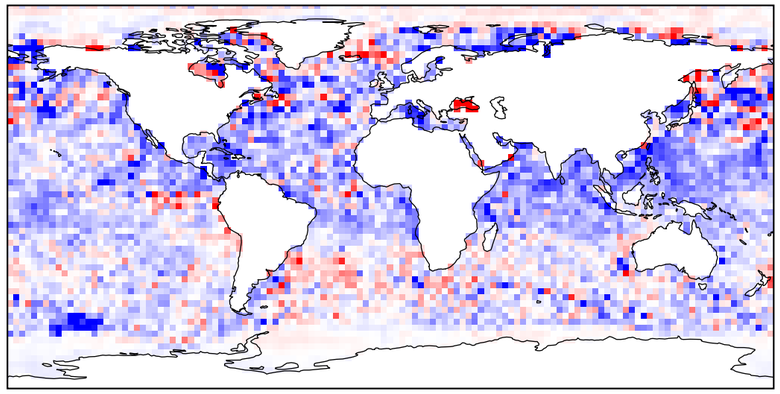
Actual — Step 1 (°C)



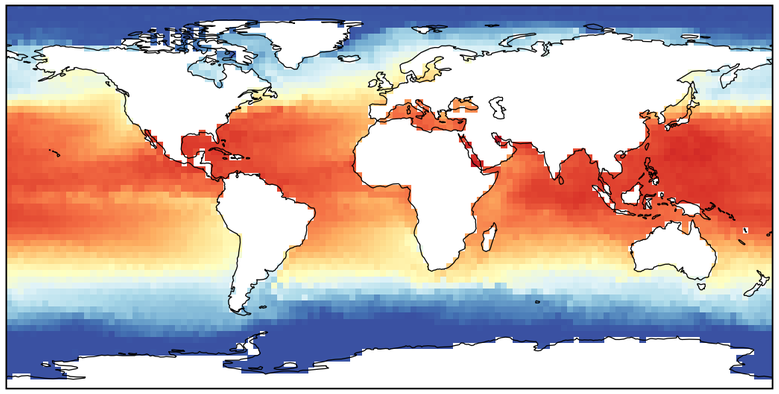
SFNO — Step 1 (°C)



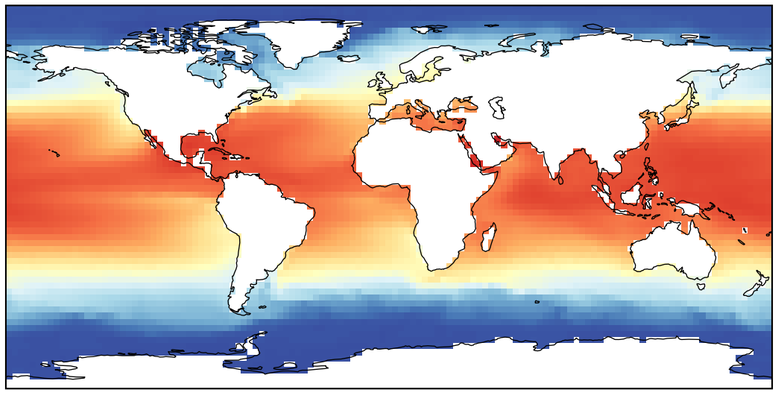
Error — Step 1 (°C)



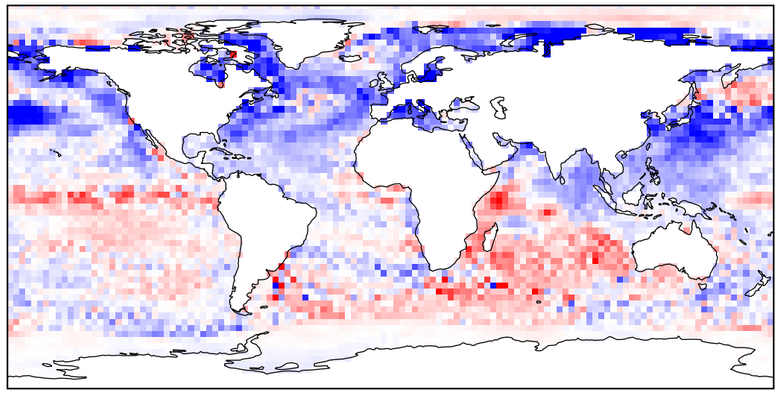
Actual — Step 46 (°C)



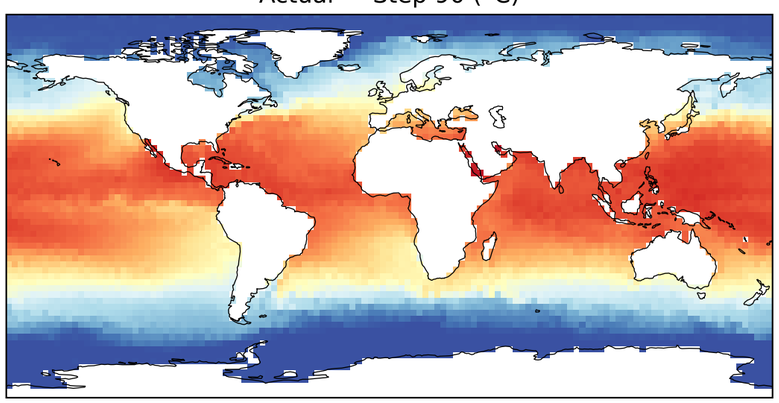
SFNO — Step 46 (°C)



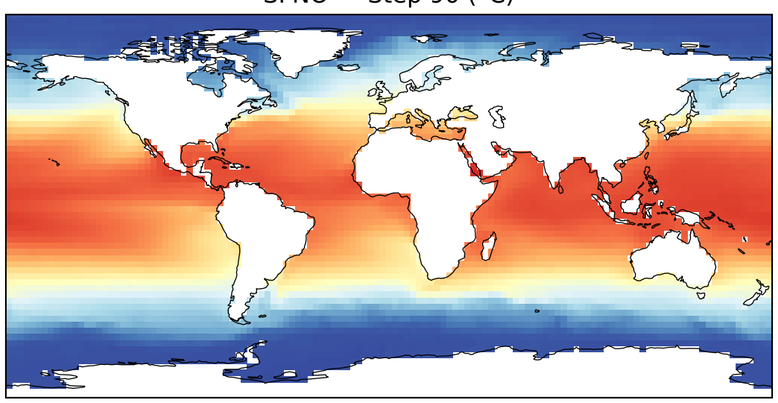
Error — Step 46 (°C)



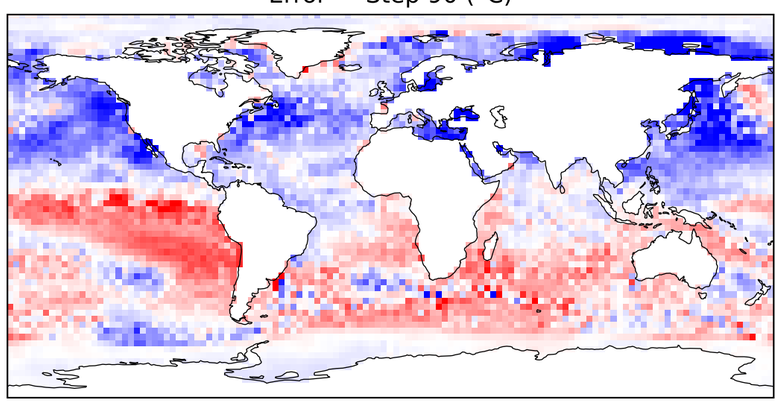
Actual — Step 90 (°C)

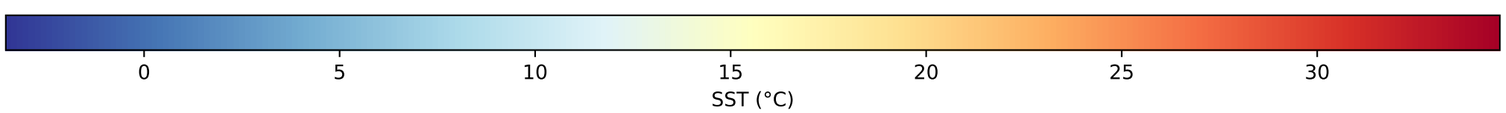


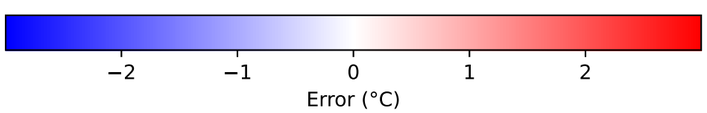
SFNO — Step 90 (°C)



Error — Step 90 (°C)

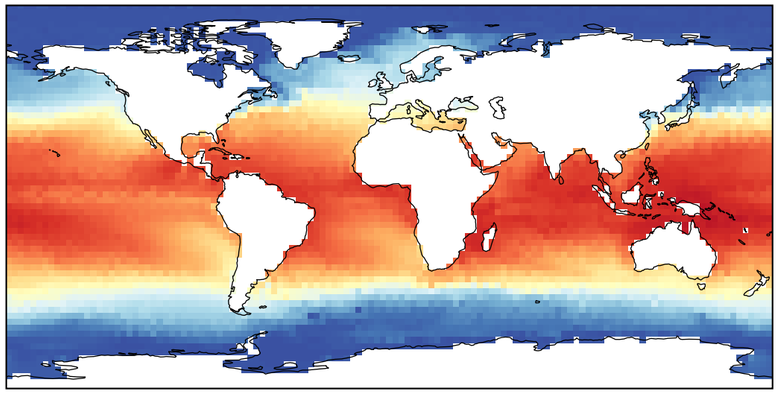




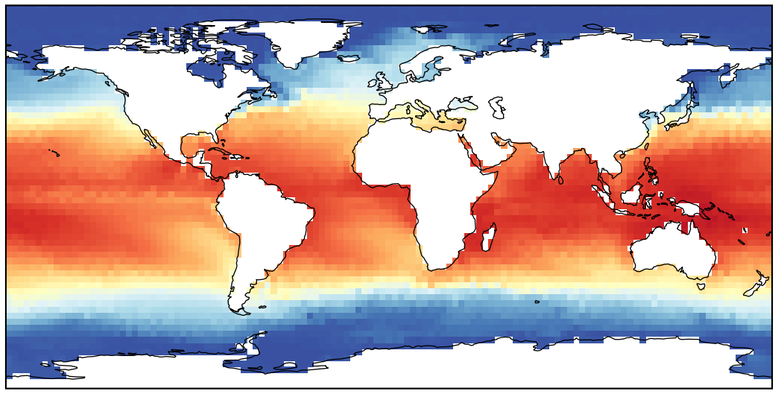


Chain init: 2021-01-06 — Test (°C)

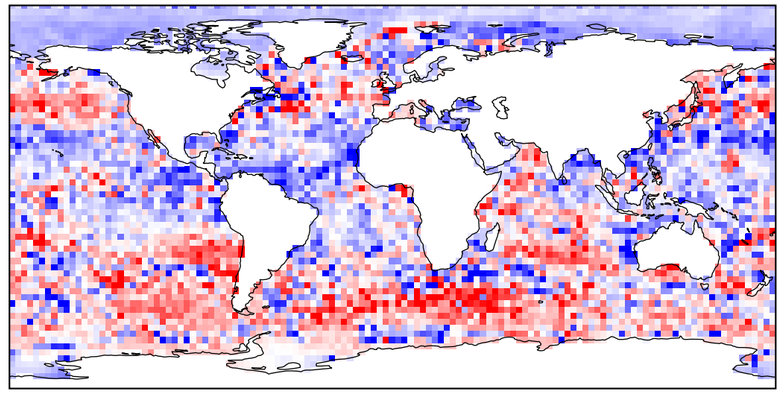
Actual — Step 1 (°C)



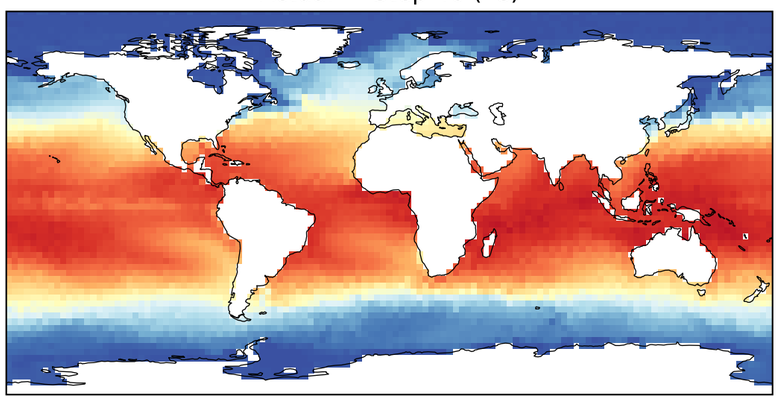
SFNO — Step 1 (°C)



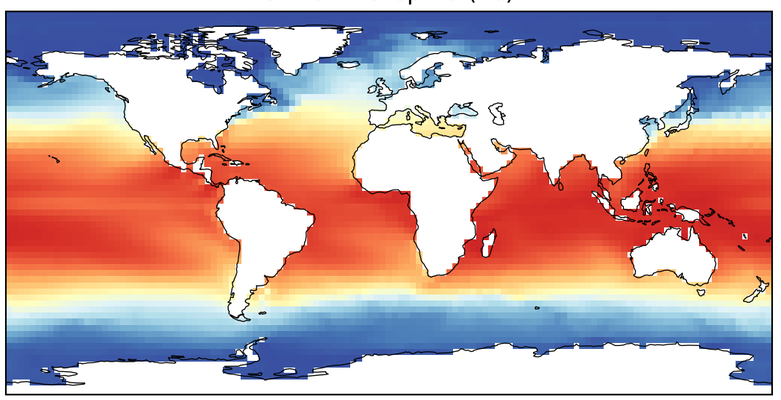
Error — Step 1 (°C)



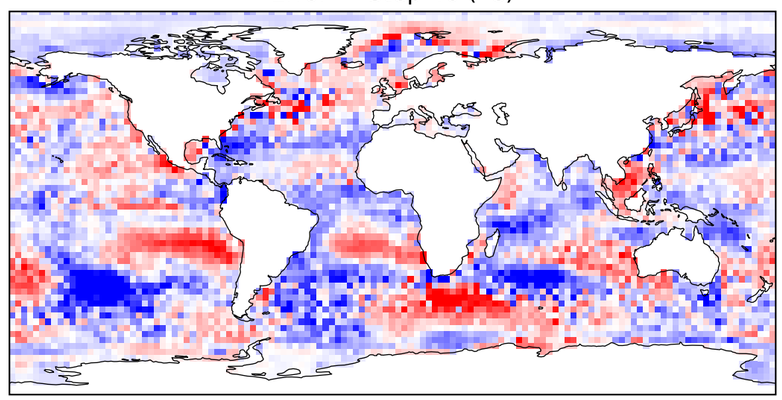
Actual — Step 46 (°C)



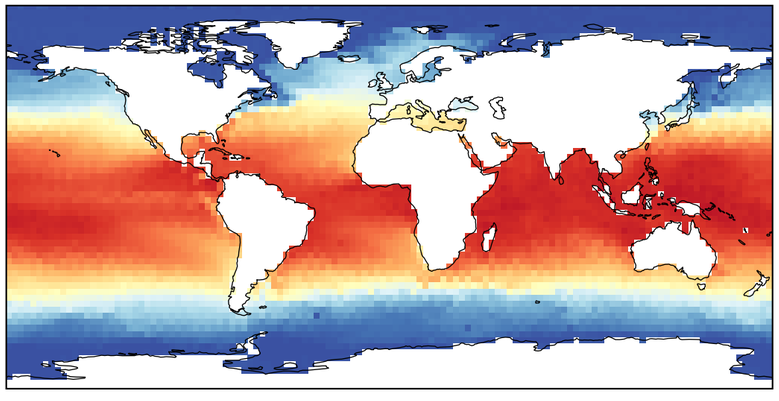
SFNO — Step 46 (°C)



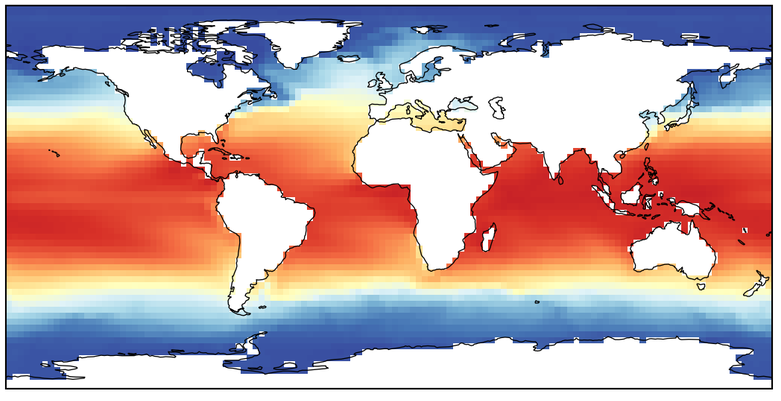
Error — Step 46 (°C)



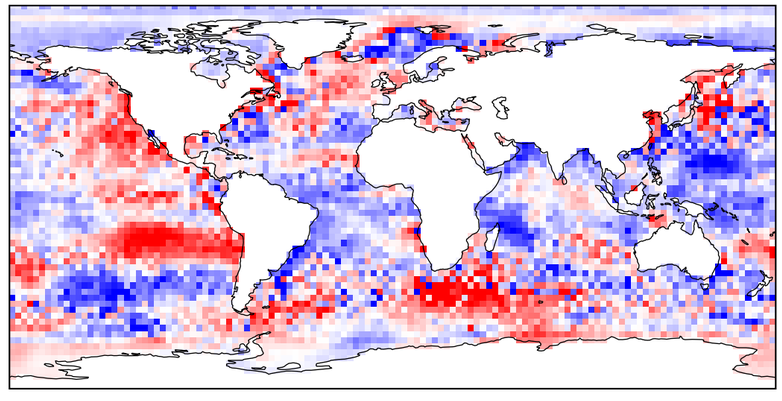
Actual — Step 90 (°C)

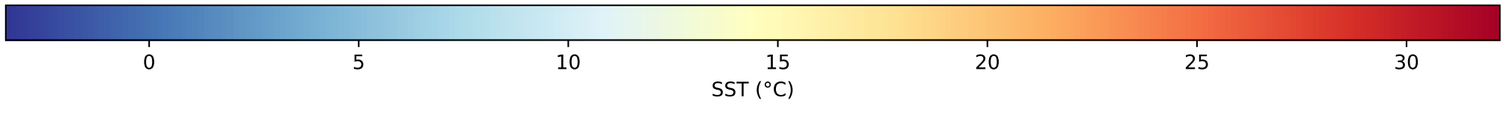


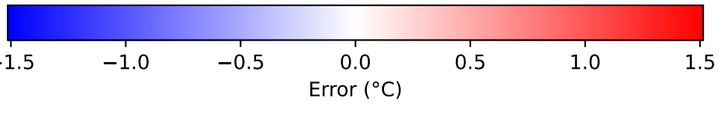
SFNO — Step 90 (°C)



Error — Step 90 (°C)

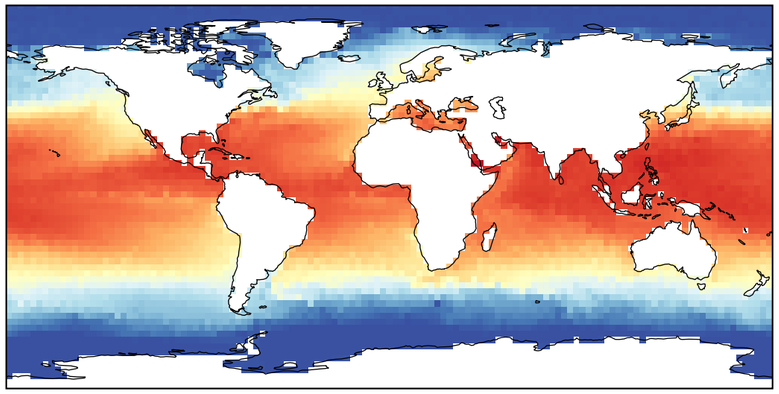




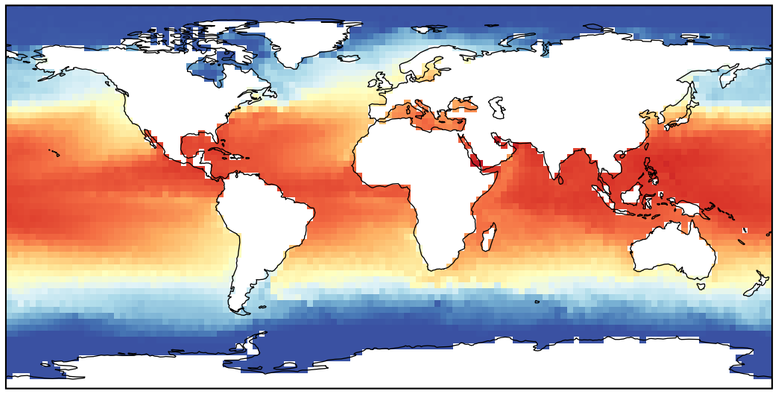


Chain init: 2021-07-05 — Test (°C)

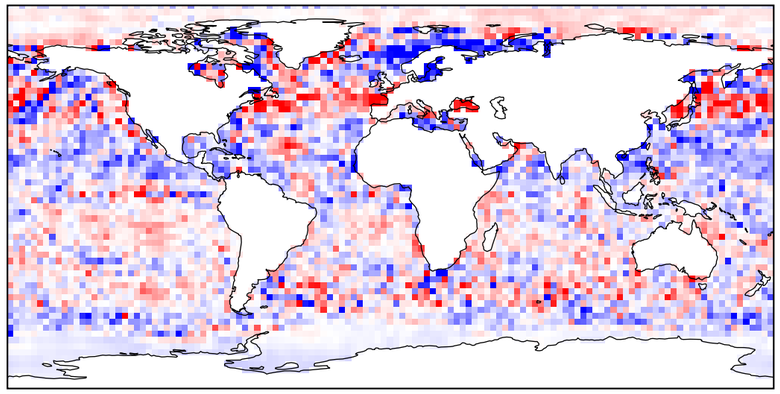
Actual — Step 1 (°C)



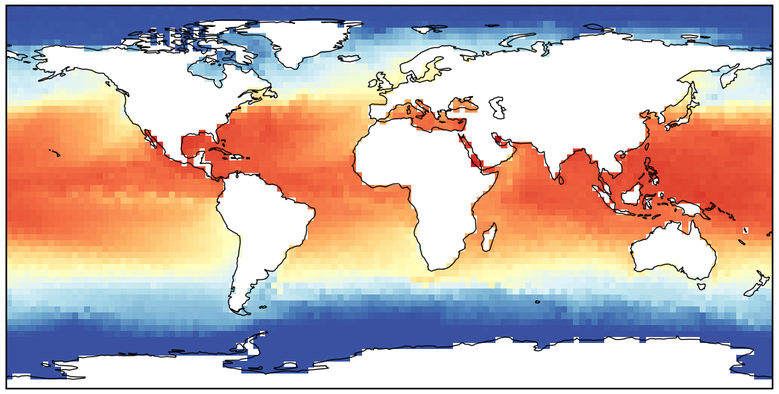
SFNO — Step 1 (°C)



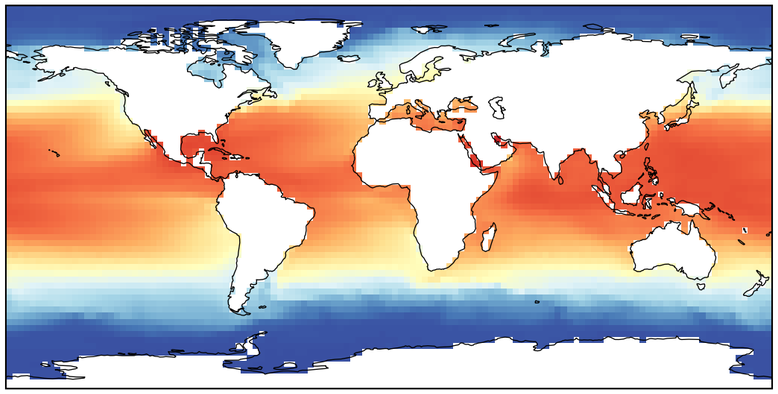
Error — Step 1 (°C)



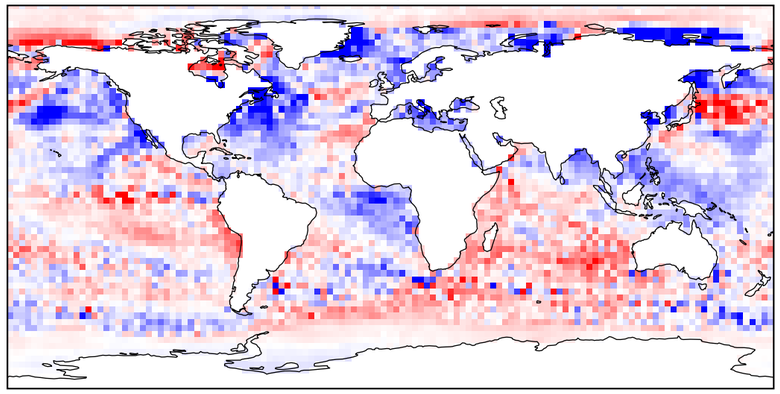
Actual — Step 46 (°C)



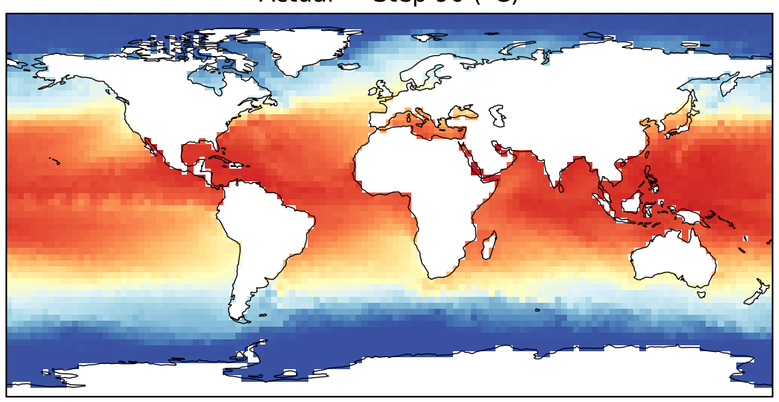
SFNO — Step 46 (°C)



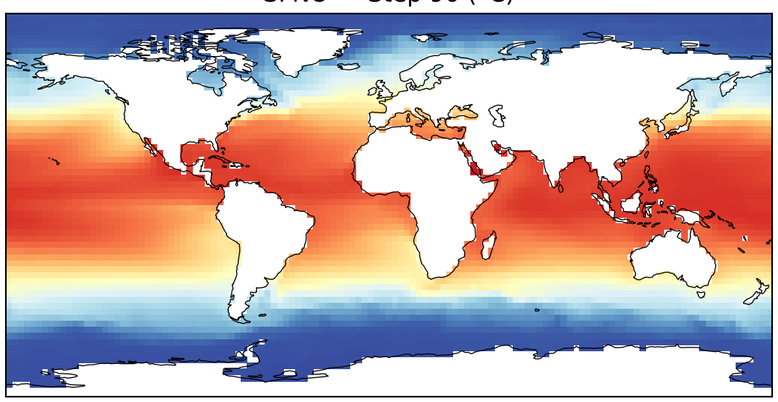
Error — Step 46 (°C)



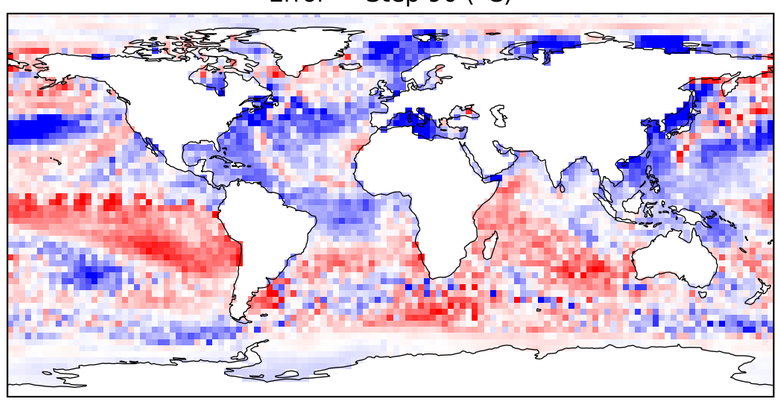
Actual — Step 90 (°C)

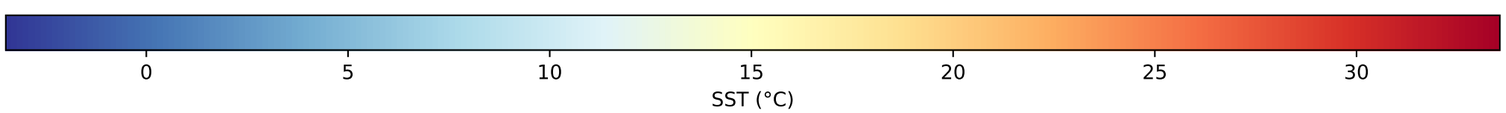


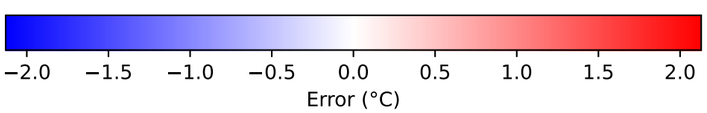
SFNO — Step 90 (°C)



Error — Step 90 (°C)

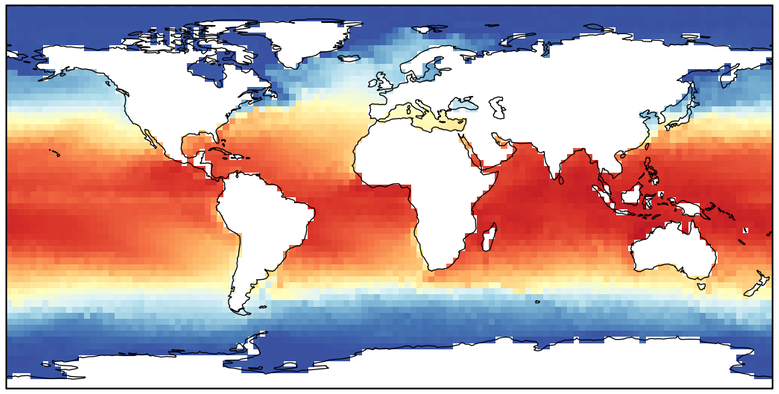




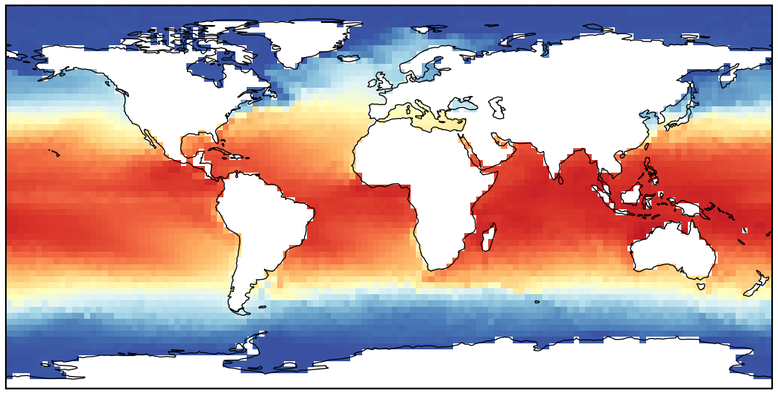


Chain init: 2022-04-01 — Test (°C)

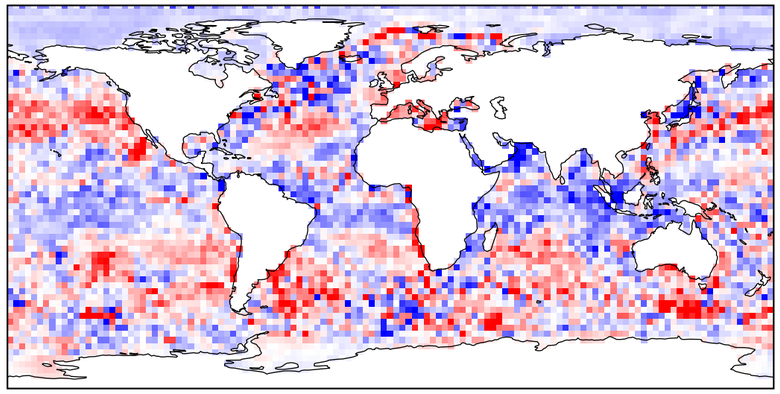
Actual — Step 1 (°C)



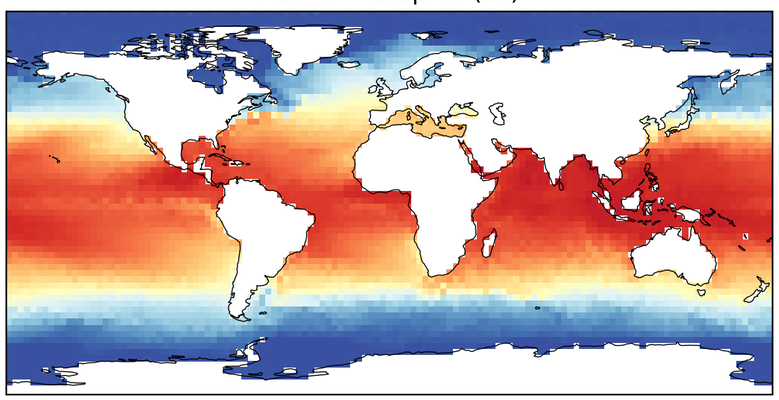
SFNO — Step 1 (°C)



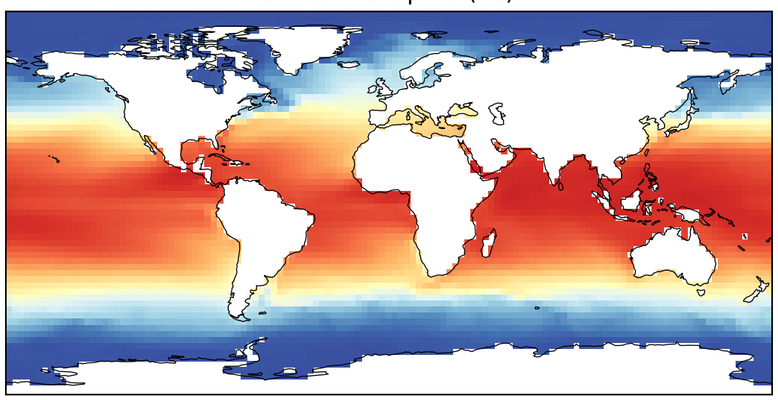
Error — Step 1 (°C)



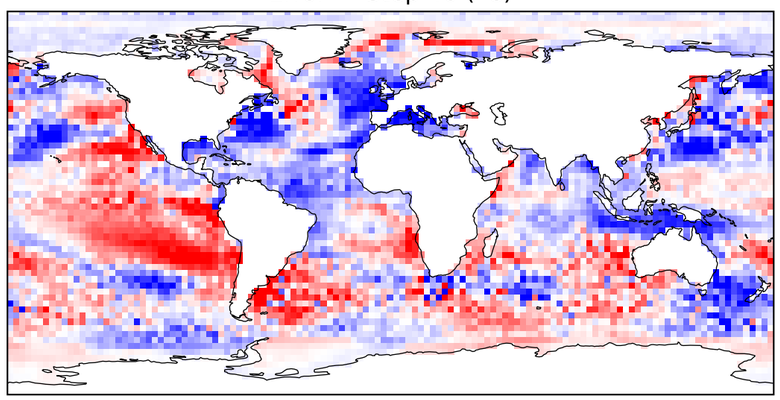
Actual — Step 46 (°C)



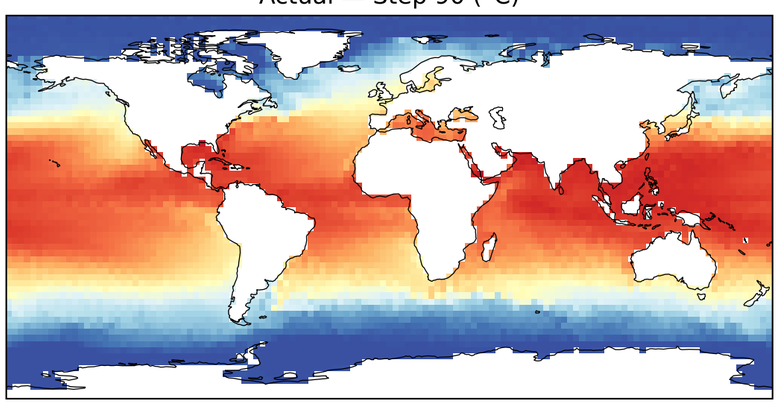
SFNO — Step 46 (°C)



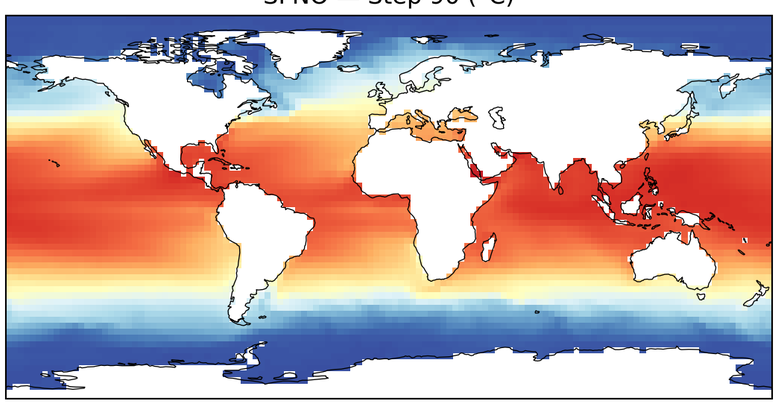
Error — Step 46 (°C)



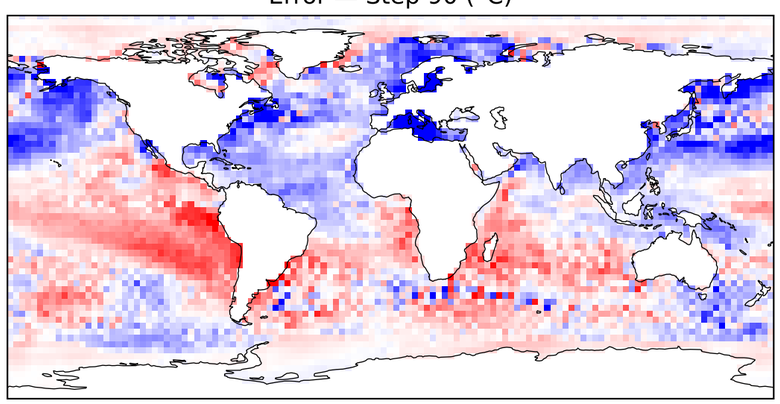
Actual — Step 90 (°C)

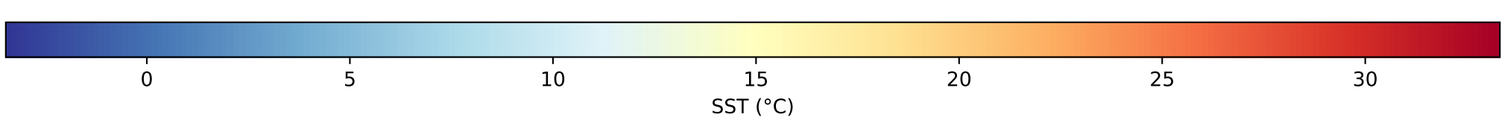


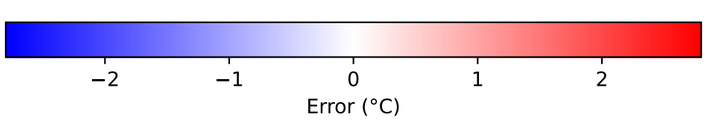
SFNO — Step 90 (°C)



Error — Step 90 (°C)

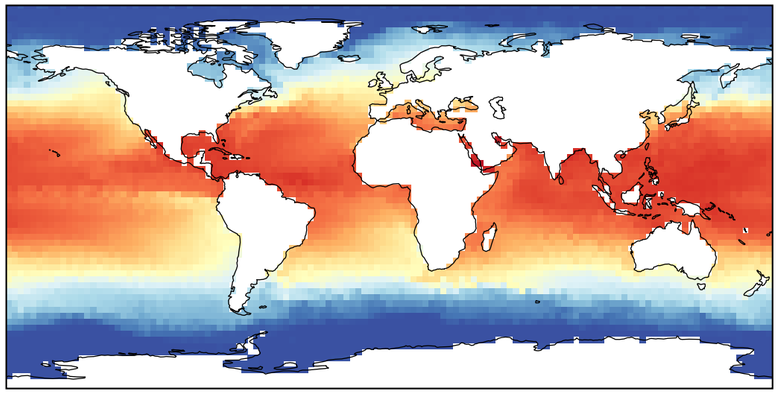




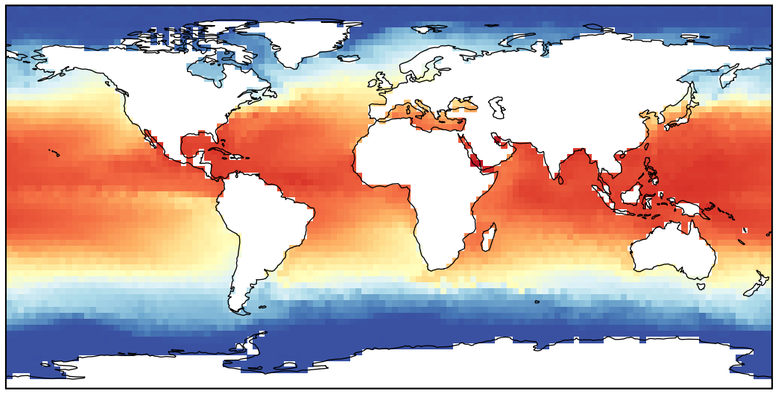


Chain init: 2022-09-28 — Test (°C)

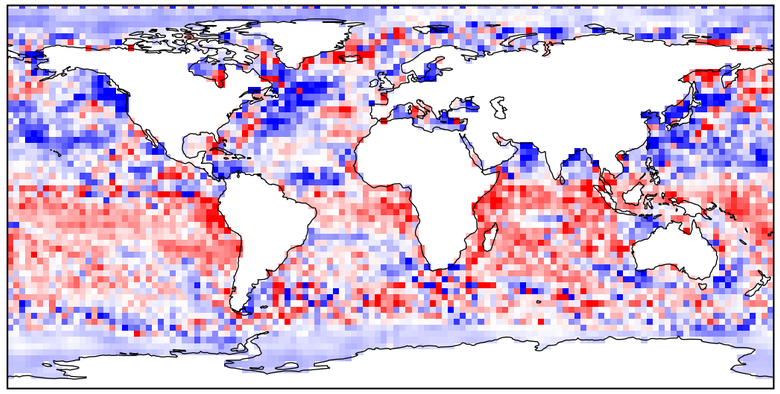
Actual — Step 1 (°C)



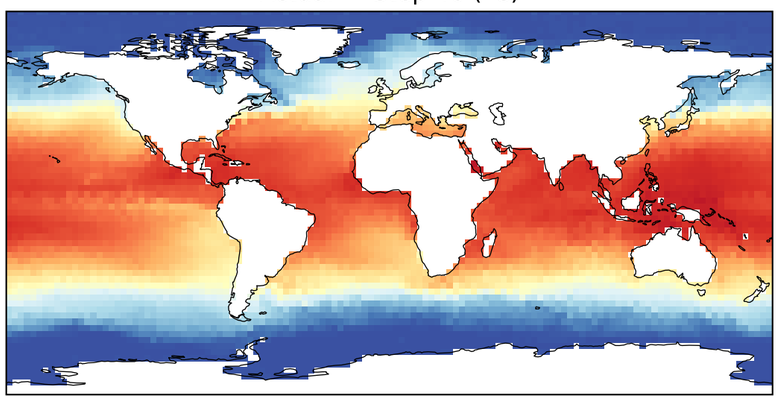
SFNO — Step 1 (°C)



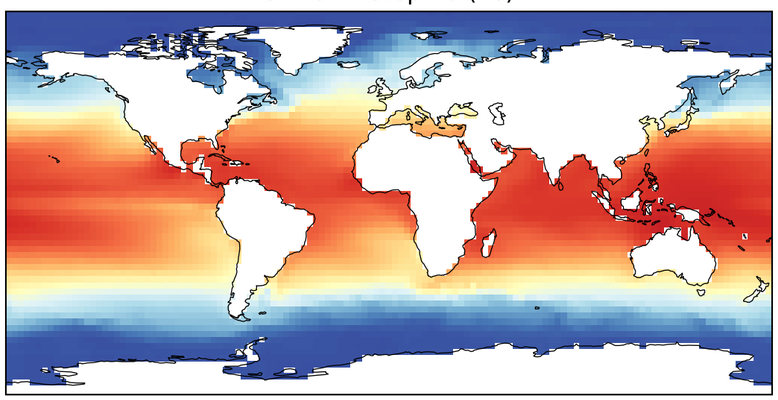
Error — Step 1 (°C)



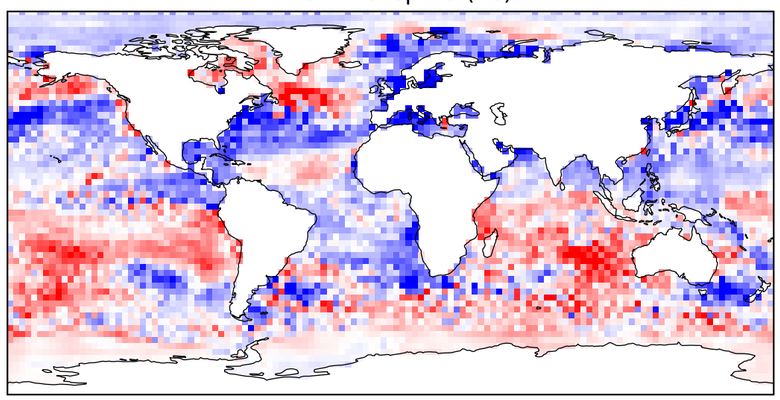
Actual — Step 46 (°C)



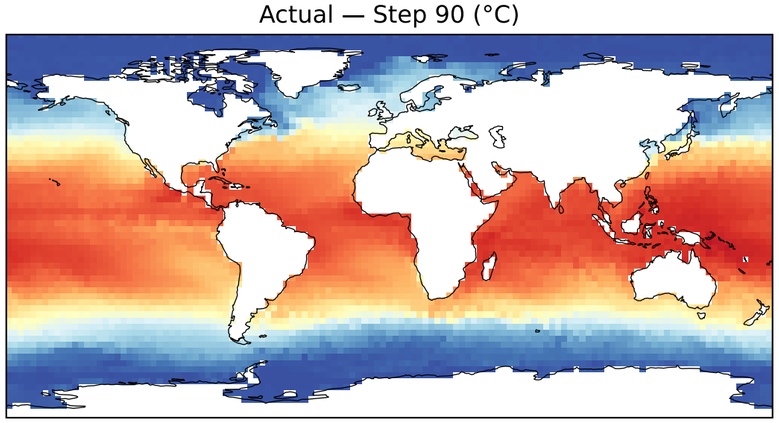
SFNO — Step 46 (°C)



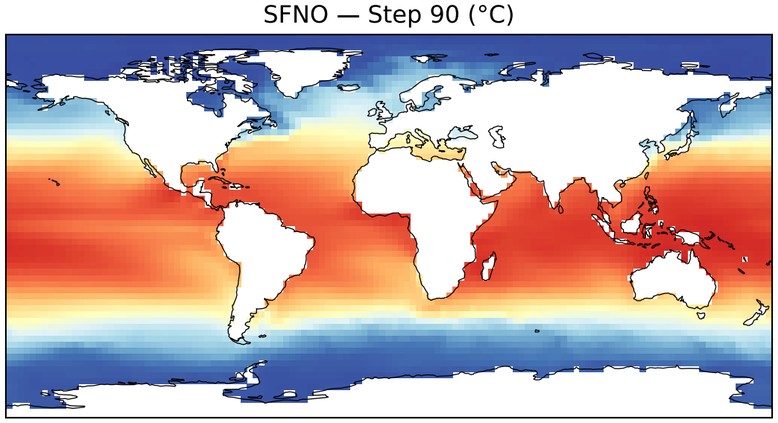
Error — Step 46 (°C)



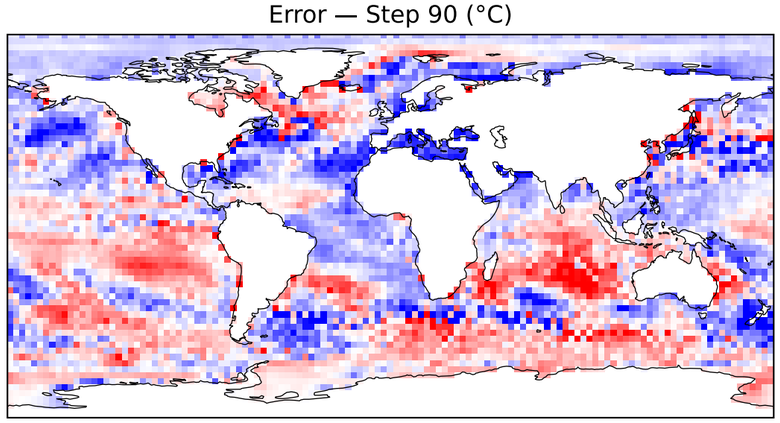
Actual — Step 90 (°C)

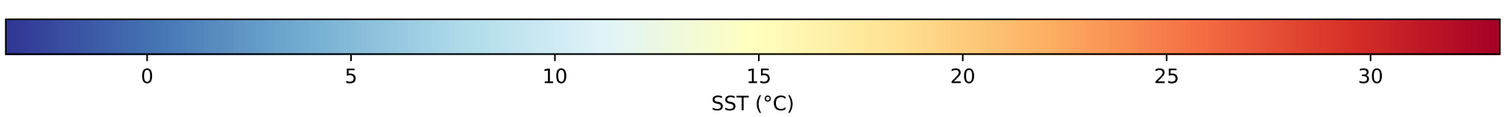


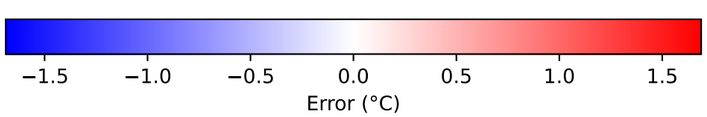
SFNO — Step 90 (°C)



Error — Step 90 (°C)

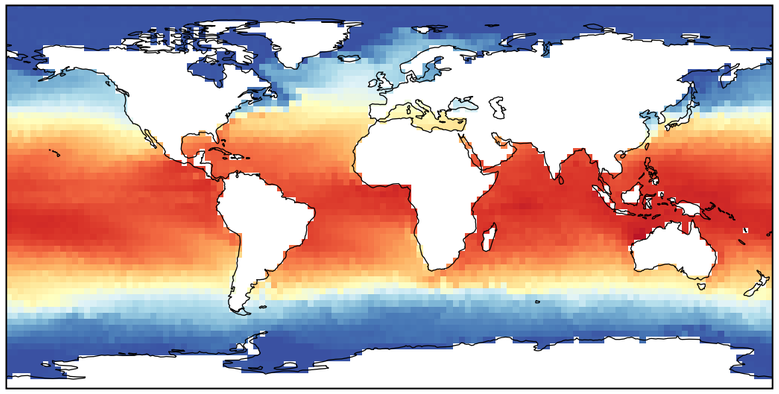




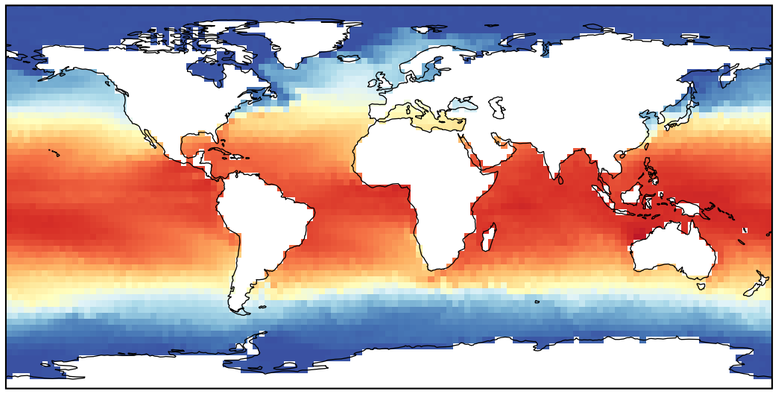


Chain init: 2023-03-27 — Test (°C)

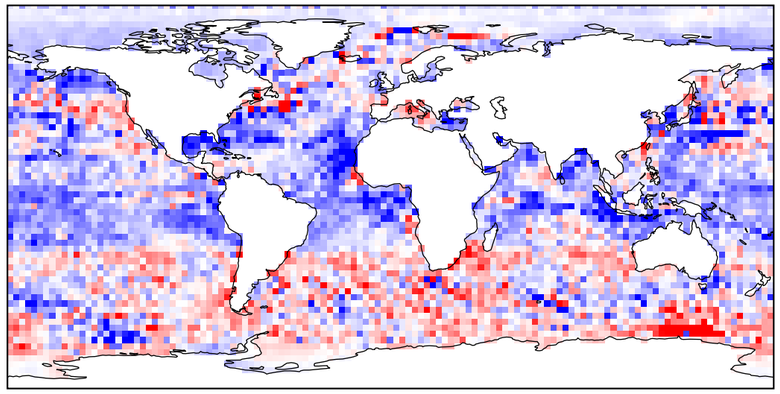
Actual — Step 1 (°C)



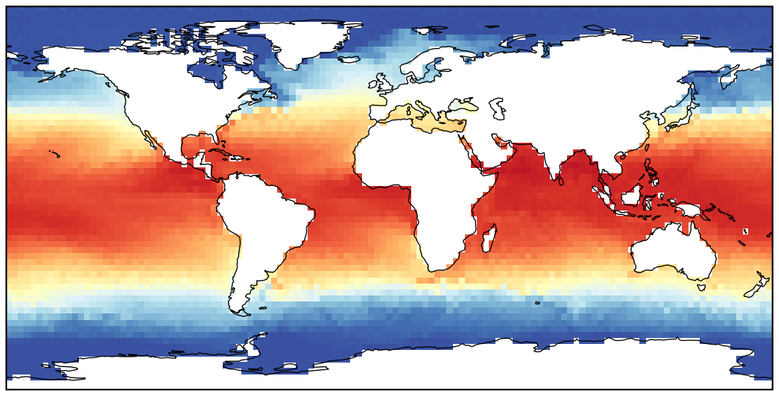
SFNO — Step 1 (°C)



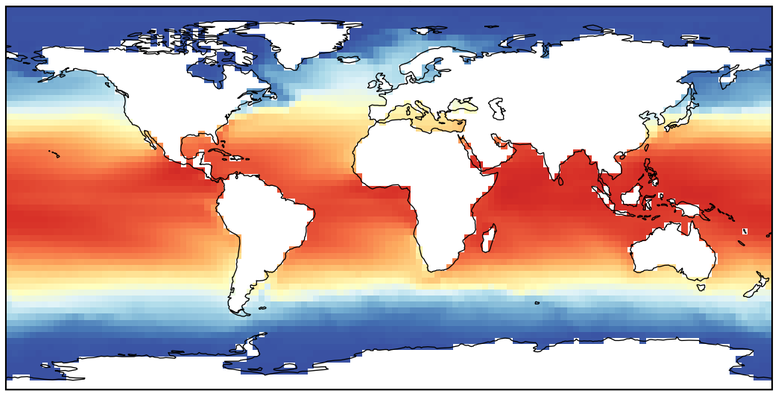
Error — Step 1 (°C)



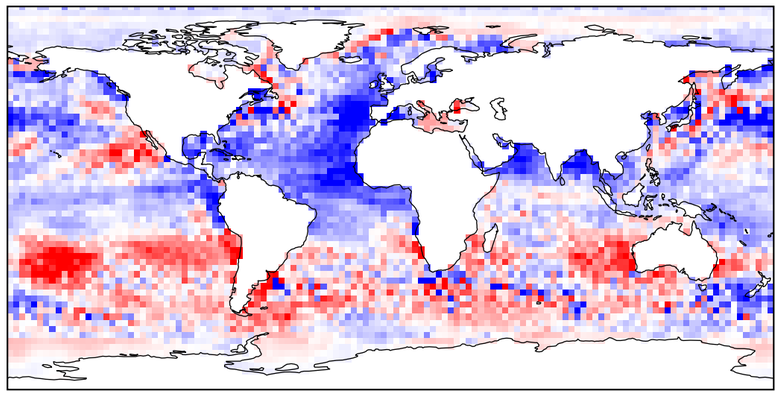
Actual — Step 46 (°C)



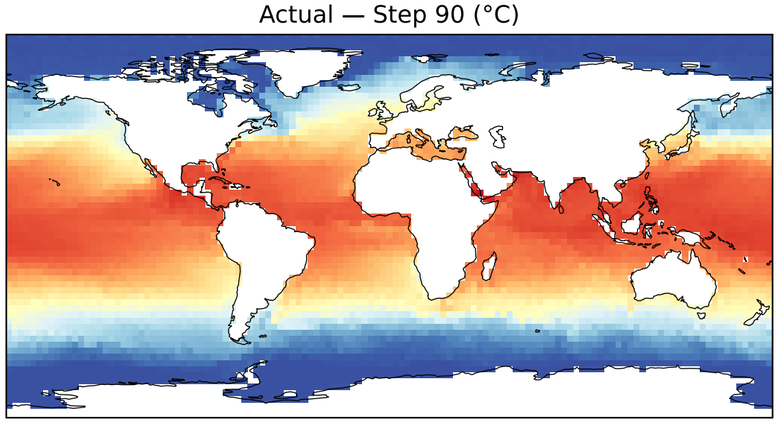
SFNO — Step 46 (°C)



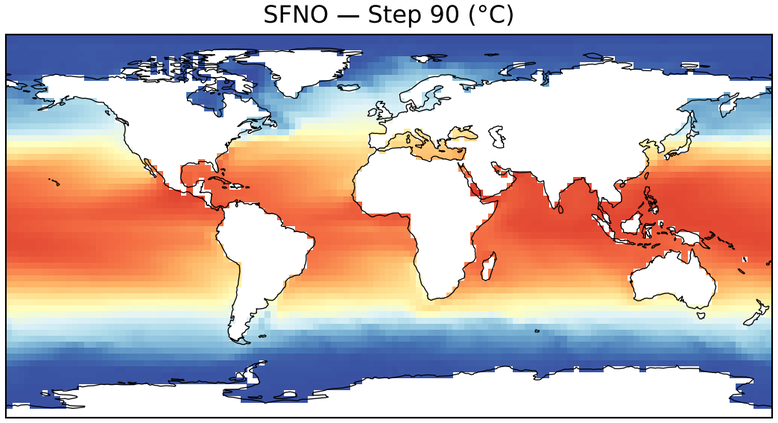
Error — Step 46 (°C)



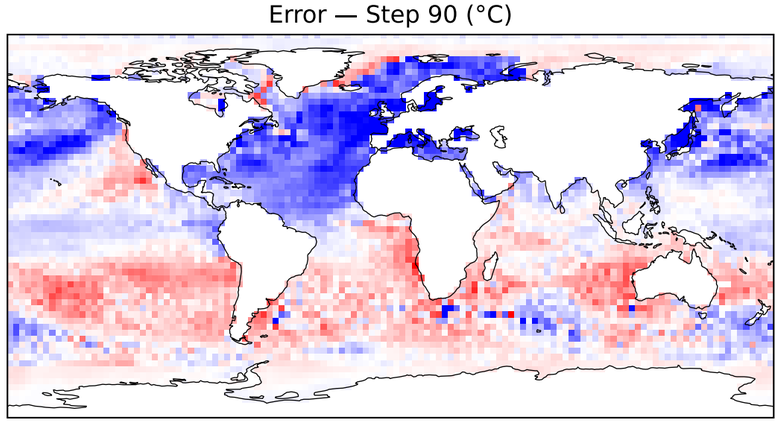
Actual — Step 90 (°C)

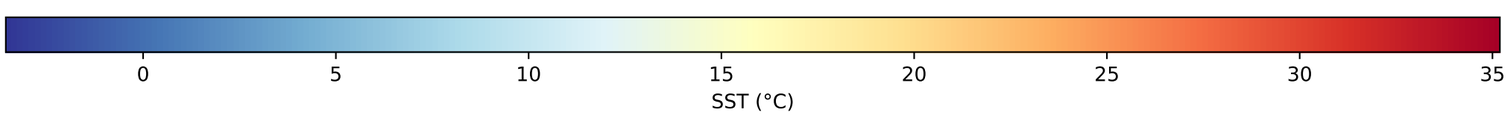


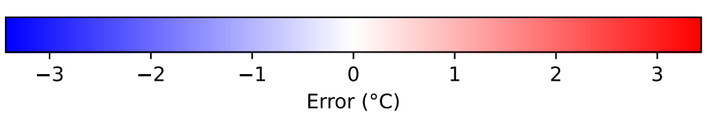
SFNO — Step 90 (°C)



Error — Step 90 (°C)

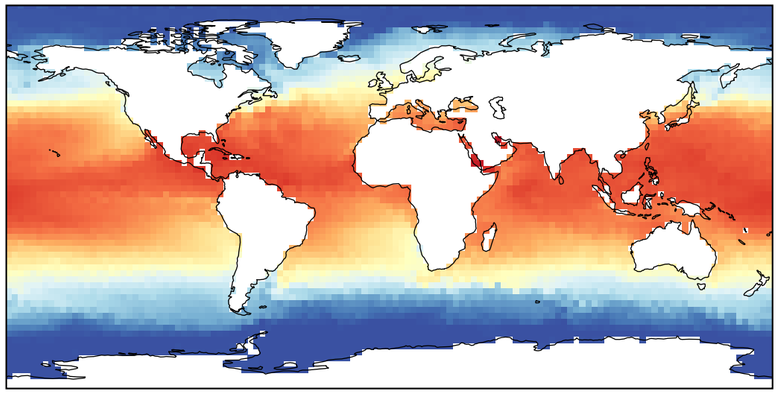




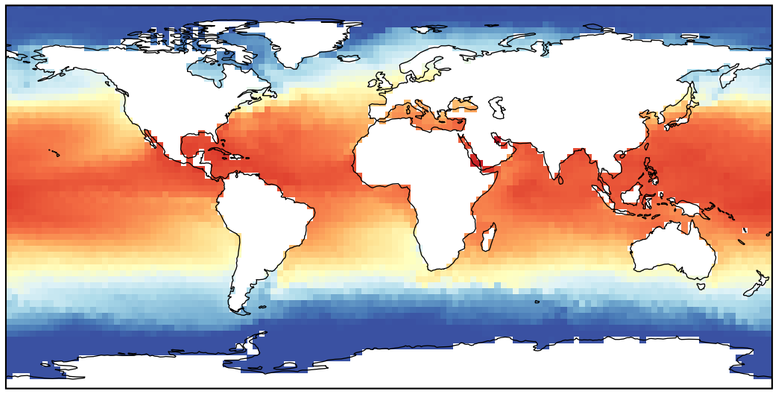


Chain init: 2023-09-23 — Test (°C)

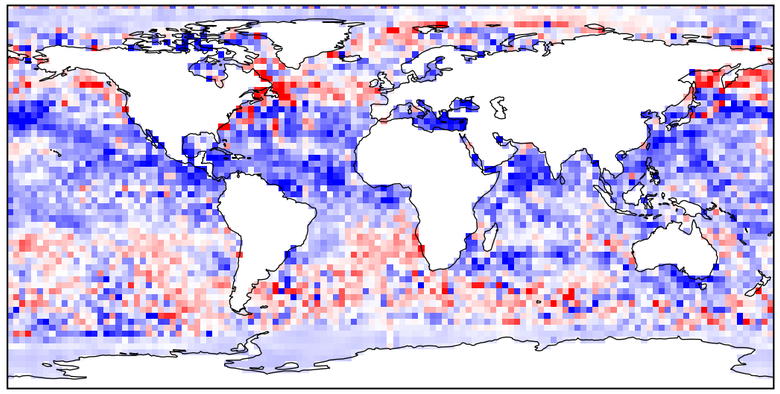
Actual — Step 1 (°C)



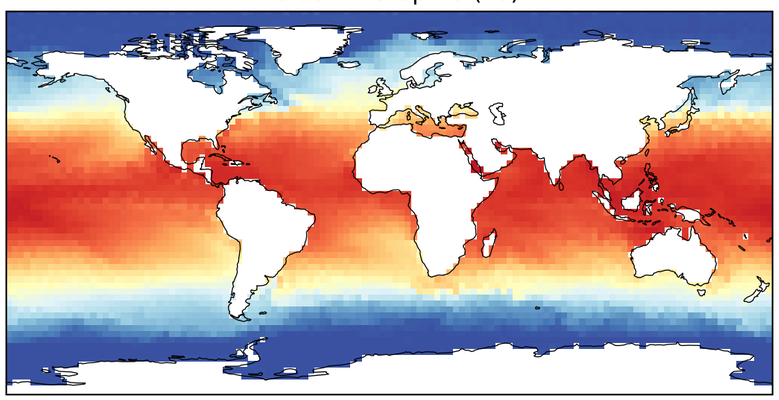
SFNO — Step 1 (°C)



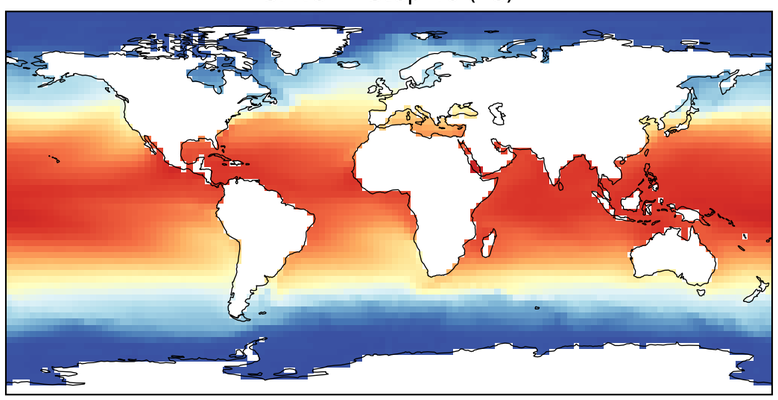
Error — Step 1 (°C)



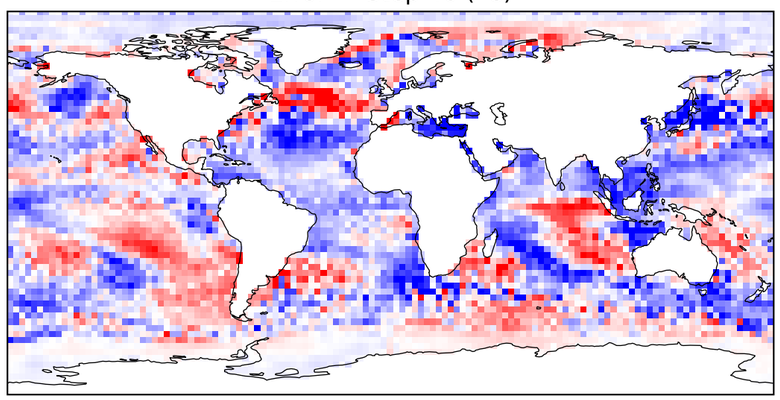
Actual — Step 46 (°C)



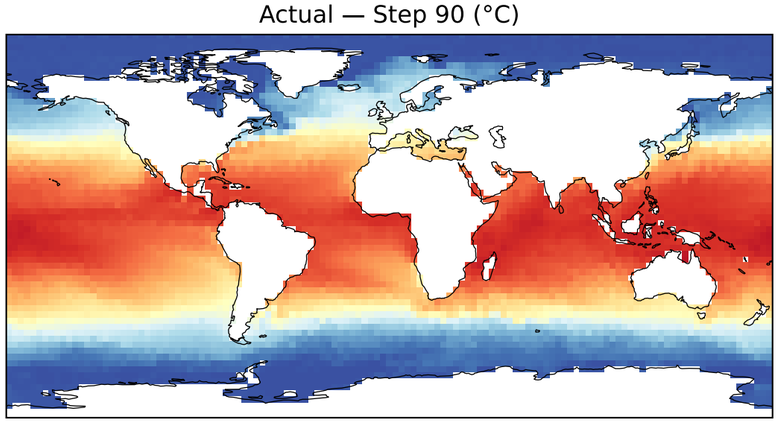
SFNO — Step 46 (°C)



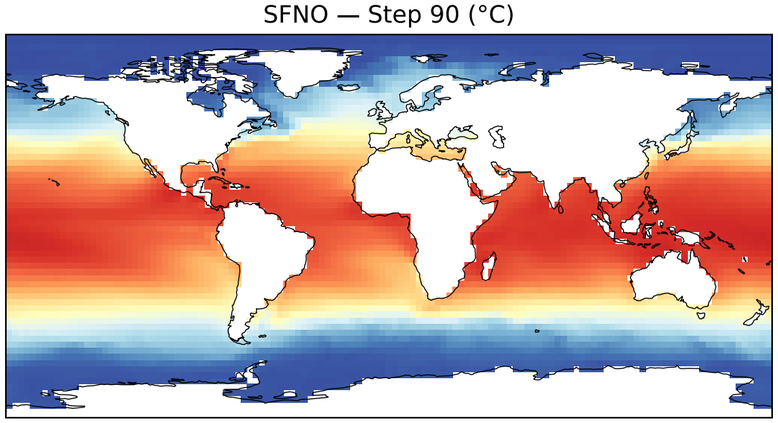
Error — Step 46 (°C)



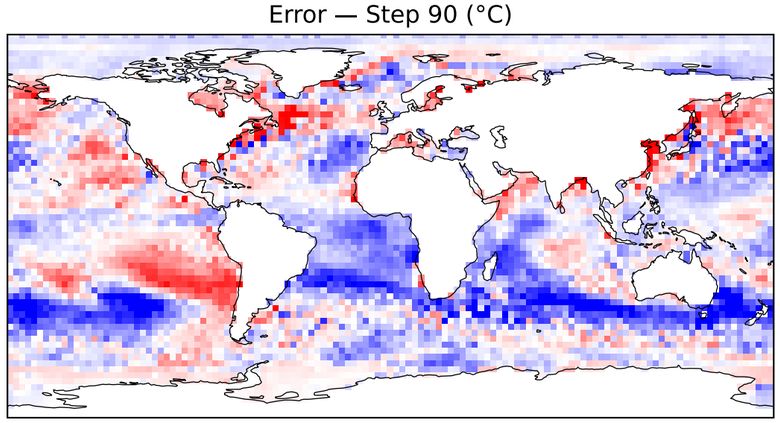
Actual — Step 90 (°C)

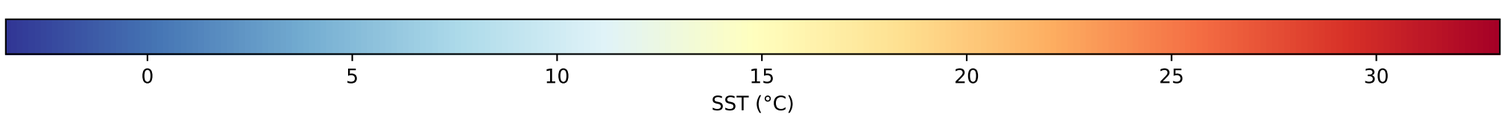


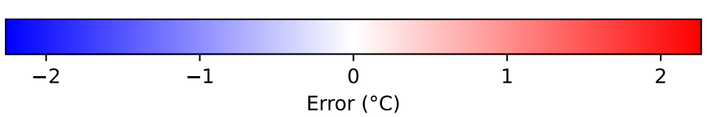
SFNO — Step 90 (°C)



Error — Step 90 (°C)

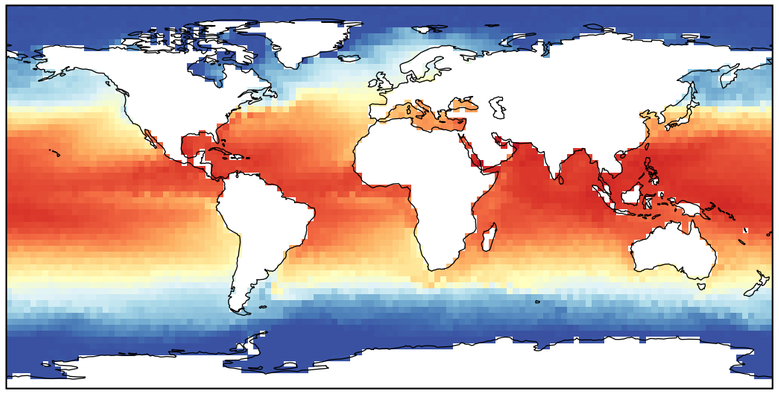




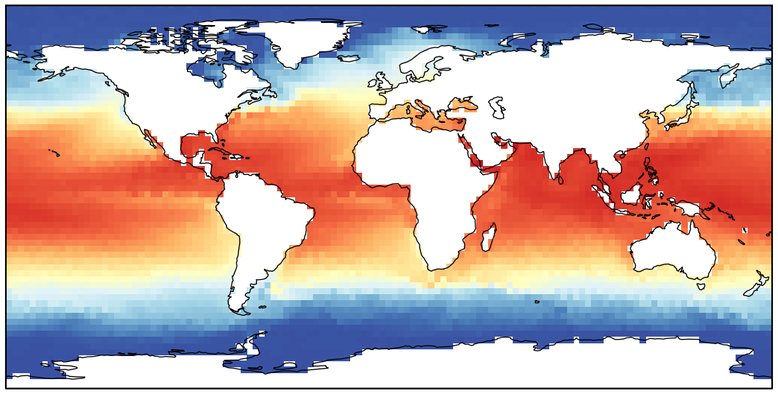


Chain init: 2024-06-19 — Test (°C)

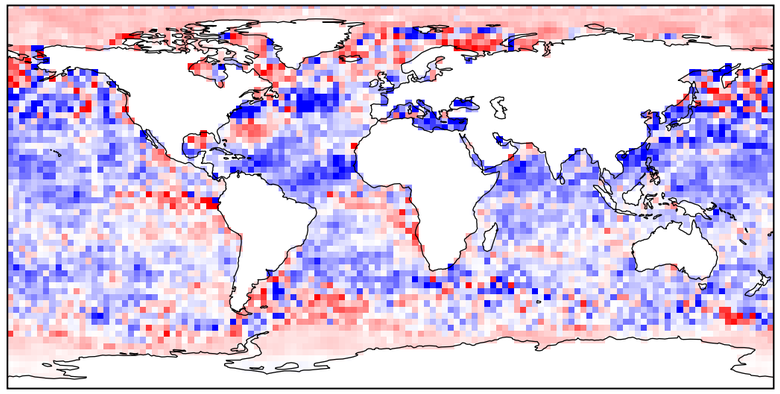
Actual — Step 1 (°C)



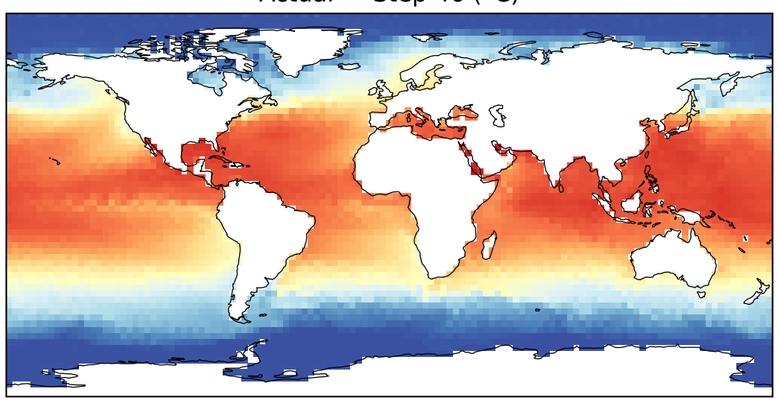
SFNO — Step 1 (°C)



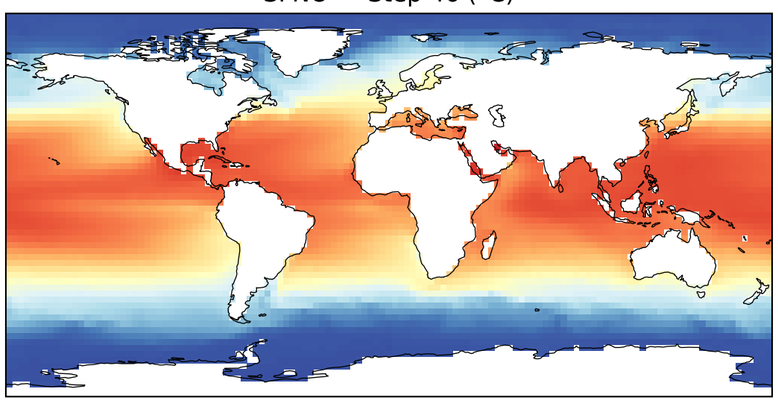
Error — Step 1 (°C)



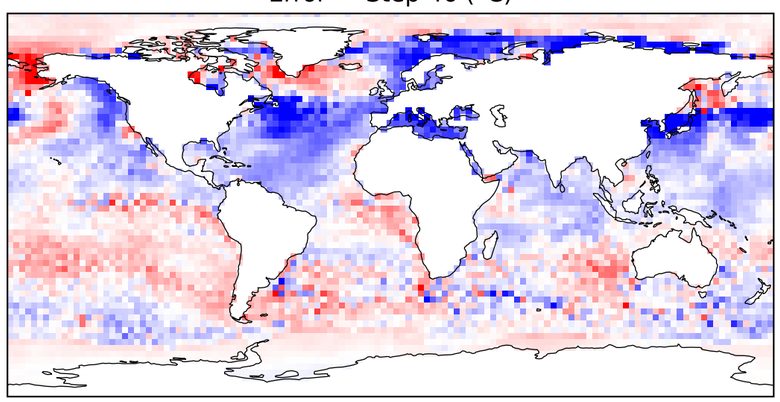
Actual — Step 46 (°C)



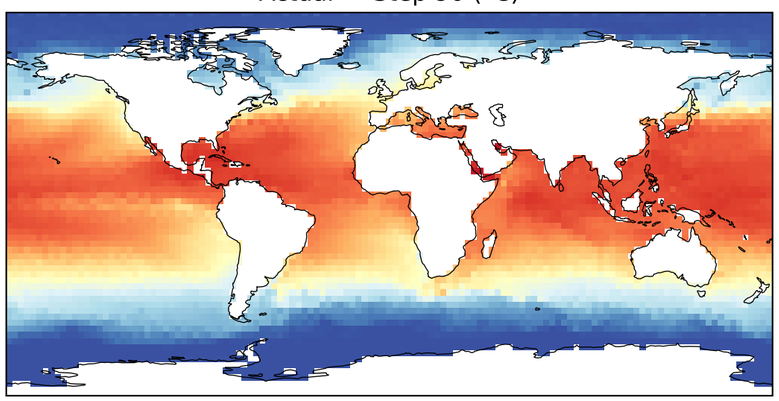
SFNO — Step 46 (°C)



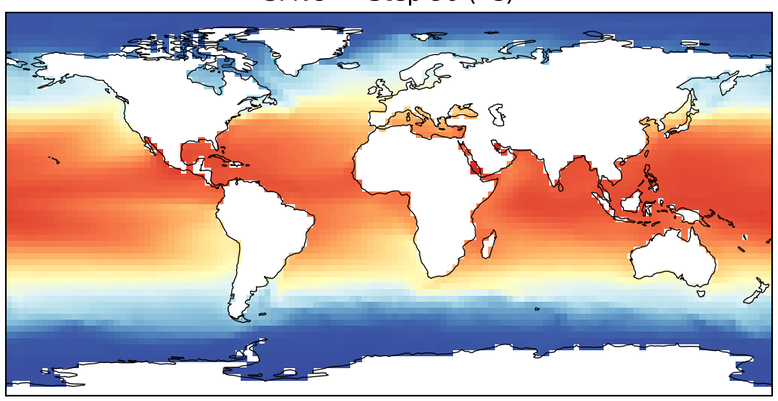
Error — Step 46 (°C)



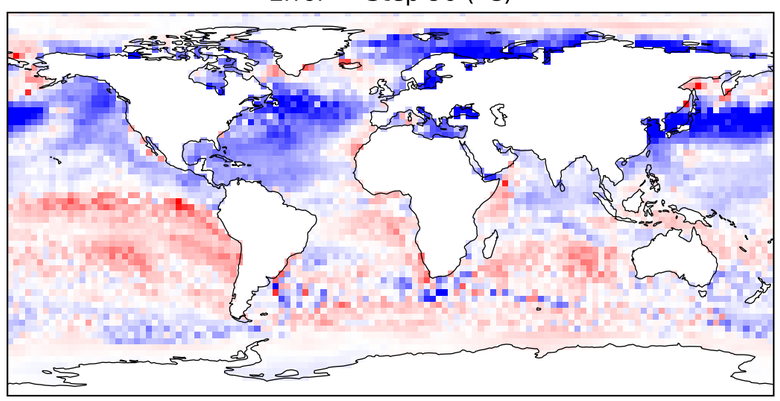
Actual — Step 90 (°C)

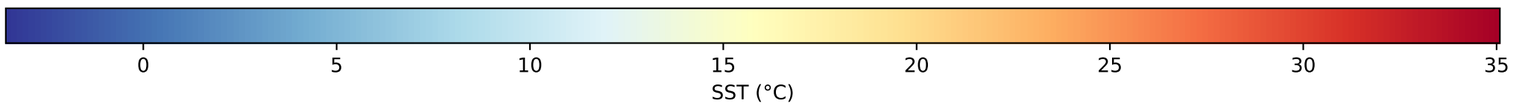


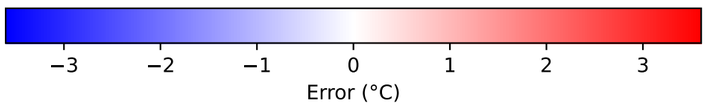
SFNO — Step 90 (°C)



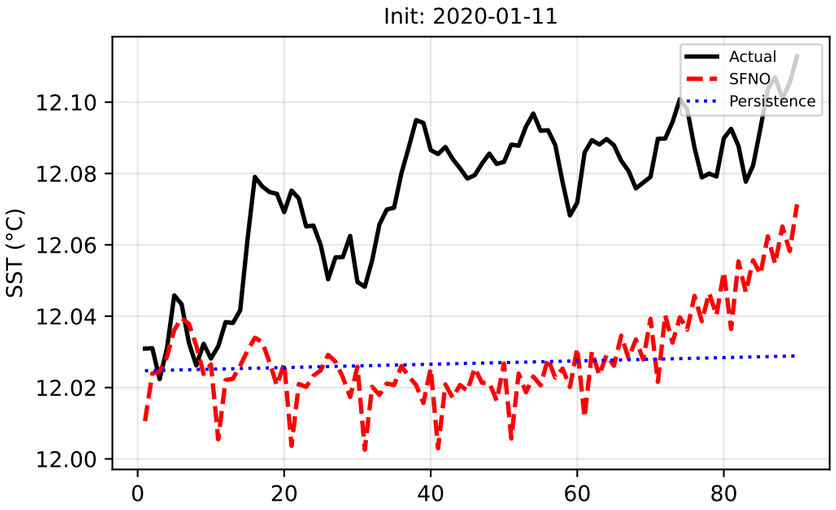
Error — Step 90 (°C)

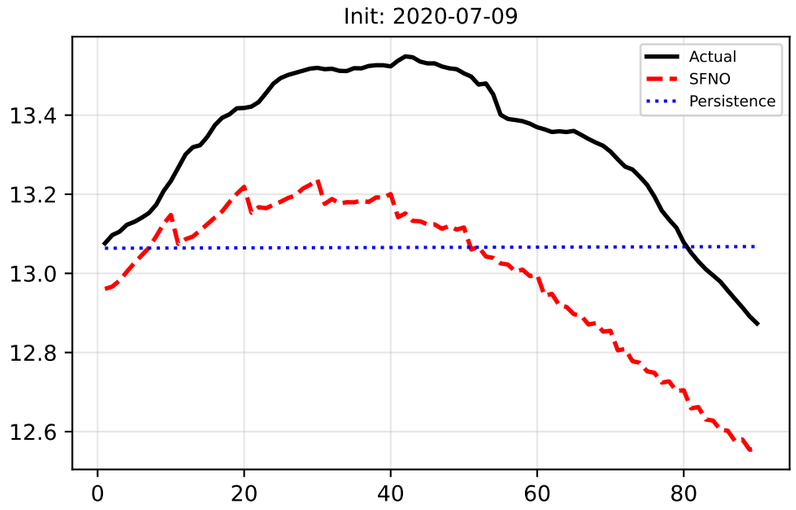


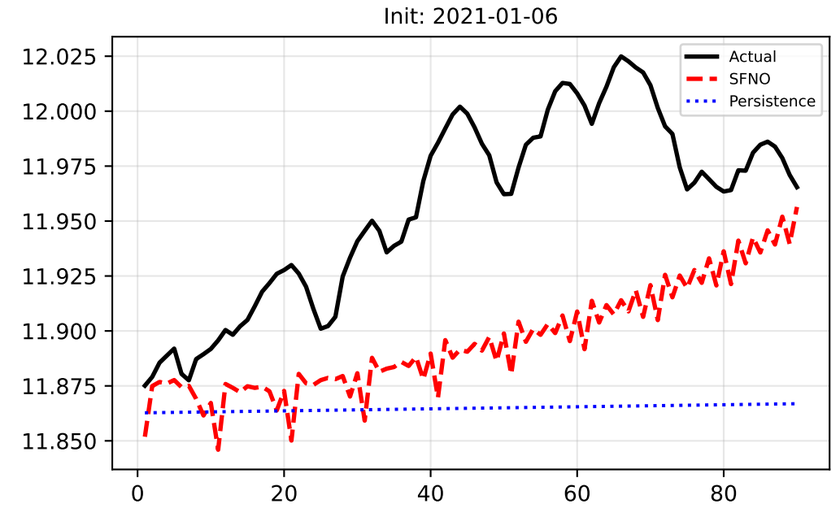


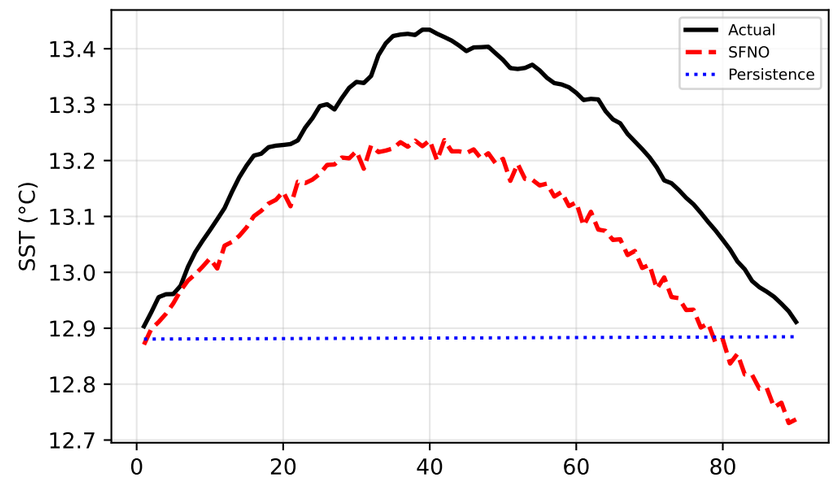


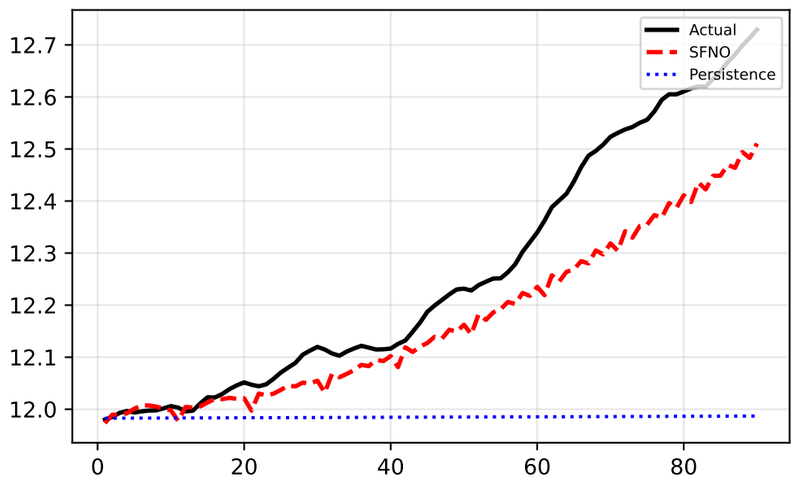
Global-Mean SST — Test (°C)

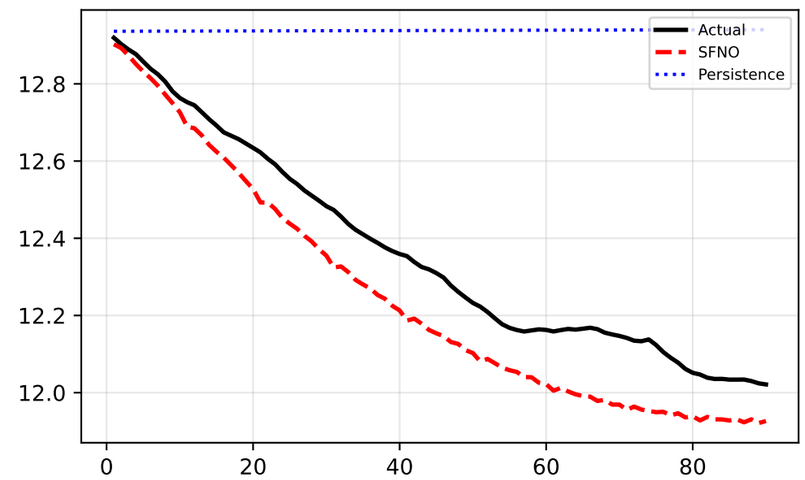


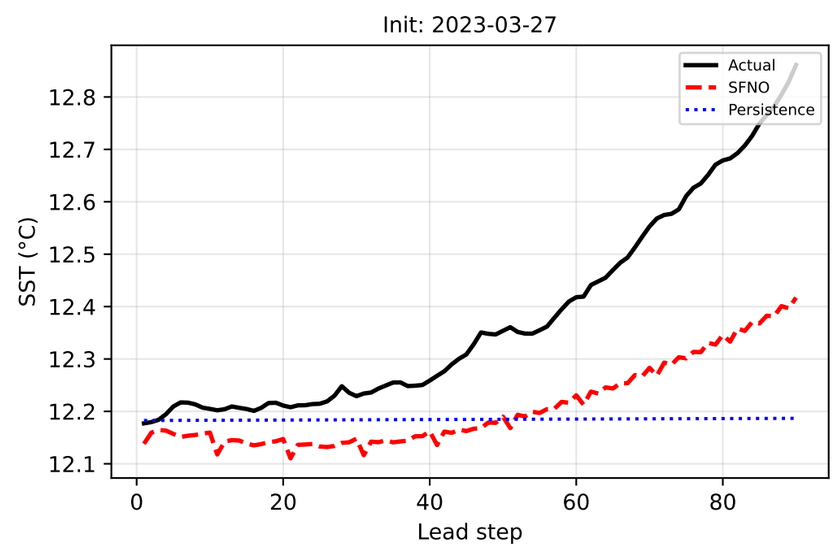


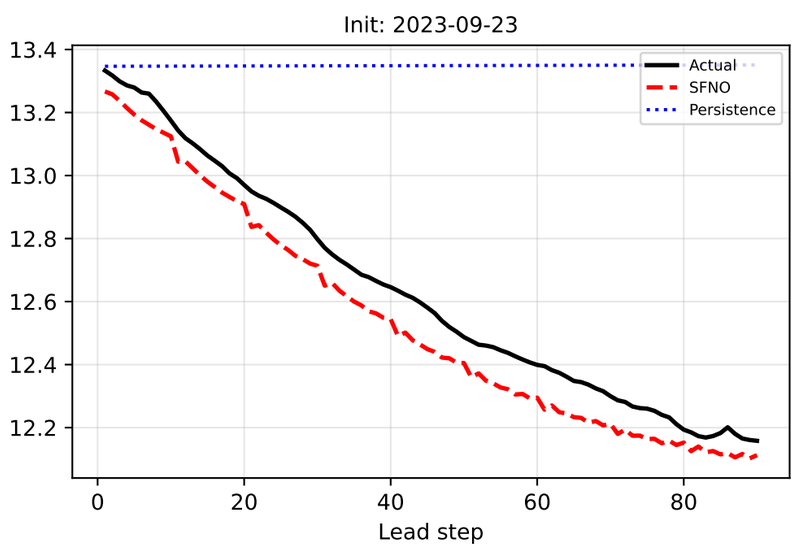


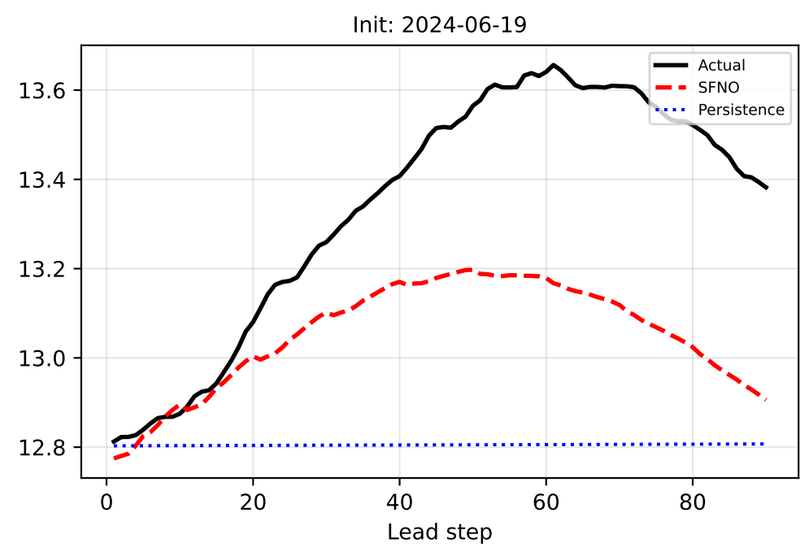








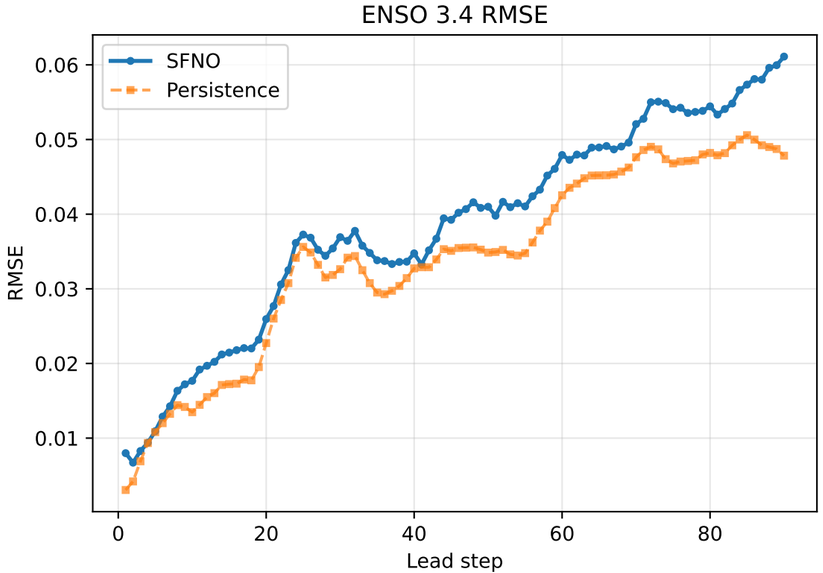


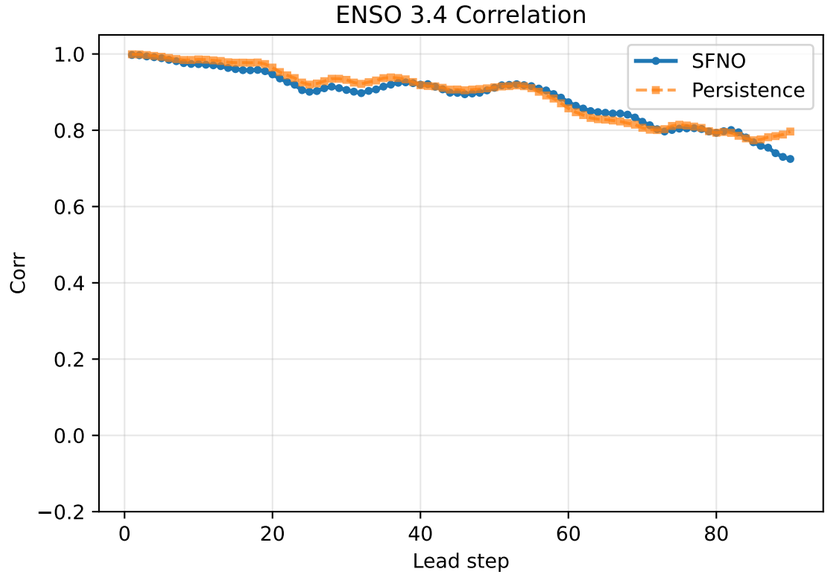


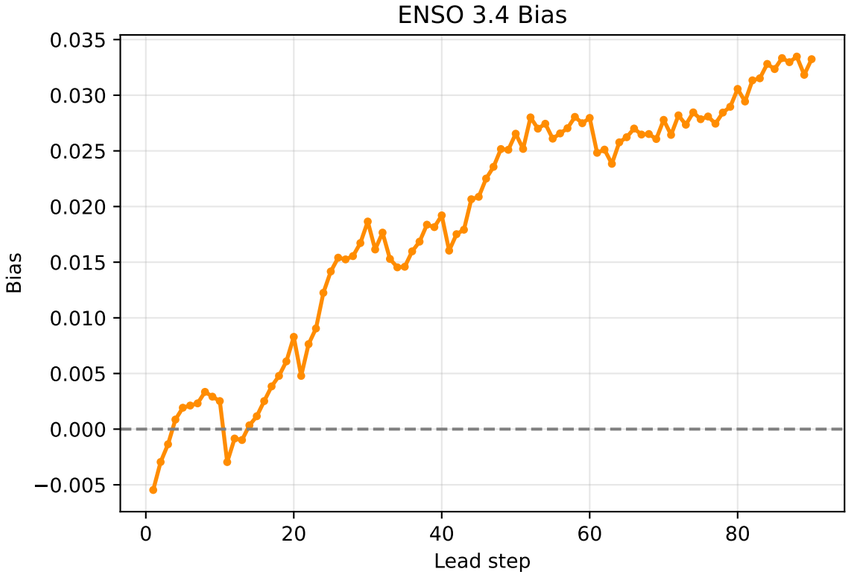
ENSO 3.4 Metrics — Test

Lead Step	SFNO RMSE	Persist RMSE	SFNO Corr	Persist Corr	SFNO Bias
1	0.0080	0.0031	0.9974	0.9992	-0.0055
2	0.0067	0.0042	0.9967	0.9989	-0.0030
3	0.0083	0.0069	0.9936	0.9972	-0.0014
4	0.0094	0.0093	0.9916	0.9950	0.0009
5	0.0109	0.0108	0.9890	0.9927	0.0019
6	0.0129	0.0120	0.9847	0.9896	0.0021
7	0.0143	0.0133	0.9811	0.9867	0.0023
8	0.0163	0.0144	0.9761	0.9842	0.0033
9	0.0172	0.0142	0.9739	0.9843	0.0029
10	0.0177	0.0135	0.9734	0.9857	0.0025
11	0.0192	0.0145	0.9718	0.9850	-0.0030
12	0.0197	0.0155	0.9703	0.9833	-0.0008
13	0.0202	0.0160	0.9682	0.9817	-0.0010
14	0.0212	0.0171	0.9633	0.9784	0.0003
15	0.0215	0.0172	0.9604	0.9772	0.0012
16	0.0218	0.0173	0.9576	0.9774	0.0025
17	0.0221	0.0178	0.9572	0.9770	0.0038
18	0.0220	0.0177	0.9582	0.9777	0.0048
19	0.0232	0.0195	0.9549	0.9736	0.0061
20	0.0259	0.0227	0.9466	0.9644	0.0083
21	0.0277	0.0260	0.9358	0.9526	0.0048
22	0.0306	0.0285	0.9265	0.9440	0.0076
23	0.0325	0.0308	0.9194	0.9369	0.0090
24	0.0361	0.0341	0.9052	0.9251	0.0122
25	0.0373	0.0356	0.9009	0.9197	0.0142
26	0.0368	0.0349	0.9028	0.9226	0.0154
27	0.0352	0.0332	0.9101	0.9287	0.0153
28	0.0344	0.0315	0.9145	0.9351	0.0155
29	0.0354	0.0319	0.9105	0.9350	0.0167
30	0.0369	0.0326	0.9056	0.9322	0.0187
31	0.0364	0.0342	0.9012	0.9251	0.0161
32	0.0378	0.0344	0.8974	0.9217	0.0177
33	0.0358	0.0325	0.9032	0.9261	0.0153
34	0.0348	0.0308	0.9071	0.9305	0.0145
35	0.0338	0.0295	0.9145	0.9367	0.0146
36	0.0337	0.0293	0.9197	0.9386	0.0160
37	0.0333	0.0297	0.9245	0.9370	0.0168
38	0.0336	0.0304	0.9256	0.9336	0.0184
39	0.0336	0.0314	0.9235	0.9265	0.0182
40	0.0347	0.0327	0.9196	0.9182	0.0192
41	0.0333	0.0329	0.9210	0.9161	0.0160
42	0.0352	0.0329	0.9142	0.9151	0.0175
43	0.0367	0.0339	0.9073	0.9113	0.0179
44	0.0394	0.0353	0.8984	0.9066	0.0207
45	0.0392	0.0351	0.8979	0.9069	0.0209
46	0.0402	0.0355	0.8944	0.9049	0.0225
47	0.0407	0.0355	0.8964	0.9067	0.0236
48	0.0416	0.0355	0.8983	0.9080	0.0252
49	0.0408	0.0352	0.9045	0.9099	0.0251
50	0.0410	0.0349	0.9113	0.9127	0.0265
51	0.0398	0.0349	0.9179	0.9148	0.0252
52	0.0417	0.0352	0.9190	0.9157	0.0280
53	0.0409	0.0346	0.9209	0.9176	0.0270
54	0.0415	0.0345	0.9182	0.9155	0.0274
55	0.0410	0.0348	0.9157	0.9104	0.0261
56	0.0424	0.0362	0.9093	0.9010	0.0266
57	0.0433	0.0378	0.9044	0.8912	0.0270
58	0.0452	0.0390	0.8949	0.8831	0.0281
59	0.0461	0.0408	0.8856	0.8706	0.0275
60	0.0479	0.0425	0.8734	0.8575	0.0280
61	0.0473	0.0435	0.8642	0.8473	0.0248
62	0.0480	0.0441	0.8570	0.8398	0.0251
63	0.0479	0.0448	0.8503	0.8324	0.0238
64	0.0489	0.0452	0.8476	0.8290	0.0258
65	0.0489	0.0452	0.8459	0.8279	0.0262
66	0.0491	0.0452	0.8440	0.8256	0.0270
67	0.0487	0.0453	0.8439	0.8230	0.0265
68	0.0491	0.0457	0.8410	0.8187	0.0265
69	0.0496	0.0462	0.8334	0.8146	0.0261
70	0.0521	0.0476	0.8223	0.8065	0.0278
71	0.0528	0.0486	0.8132	0.8014	0.0264
72	0.0550	0.0490	0.8027	0.8005	0.0282
73	0.0551	0.0487	0.7967	0.8030	0.0274
74	0.0549	0.0474	0.8007	0.8113	0.0285
75	0.0541	0.0468	0.8043	0.8146	0.0278
76	0.0543	0.0470	0.8050	0.8124	0.0281
77	0.0536	0.0471	0.8054	0.8097	0.0274
78	0.0537	0.0472	0.8033	0.8062	0.0284
79	0.0538	0.0480	0.7969	0.7976	0.0290
80	0.0544	0.0482	0.7933	0.7935	0.0306
81	0.0533	0.0479	0.7975	0.7956	0.0294
82	0.0541	0.0482	0.8006	0.7931	0.0313
83	0.0548	0.0492	0.7947	0.7855	0.0315
84	0.0566	0.0500	0.7811	0.7786	0.0328
85	0.0574	0.0506	0.7686	0.7738	0.0324
86	0.0581	0.0500	0.7592	0.7761	0.0333
87	0.0580	0.0492	0.7544	0.7818	0.0330
88	0.0596	0.0490	0.7399	0.7847	0.0335
89	0.0600	0.0487	0.7301	0.7888	0.0318
90	0.0611	0.0478	0.7248	0.7967	0.0332

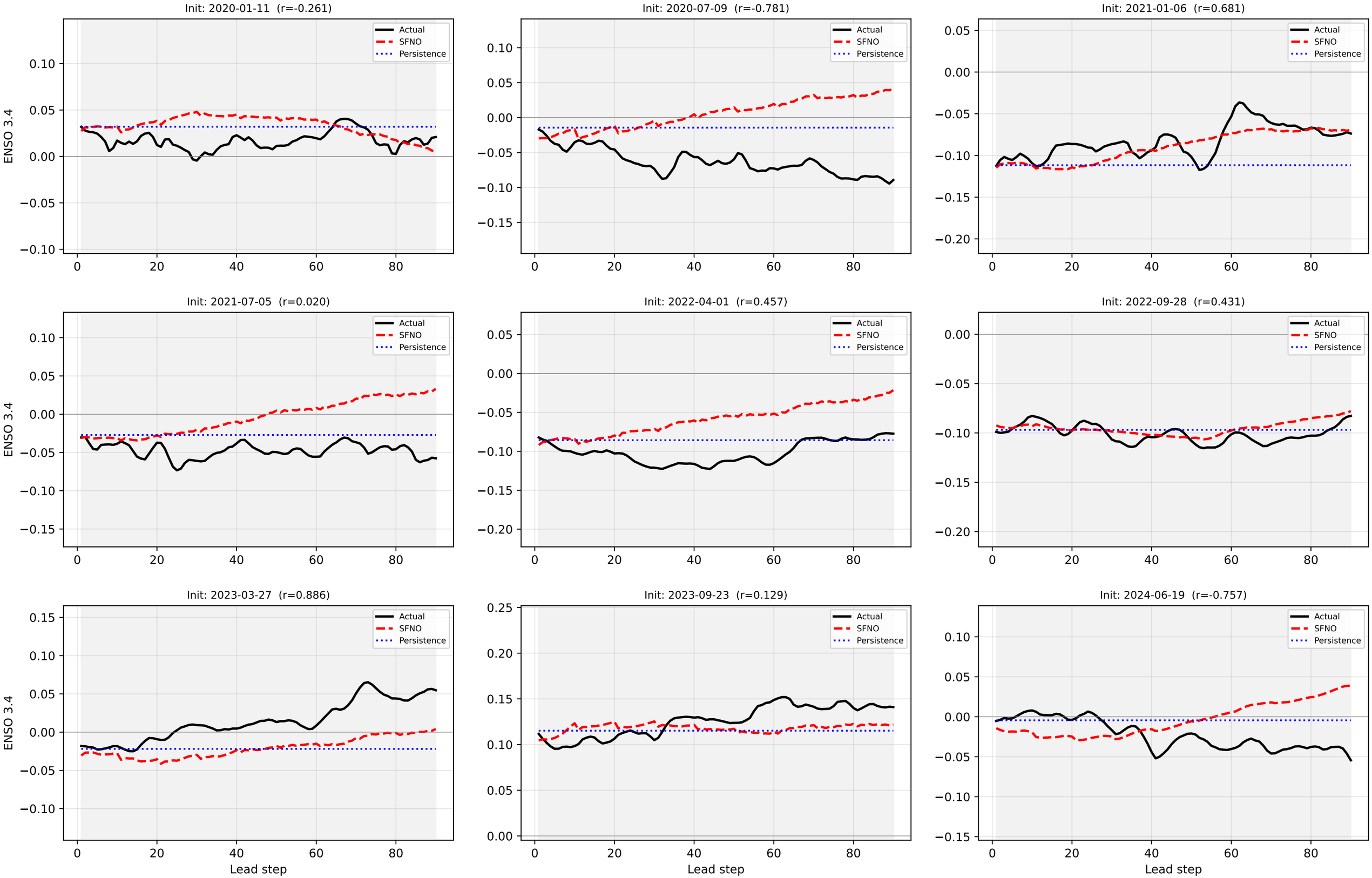
ENSO 3.4 Forecast Skill — Test







ENSO 3.4 Index — Test



ENSO 3.4 Scatter — Test